

IMPERIAL

GENERAL RELATIVITY

IMPERIAL COLLEGE LONDON

DEPARTMENT OF PHYSICS

Comprehensive Lecture Summaries on General Relativity

*Based on Lectures by
Prof. Claudia de Rham*

Author:

Eugene (Yoogeun) Song

Date: May 4, 2025

These summaries draw upon the lecturer's outstanding presentations and are further enhanced by my own insights and carefully curated materials from respected sources, including:

- Introducing Einstein's Relativity by Ray d'Inverno
- Spacetime and Geometry: An Introduction to General Relativity by Sean Carroll
- A First Course in General Relativity by Bernard Schutz

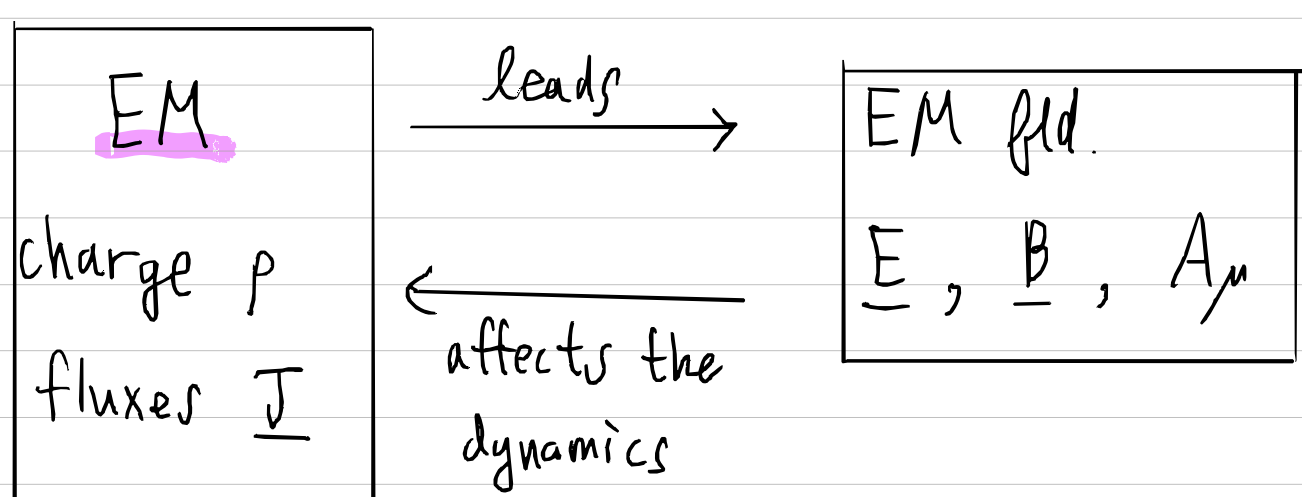
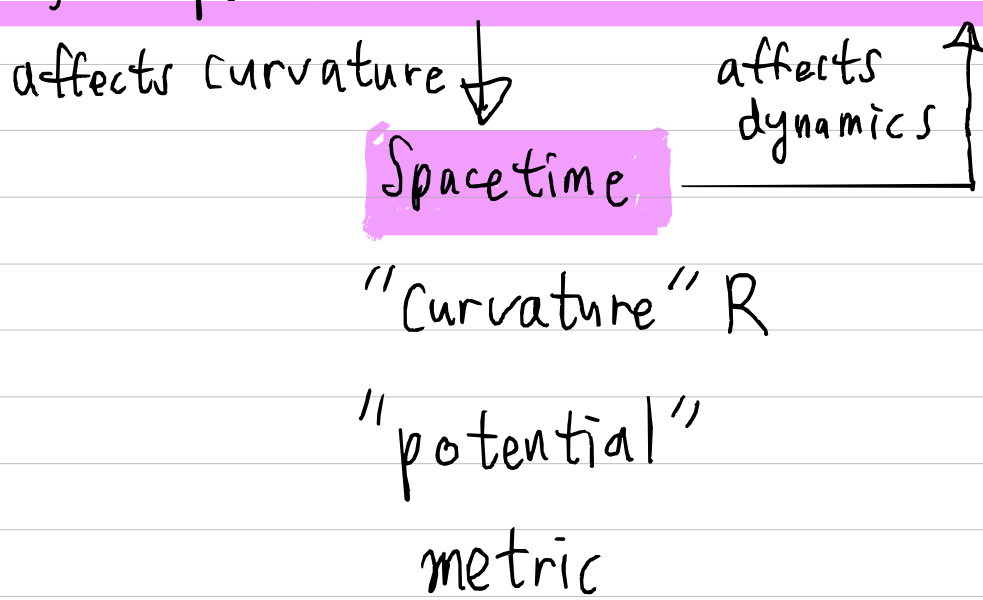
I hope these compiled summaries prove beneficial and insightful for students who aspire to learn GR.

– Yoogeun Song.

* 1 way to phrase Equivalence Principle:

In a grav. fld., a free-falling (small enough) region of sp.-tm. (i.e., in a "small lab" for a short period of time), the laws of physics are the same as in an inertial frame in the absence of gravity.

Anything living in the universe : particles, Energy, Pressure, Temperature.

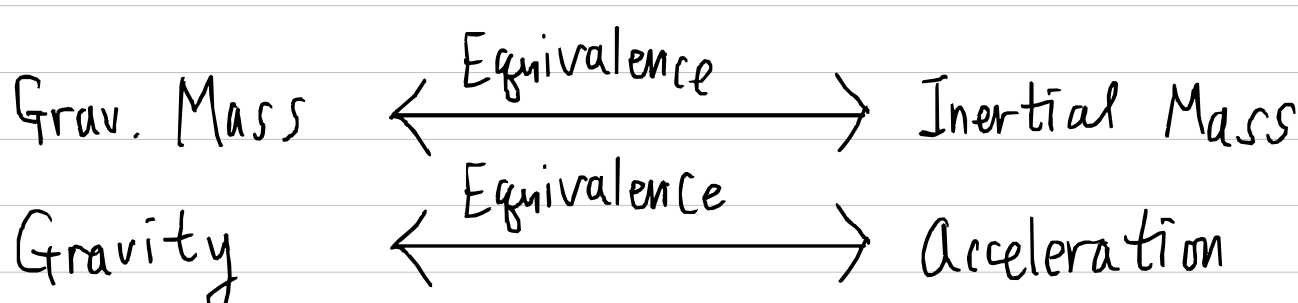
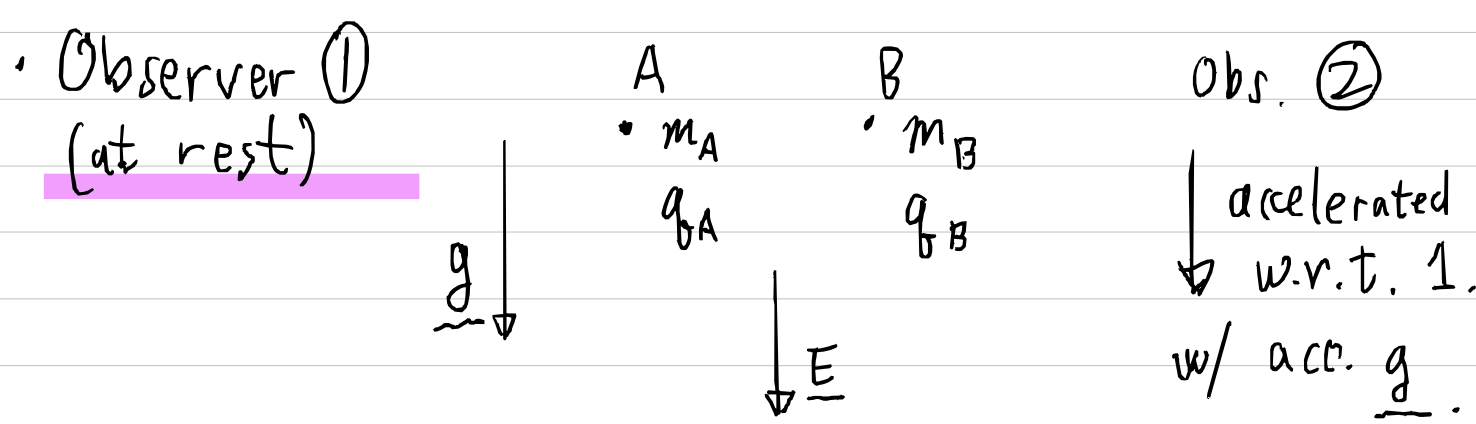


photon (γ) : massless spin-1 fld.

graviton : massless spin-2 fld.

Chap. 1 : Curved Spacetime

§1.1. Gravity & Spacetime Geometry (Acceleration)

(1) Equivalence Principle* 2 massive & charged particles A, B in a gravitational & Elec. fld. (parallel to e.o.).

According to ① :

$$\left\{ \begin{array}{l} m_A \underline{\ddot{r}}_{A,1} = q_A \underline{E} + m_A \underline{g} \\ m_B \underline{\ddot{r}}_{B,2} = q_B \underline{E} + m_B \underline{g} \end{array} \right\}$$

$$\underline{\ddot{r}}_2 = \underline{\ddot{r}}_1 - \underline{g} \quad \text{w.r.t. observer ②}$$

$$\begin{aligned} m_A \underline{\ddot{r}}_{A,2} &= m_A \underline{\ddot{r}}_{A,1} - m_A \underline{g} \\ &= q_A \underline{E} \end{aligned}$$

$$m_B \underline{\ddot{r}}_{B,2} = q_B \underline{E} \quad (\text{For the observer B, there's no such thing as a grav. force})$$

there's no such thing

as a grav. force)

$\underline{E}$ remains the same since $\underline{E} \parallel \underline{g}$

②

Gravity is NOT equal to the other forces of nature because of the equivalence principle.

* A 3rd observer. "③" $\rightarrow$ ③ is also accelerated w.r.t. ①

$$q_A \underline{E} = m_A \underline{\ddot{R}}_A$$

$$\underline{\ddot{R}}_{3,A,B} = \underline{\ddot{R}}_{1,A,B} - \frac{q_A}{m_A} \underline{E}$$

For obs. ③,

$$m_A \underline{\ddot{R}}_{3,A} = m_A \underline{\ddot{R}}_{1,A} - \cancel{m_A} \frac{q_A}{\cancel{m_A}} \underline{E} = m_A \underline{g}$$

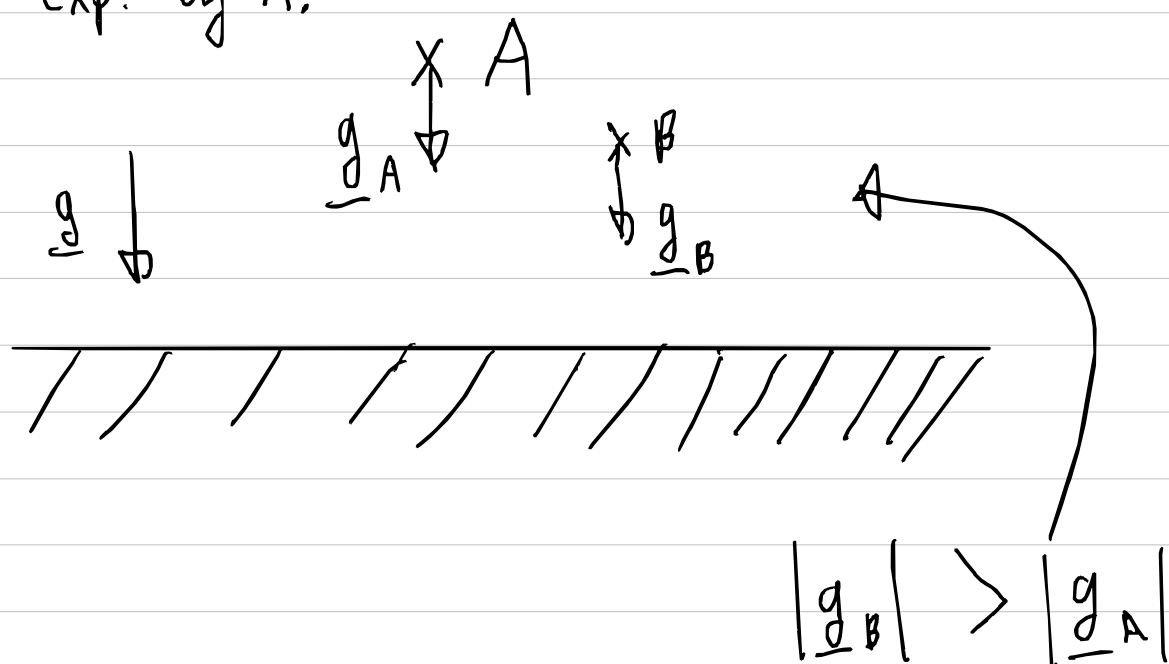
$$m_B \underline{\ddot{R}}_{3,B} = \left(q_B - \frac{m_B}{m_A} q_A \right) \underline{E} + \underbrace{m_B \underline{g}}_{\substack{\uparrow \\ \text{still present.}}}$$

(2) Tidal Forces

Inability to distinguish b/t gravity & acc. is $\left(\underline{g} \quad \& \quad \underline{a} \right)$

only true locally. For a small region of sp.-tm.

$\hookrightarrow$ grav. fld. or acc. will be stronger than that exp. by A.



A. x A'
 B. x B'

$t=0$ $t \rightarrow$

Separation b/t A & B increases over time

* Tidal Force $\rightarrow$ a manifestation of the grav. F.
 GWs are Tidal F.s.

(3)

(3) Natural Units

Spd. of light in vacuum : $c = 2.997 \times 10^8 \text{ m/s}$

1 m = dist. travelled by light in vacuum
in 299792458 m/s.

$$c = h = 1 \quad \rightarrow \quad \overset{L}{[\text{dist.}]} = \overset{T}{[\text{time}]} = \overset{\text{Hz}}{[\text{freq.}]^{-1}}$$

$c \equiv 1$

$$\hbar = 1 = \frac{h}{2\pi} = 6.58 \times 10^{-16} \text{ eV}\cdot\text{s}$$

↑
reduced Planck scale

$$[E_n] = [\text{Time}]^{-1} = [\text{Freq.}] = [\text{dist.}]^{-1} = [WL]^{-1}$$

$$E = mc^2 \quad \rightarrow \quad E = m$$

$$[\text{spd.}] = 0 \text{ (dim.less)}$$

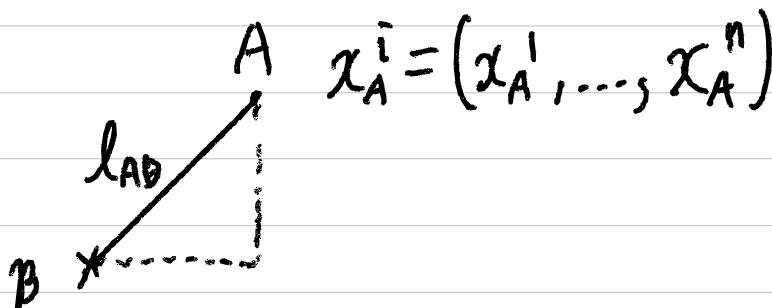
§1.3. Manifolds

A manifold is a sp., which looks locally like an n -dim. Euclidean sp. $\mathbb{R}^n$.

§2.1. Metric in Flat Euclidean Sp.

$$x_A^{\bar{i}} = (x_A^1, \dots, x_A^n) \quad \& \quad x_B^{\bar{i}} = (x_B^1, \dots, x_B^n) \\ (\bar{i} = 1, \dots, n)$$

$$x_B^{\bar{i}} = x_A^{\bar{i}} + \Delta x^{\bar{i}}$$



$$x_B^{\bar{i}} = (x_B^1, \dots, x_B^n) = (x_A^1 + \Delta x^1, \dots, x_A^n + \Delta x^n)$$

In flat sp., the dist. l_{AB} b/t the 2 pt.s A & B $\rightarrow$ given by

$$l_{AB}^2 = (\Delta x^1)^2 + \dots + (\Delta x^n)^2 = \sum_{\bar{i}, \bar{j}=1}^n \delta_{\bar{i}\bar{j}} \Delta x^{\bar{i}} \Delta x^{\bar{j}}$$

$$l_{AB}^2 = \delta_{\bar{i}\bar{j}} \Delta x^{\bar{i}} \Delta x^{\bar{j}}$$

$$V \cdot W = V^1 W^1 + V^2 W^2 + \dots + V^n W^n = \delta_{\bar{i}\bar{j}} V^{\bar{i}} W^{\bar{j}}$$

(2)

If V^i & $W^i : 2$ vec.s

$$V^i = (V^1, \dots, V^n) \quad \& \quad W^i = (W^1, \dots, W^n).$$

$$V \cdot W = \sum_{i=1}^n V^i W^i = \sum_{i,j=1}^n \delta_{ij} V^i W^j$$

GR L3

$$ds^2 = g_{ab}(x) dx^a dx^b$$

(Eq. (1-24) LN)

$$V \cdot W = g_{ab}(x) V^a W^b$$

$$x^a \rightarrow \tilde{x}^a \quad \text{w/} \quad x^a = x^a(\tilde{x})$$

$$ds^2 = g_{rr} dr^2 + 2 g_{r\theta} dr d\theta + g_{\theta\theta} d\theta^2$$

$$\downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow$$

$$= dr^2 + 0 + r^2 d\theta^2$$

$$g_{rr} = 1$$

$$g_{r\theta} = g_{\theta r} = 0$$

$$g_{\theta\theta} = r^2$$

$$\left\{ \begin{array}{l} \text{Matrix notation} \\ g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \\ = g_{ij}(r, \theta) \end{array} \right\}$$

At the origin, (where $r = 0$)

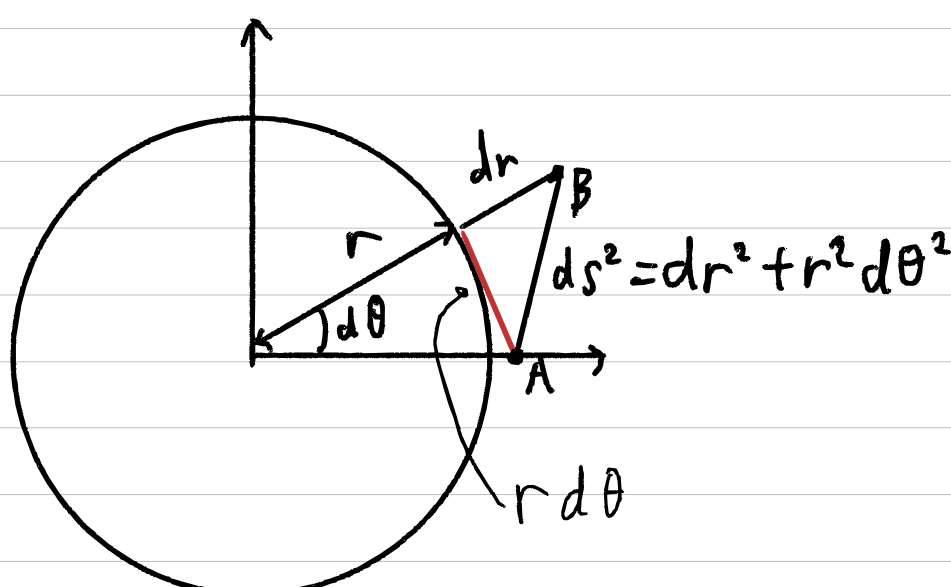
the metric in polar coord.s is singular:

$$g_{ij}(r=0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Coordinate singularity not a physical singularity.

§2.2. Metric in a general manifold

(2)



The infinitesimal dist. b/t the 2 pt.s :

$$ds^2 = dr^2 + r^2 d\theta^2 = \tilde{g}_{ij}^{(\text{polar})} d\tilde{x}^i d\tilde{x}^j$$

$$\tilde{g}_{ij}^{(\text{polar})}(r, \theta) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}.$$

(EX) at the origin, the metric in polar coord.s :

$$\tilde{g}_{ij}^{(\text{polar})}(0) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

(b) General Coord. tr.s.

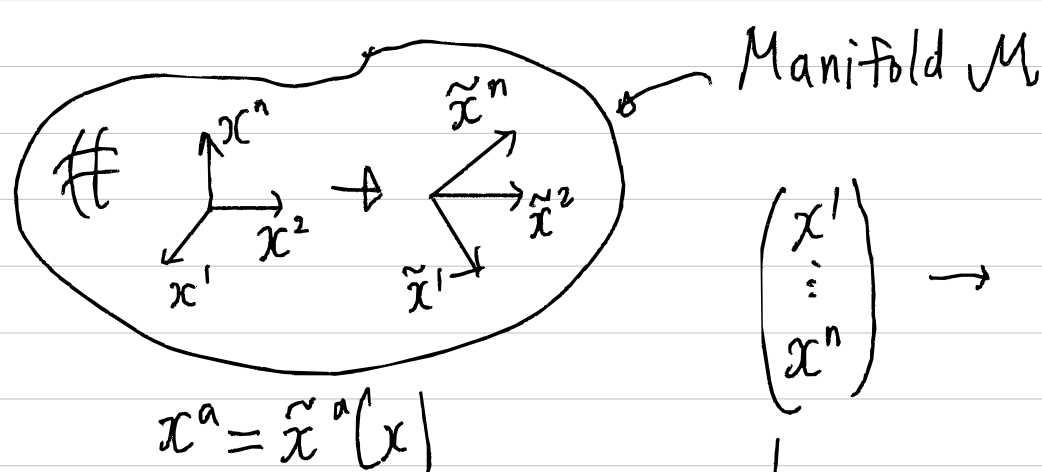
$$ds^2 = g_{ab}(x) dx^a dx^b.$$

$$x^a \rightarrow \tilde{x}^a \quad \text{w/} \quad x^a = x^a(\tilde{x})$$

The infinitesimal dist. $d\tilde{s}$ b/t 2 pt.s

infinitesimally close to one another :

$$d\tilde{s}^2 = \tilde{g}_{ab}(\tilde{x}) d\tilde{x}^a d\tilde{x}^b$$



$$\begin{pmatrix} x^1 \\ \vdots \\ x^n \end{pmatrix} \rightarrow \begin{pmatrix} \tilde{x}^1 \\ \vdots \\ \tilde{x}^n \end{pmatrix}$$

metric $g_{ab}(x)$ metric $\tilde{g}_{ab}(\tilde{x})$

$$\tilde{g}_{ab}(\tilde{x}) = g_{cd} \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b}$$

$$ds^2 = g_{ab}(x) dx^a dx^b \equiv \tilde{g}_{ab}(\tilde{x}) d\tilde{x}^a d\tilde{x}^b = d\tilde{s}^2$$

$$dx^a = \frac{\partial x^a(\tilde{x})}{\partial \tilde{x}^c} d\tilde{x}^c$$

$$ds = g_{ab}(x) dx^a dx^b = g_{ab}(x) \frac{\partial x^a(\tilde{x})}{\partial \tilde{x}^c} d\tilde{x}^c \times \frac{\partial x^b(\tilde{x})}{\partial \tilde{x}^d} d\tilde{x}^d$$

$$= \underbrace{g_{ab}(x(\tilde{x})) \frac{\partial x^a(\tilde{x})}{\partial \tilde{x}^c} \frac{\partial x^b(\tilde{x})}{\partial \tilde{x}^d}}_{= \tilde{g}_{cd}(\tilde{x})} d\tilde{x}^c d\tilde{x}^d$$

$$\tilde{g}_{ab}(\tilde{x}) = g_{cd}(x(\tilde{x})) \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b}$$

(3)

$$ds^2 = g_{ab}(x) dx^a dx^b \longrightarrow d\tilde{s}^2 = \tilde{g}_{ab}(\tilde{x}) d\tilde{x}^a d\tilde{x}^b$$

$$ds = d\tilde{s}$$

renaming the dummy indices.

$$ds^2 = g_{ab}(x) dx^a dx^b = g_{cd}(x) dx^c dx^d$$

$$x^c = x^c(\tilde{x}) \longrightarrow dx^c = \frac{\partial x^c}{\partial \tilde{x}^a} d\tilde{x}^a$$

$$= (\partial_a x^c) d\tilde{x}^a$$

$$= \sum_{a=1}^n \frac{\partial x^c}{\partial \tilde{x}^a}(\tilde{x}) d\tilde{x}^a$$

$$dx^d = \frac{\partial x^d}{\partial \tilde{x}^b} d\tilde{x}^b$$

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ dx = \cos \theta dr - r \sin \theta d\theta \\ dy = \sin \theta dr + r \cos \theta d\theta \end{cases}$$

$$ds^2 = dx^2 + dy^2 = (\cos \theta dr - r \sin \theta d\theta)^2 + (\sin \theta dr + r \cos \theta d\theta)^2$$

$$= (dr)^2 + r^2 \sin^2 \theta d\theta^2 - \cancel{2r \cos \theta \sin \theta dr d\theta}$$

$$+ \cancel{2r \cos \theta \sin \theta dr d\theta} + r^2 \cos^2 \theta d\theta^2$$

$$= dr^2 + r^2 d\theta^2$$

$$ds^2 = g_{cd} dx^c dx^d = g_{cd} \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b} d\tilde{x}^a d\tilde{x}^b$$

$$= g_{cd} \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b} d\tilde{x}^a d\tilde{x}^b$$



$$d\tilde{s}^2 = \tilde{g}_{ab}(\tilde{x}) d\tilde{x}^a d\tilde{x}^b$$

(4)

* Under a coord. tr.:

$$\tilde{x}^a \rightarrow \tilde{x}^a(x) \rightarrow \text{Inverse: } x^a = x^a(\tilde{x})$$

$$ds \rightarrow d\tilde{s} = ds$$

$$g_{ab}(x) \longrightarrow \tilde{g}_{ab}(\tilde{x}) = g_{cd}(x(\tilde{x})) \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b}$$

The metric does transform.

$$x^i = \{x, y, z\} \longrightarrow \tilde{x}^i = \{r, \theta, \varphi\}$$

$$x^1 = x = r \cos \theta$$

$$x^2 = y = r \sin \theta \cos \varphi$$

$$x^3 = z = r \sin \theta \sin \varphi$$

* Metric in spherical coord.s

$$\tilde{g}_{rr} = g_{ij} \frac{\partial x^i}{\partial r} \frac{\partial x^j}{\partial r} = \delta_{ij} \frac{\partial x^i}{\partial r} \frac{\partial x^j}{\partial r}$$

$g_{ij} = \delta_{ij}$ in Cartesian coord.s.

$$\tilde{g}_{rr} = \left(\frac{\partial x}{\partial r} \right)^2 + \left(\frac{\partial y}{\partial r} \right)^2 + \left(\frac{\partial z}{\partial r} \right)^2$$

$$= \cos^2 \theta + \underbrace{\sin^2 \theta \cos^2 \varphi + \sin^2 \theta \sin^2 \varphi}_{= \sin^2 \theta} = 1.$$

$$\tilde{g}_{\theta\theta} = \left(\frac{\partial x}{\partial \theta} \right)^2 + \left(\frac{\partial y}{\partial \theta} \right)^2 + \left(\frac{\partial z}{\partial \theta} \right)^2$$

$$= r^2 \sin^2 \theta + r^2 \cos^2 \theta \cos^2 \varphi + r^2 \cos^2 \theta \sin^2 \varphi$$

$$= r^2$$

$$\tilde{g}_{\varphi\varphi} = \left(\frac{\partial x}{\partial \varphi} \right)^2 + \left(\frac{\partial y}{\partial \varphi} \right)^2 + \left(\frac{\partial z}{\partial \varphi} \right)^2$$

$$= 0 + r^2 \sin^2 \theta \sin^2 \varphi + r^2 \sin^2 \theta \cos^2 \varphi = r^2 \sin^2 \theta$$

$$\tilde{g}_{r\theta} = \frac{\partial x}{\partial r} \frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r} \frac{\partial y}{\partial \theta} + \frac{\partial z}{\partial r} \frac{\partial z}{\partial \theta}$$

$$= -r \cos \theta \sin \theta + r \sin \theta \cos \theta \cos^2 \varphi + r \sin \theta \cos \theta \sin^2 \varphi$$

$$= -r \cos \theta \sin \theta + r \sin \theta \cos \theta = 0$$

$$\tilde{g}_{ab} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}$$

• line elem. : $ds^2 = g_{ab} dx^a dx^b$

• inner prod. : $U \cdot V = g_{ab} U^a V^b$

• change of coord.s : $x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$

$$x^a = x^a(\tilde{x})$$

$$g_{ab}(x) \rightarrow \tilde{g}_{ab}(\tilde{x}) = g_{cd}(x(\tilde{x})) \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b}$$

Flat Sp. in Cartesian $g_{ij} = \delta_{ij} = \mathbb{1}$

$$ds^2 = \delta_{ij} dx^i dx^j$$

Flat Sp. in Spherical $g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}$

$$g^{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^{-2} & 0 \\ 0 & 0 & r^{-2} \sin^{-2} \theta \end{pmatrix}$$

Minkowski metric of 4D flat Sp-Tm.

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & & 0 \\ & 1 & \\ 0 & & 1 \\ & & & 1 \end{pmatrix}$$

$$ds^2 = -dt^2 + \delta_{ij} dx^i dx^j$$

$$\eta^{\mu\nu} = \begin{pmatrix} -1 & & 0 \\ \hline & & \delta^{ij} \\ 0 & & \end{pmatrix}$$

Cl, §3. Geodesics

$$ds^2(\lambda) = g_{ab}(x(\lambda)) \underbrace{dx^a dx^b}_{\text{along the curve}}$$

along the curve

$$dx^a = \frac{dx^a(\lambda)}{d\lambda} d\lambda$$

$$ds^2(\lambda) = g_{ab}(x(\lambda)) \frac{dx^a(\lambda)}{d\lambda} \frac{dx^b(\lambda)}{d\lambda} d\lambda^2$$

At $\forall$ pt. along the curve, we can define the tan. vec. $T^a(\lambda)$:

$$T^a(\lambda) = \frac{dx^a(\lambda)}{d\lambda}$$

$$ds^2(\lambda) = \underbrace{T \cdot T(\lambda)}_{\text{"norm" of the tan. vec.}} d\lambda^2$$

$$T \cdot T(\lambda) = g_{ab}(x(\lambda)) T^a(\lambda) T^b(\lambda)$$

• Curve C has, for tot. leng. $S[x^a(\lambda)]$:

$$S[x^a(\lambda)] = \int_{\lambda_i}^{\lambda_f} ds(\lambda) = \int_{\lambda_i}^{\lambda_f} \sqrt{g_{ab}(x(\lambda)) \frac{dx^a(\lambda)}{d\lambda} \frac{dx^b(\lambda)}{d\lambda}} d\lambda$$

$$= \int_{\lambda_i}^{\lambda_f} \sqrt{g_{ab}(x(\lambda)) \frac{dx^a(\lambda)}{d\lambda} \frac{dx^b(\lambda)}{d\lambda}} d\lambda$$

②

(2) Affine parameter & Proper time

Curve C at $\lambda \rightarrow$ it's null at that pt.

$$T \cdot T = 0.$$

C is timelike at λ if $T(\lambda) \cdot T(\lambda) < 0$.

spacelike at λ if $T(\lambda) \cdot T(\lambda) > 0$.

GR L5

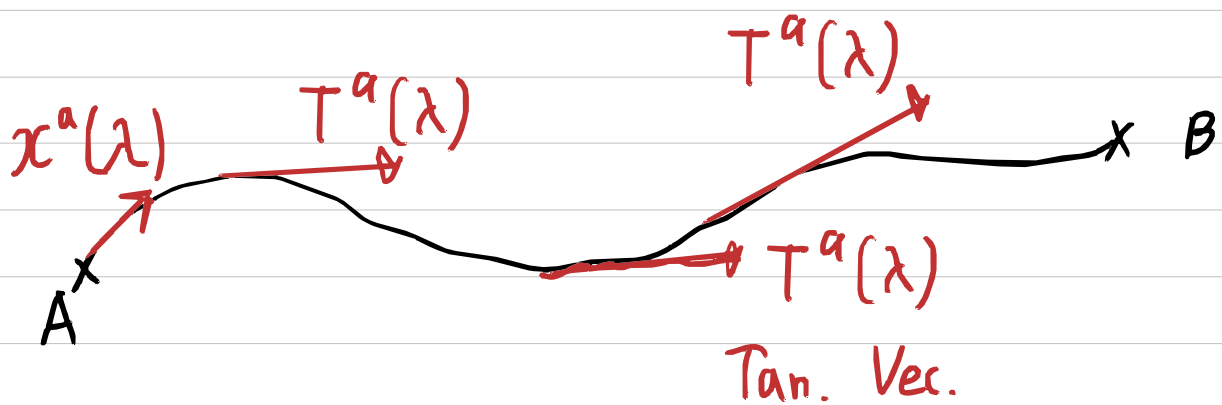
Ch. 1. Curved Sp. - Tm.

I. Gravity & Sp. - Tm. Geometry

II. Metric

III. Geodesics

① Distance along a curve



Curve $x^a = x^a(\lambda)$ w/ parameter λ
along the curve $\lambda_i \leq \lambda \leq \lambda_f$.

Tan. Vec. : $T^a(\lambda) = \frac{dx^a}{d\lambda}$

Length of curve : $S[x^a(\lambda)] = \int_{\lambda_i}^{\lambda_f} \sqrt{T \cdot T} d\lambda$

$$T \cdot T = g_{ab} T^a T^b$$

(2) Affine Parameter

$\lambda = \tau$ is an affine parameter if $T \cdot T = \text{const.}$

$\Rightarrow \frac{d}{d\tau}(T \cdot T) = 0$, in terms of an affine parameter τ .

$$S[x(\tau)] = \int_{\tau_i}^{\tau_f} L(x^a(\tau), \dot{x}^a(\tau)) d\tau$$

$$\dot{x}^a = \frac{dx^a(\tau)}{d\tau} = T^a(\tau)$$

$$L = \sqrt{T \cdot T}$$

τ : affine parameter $\rightarrow \frac{d}{d\tau}(T \cdot T) = 0$

$\leadsto \frac{dL}{d\tau} = 0$

$\rightarrow$ Can use $\tilde{L} = L^2$

[Eq. (1.60)]

[Eq. (1.59)
LN]

②

$$\tilde{L} = \tilde{L} \{x^a(\tau), \dot{x}^a(\tau)\} = g_{ab}(x(\tau)) \dot{x}^a(\tau) \dot{x}^b(\tau)$$

$$\dot{x}^a = T^a = \frac{dx^a}{d\tau}$$

$$\begin{aligned} \frac{\partial \tilde{L}}{\partial x^d} &= \frac{\partial g_{ab}}{\partial x^d} \dot{x}^a \dot{x}^b = (\partial_d g_{ab}) \dot{x}^a \dot{x}^b \\ &= g_{ab,d} \dot{x}^a \dot{x}^b \end{aligned}$$

$$\begin{aligned} \frac{\partial \tilde{L}}{\partial \dot{x}^d} &= g_{ab} \delta_d^a \dot{x}^b + g_{ab} \dot{x}^a \delta_d^b \\ &= g_{db}(x(\tau)) \dot{x}^b(\tau) + g_{ad}(x(\tau)) \dot{x}^a(\tau) \end{aligned}$$

$$\frac{d}{d\tau} \left(\frac{\partial \tilde{L}}{\partial \dot{x}^d} \right) = \partial_a g_{db} \dot{x}^a \dot{x}^b + \partial_b g_{ad} \dot{x}^a \dot{x}^b + 2 g_{de} \ddot{x}^e$$

$$\frac{d}{d\tau} (g_{db}) = \frac{\partial}{\partial x_a} g_{db} \dot{x}^a = \frac{\partial}{\partial x_c} g_{db} \dot{x}^c$$

$a \leftrightarrow c$

$$\frac{d}{d\tau} (g_{ad}) = \frac{\partial}{\partial x^b} g_{ad} \dot{x}^b$$

$$2 g_{de} \ddot{x}^e + [\partial_a g_{db} + \partial_b g_{ad} - \partial_d g_{ab}] \dot{x}^a \dot{x}^b = 0$$

Geodesic Eqn.

$$g_{de} \ddot{x}^e + \frac{1}{2} [\partial_a g_{db} + \partial_b g_{ad} - \partial_d g_{ab}] \dot{x}^a \dot{x}^b = 0$$

* Multiplying w/ the inverse g_{ed} :

$$g^{cd} g_{de} = \delta^c_e$$

$$g^{cd} g_{de} \ddot{x}^e + \frac{1}{2} g^{cd} [\partial_a g_{db} + \partial_b g_{ad} - \partial_d g_{ab}] \dot{x}^a \dot{x}^b = 0$$

$\delta^c_e \ddot{x}^e$

$$\ddot{x}^c + \Gamma_{ab}^c \dot{x}^a \dot{x}^b = 0 \quad \text{: Geodesic Eqn.}$$

Straight line

Put EM force on the RHS.

③

* Christoffel symbols

$$\Gamma_{ab}^c = \frac{1}{2} g^{cd} [\partial_a g_{db} + \partial_b g_{ad} - \partial_d g_{ab}]$$

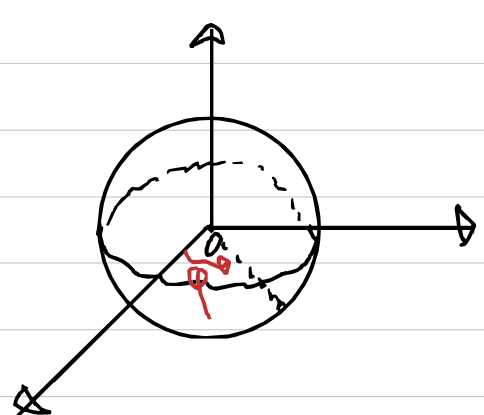
They are symmetric : $\Gamma_{ab}^c = \Gamma_{ba}^c$

(EX) Great circle on a sphere.

2-sphere

w/ radius 1

(unit sphr.)



Coords $\{\theta, \varphi\}$

$$0 \leq \theta \leq \pi$$

$$0 \leq \varphi < 2\pi$$

Metric $g_{ab}^{S^2} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$

* Straight lines on the sphere have the eq.:

$$\begin{cases} \ddot{\theta} + \Gamma_{\theta\theta}^{\theta} \dot{\theta}^2 + 2 \Gamma_{\theta\varphi}^{\theta} \dot{\theta} \dot{\varphi} + \Gamma_{\varphi\varphi}^{\theta} \dot{\varphi}^2 = 0 \\ \ddot{\varphi} + \Gamma_{\theta\theta}^{\varphi} \dot{\theta}^2 + 2 \Gamma_{\theta\varphi}^{\varphi} \dot{\theta} \dot{\varphi} + \Gamma_{\varphi\varphi}^{\varphi} \dot{\varphi}^2 = 0 \end{cases}$$

(EX) Equator $\theta = \frac{\pi}{2}$, $\dot{\theta} = 0 = \ddot{\theta}$

$$\varphi(\tau) = \tau , \quad \dot{\varphi} = 1 , \quad \ddot{\varphi} = 0$$

To check whether the equator is a straight line, we need to compute $\Gamma_{\varphi\varphi}^{\theta}$ & $\Gamma_{\varphi\varphi}^{\varphi}$:

$$\Gamma_{\varphi\varphi}^{\theta} = \frac{1}{2} g^{\theta d} (\partial_{\varphi} g_{d\varphi} + \partial_{\varphi} g_{\varphi d} - \partial_d g_{\varphi\varphi})$$

$$g_{d\varphi} = 0, \quad g_{\varphi d} = 0$$

only potentially $\neq 0$ if $d = \theta$

$$= -\frac{1}{2} g^{\theta\theta} \partial_{\theta} \sin^2 \theta = -\cos \theta \sin \theta$$

$$g_{\varphi\varphi} = \sin^2 \theta$$

At the equator.

$$\text{At } \theta = \frac{\pi}{2}, \quad \Gamma_{\varphi\varphi}^{\theta} = 0$$

$$\ddot{\theta} + \Gamma_{\theta\theta}^{\theta} \dot{\theta}^2 + 2 \Gamma_{\theta\varphi}^{\theta} \dot{\theta} \dot{\varphi} + \Gamma_{\varphi\varphi}^{\theta} \dot{\varphi}^2 = 0$$

$$\ddot{\varphi} + \Gamma_{\theta\theta}^{\varphi} \dot{\theta}^2 + 2 \Gamma_{\theta\varphi}^{\varphi} \dot{\theta} \dot{\varphi} + \Gamma_{\varphi\varphi}^{\varphi} \dot{\varphi}^2 = 0$$

* Similarly,

$$\Gamma_{\varphi\varphi}^{\varphi} = \frac{1}{2} g^{\varphi d} [\partial_{\varphi} g_{d\varphi} + \partial_{\varphi} g_{\varphi d} - \partial_d g_{\varphi\varphi}]$$

$$\left(d = \varphi \text{ so } \neq 0 \right)$$

$$= -\frac{1}{2} \frac{1}{\sin^2 \theta} \partial_{\varphi} g_{\varphi\varphi} = 0$$

So, both geodesic eqns are satisfied along the equator:

$$\ddot{\theta} + \Gamma_{ab}^{\theta} \dot{x}^a \dot{x}^b = 0 \Big|_{\theta = \frac{\pi}{2}}$$

$$\& \ddot{\varphi} + \Gamma_{ab}^{\varphi} \dot{x}^a \dot{x}^b = 0$$

→ The eqn. is a straight line.

Any great circle. (that cuts a sphere in 2 equal parts) is a straight line.

IV. SR.

(3+1) flat dim. sp.-tm.

$$\text{Minkowski coord.s: } x^{\mu} = \{x^0, x^1, x^2, x^3\}$$

$$= \{t, x, y, z\}$$

$$\text{Metric } \eta_{\mu\nu} = \begin{pmatrix} -1 & & & \\ & +1 & & \\ & & +1 & \\ 0 & & & +1 \end{pmatrix}$$

(4)

Consider a massive particle w/ m follows a trajectory $x^\mu = x^\mu(\tau)$ spans a world line of the particle in the case tangent vec., $\frac{dx^\mu}{d\tau}$, is the 4-vel. vec. of the particle

$$U^\mu = \frac{dx^\mu}{d\tau}$$

If we normalize this vec.,

$$U \cdot U = \eta_{\mu\nu} U^\mu U^\nu = -1$$

$$\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = -1$$

$$ds^2 = U \cdot U d\tau^2 = -d\tau^2 \quad \text{: parameter } \tau \text{ is the proper time of the particle.}$$

* 4-vel.

$$U^\mu = \begin{pmatrix} \frac{dt}{d\tau} \\ \frac{dx}{d\tau} \\ \frac{dy}{d\tau} \\ \frac{dz}{d\tau} \end{pmatrix} = \frac{dt}{d\tau} \begin{pmatrix} 1 \\ \underline{v} \end{pmatrix}$$

$$v^2 = \delta_{ij} v^i v^j = |\underline{v}|^2$$

$$U \cdot U = -1 = \left(\frac{dt}{d\tau}\right)^2 (-1 + v^2)$$

$$\frac{dt}{d\tau} = \gamma = \frac{1}{\sqrt{1-v^2}}$$

only non-rel. limit
 $v \ll 1$
($c \rightarrow \infty$)

do we have $t \sim \tau$
generically it's not.

V. Transformation Laws

SR: $x^\mu \xrightarrow{\text{Poincaré trans.}} \tilde{x}^\mu = \Lambda^\mu_\alpha x^\alpha + \bar{x}^\mu$

↑
translation

• L.T. w/ Lorentz matrix Λ^μ_α

• inverse L.T.

$$\Lambda_\beta^\nu \equiv (\Lambda^{-1})^\nu_\beta$$

$$\Lambda^\alpha_\mu \Lambda_\alpha^\nu = \delta_\mu^\nu$$

$$\Lambda^T \eta \Lambda = \eta$$

GR. Ch1. Curved Sp. - Tm.

III Geodesics

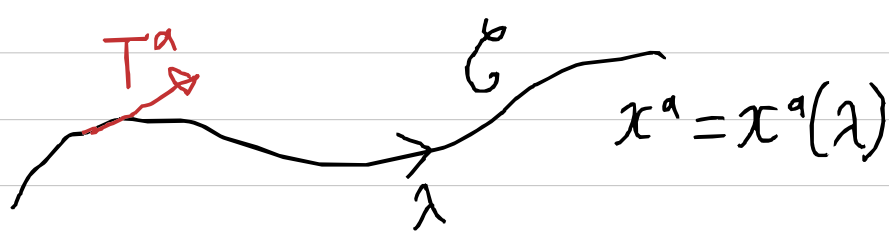
IV Recovery of Special Relativity

V Transformation Laws

(1) General Transformation

(a) Review of Poincaré transf.

$$x^a \rightarrow \tilde{x}^a = \Lambda^a_b x^b + \bar{x}^a$$

(b) Generic Coord. transf. $x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$ For curve $\mathcal{C}$: $x^a = x^a(\lambda)$ 

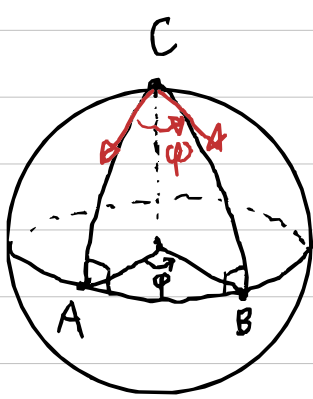
Tan. Vec. : $T^a(\lambda) = \dot{x}^a(\lambda) = \frac{dx^a}{d\lambda}$

λ : affine parameter if $T \cdot T = g_{ab} T^a T^b = \text{const.}$

Geodesic Eq. : $\ddot{x}^a(\lambda) + \Gamma^a_{bc} \dot{x}^b \dot{x}^c = 0$

(for a straight line)

Christoffel Symbols : $\Gamma^a_{bc} = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{bd} - \partial_d g_{bc})$

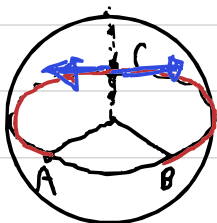


$$\theta_c \rightarrow 0$$

$$\varphi_A \rightarrow \frac{\pi}{2}$$

$$\varphi_B \rightarrow \frac{\pi}{2}$$

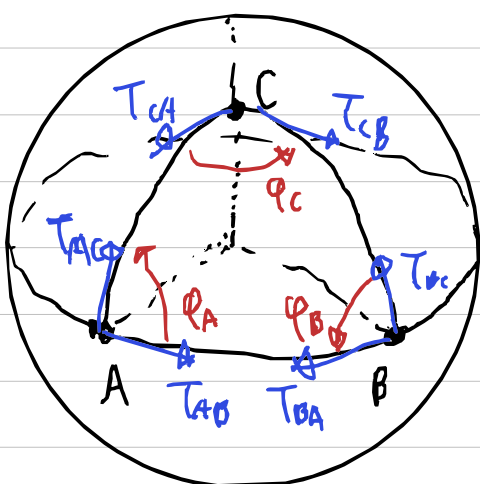
$$\varphi_c \rightarrow \varphi_B - \varphi_A$$



$$\theta_A = \theta_B = \theta_C = \frac{\pi}{2}$$

$$\varphi_A < \varphi_B < \varphi_C$$

$$\varphi_A = \varphi_B = \varphi_C = \pi$$



(2)

* Tensors under Poincaré Transf.

↳ ACP LN or GR LN Appendix B.

* LT. : $x^\mu \rightarrow \Lambda^\mu_\nu x^\nu$

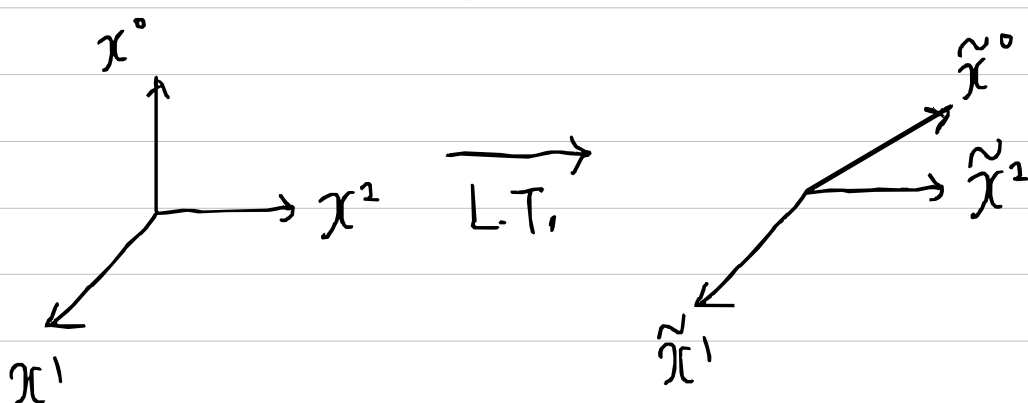
$$\Lambda^\mu_\alpha \eta_{\mu\nu} \Lambda^\nu_\beta = \eta_{\alpha\beta}$$

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu_\nu x^\nu + \bar{x}^\mu$$

$$\begin{aligned} T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}(x) &\rightarrow T'^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}(x') \\ &= \Lambda^{\mu_1}_{\alpha_1} \dots \Lambda^{\mu_m}_{\alpha_m} \Lambda_{\nu_1}^{\beta_1} \dots \Lambda_{\nu_l}^{\beta_l} \\ &\quad \times T^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_l}(x) \end{aligned}$$

(a) Poincaré Transformation

$$x^a \rightarrow \tilde{x}^a = \underbrace{\Lambda^a_b}_{\text{Lorentz boost (or rotation)}} x^b + \underbrace{x^a_0}_{\text{translation}}$$



Λ^a_b : Lorentz matrix

$$(\Lambda^{-1})^d_c = \Lambda^d_c : \text{Inverse Lorentz Matrix.}$$

S.R. boost. : Rank-(m, l) tensor.

Each contravar. index transforms w/ Lorentz matrix Λ^a_c .

Each contravar. index transforms w/ Inverse Lorentz matrix Λ_d^b .

(3)

(b) Generic Coord. Transf.

$$x^a \longrightarrow \tilde{x}^a = \tilde{x}^a(x) \quad \rightarrow \text{Inverse} \quad x^a = x^a(\tilde{x})$$

Define transformation matrix: $A^a_c = \frac{\partial \tilde{x}^a(x)}{\partial x^c}$

Define the inverse transformation matrix:

$$B^b_d(\tilde{x}) = \frac{\partial x^b(\tilde{x})}{\partial \tilde{x}^d}$$

$$\therefore A^a_b B^b_c = \frac{\partial \tilde{x}^a}{\partial x^b} \frac{\partial x^b}{\partial \tilde{x}^c} = \frac{\partial \tilde{x}^a}{\partial \tilde{x}^c} = \delta^a_c$$

 $\rightarrow B$ is indeed the inverse of A . $T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_\ell}$ is a rank- (m, ℓ) tensor.If under $x^a \rightarrow \tilde{x}^a(x)$

$$T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_\ell} \longrightarrow \tilde{T}^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_\ell}(\tilde{x})$$

$$\tilde{T}^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_\ell}(\tilde{x}) = T^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_\ell} \frac{\partial \tilde{x}^{\mu_1}}{\partial x^{\alpha_1}} \dots \frac{\partial \tilde{x}^{\mu_m}}{\partial x^{\alpha_m}} \times \frac{\partial x^{\beta_1}}{\partial \tilde{x}^{\nu_1}} \dots \frac{\partial x^{\beta_\ell}}{\partial \tilde{x}^{\nu_\ell}}$$

(EX) Metric Tensor $g_{ab}(x)$

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x) \quad \text{or inversely} \quad x^a = x^a(\tilde{x})$$

$$g_{ab}(x) \longrightarrow \tilde{g}_{ab}(\tilde{x}) = g_{cd}(x) \frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial x^d}{\partial \tilde{x}^b}$$

 g_{ab} is a rank- $(0, 2)$ tensor.

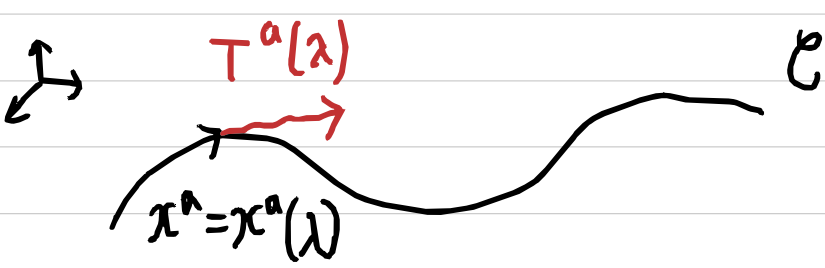
$$g \longrightarrow \tilde{g} = B \cdot g \cdot B$$

$$g^{-1} \longrightarrow \tilde{g}^{-1} = (B \cdot g \cdot B)^{-1} = (B^{-1}) \cdot g^{-1} \cdot (B^{-1}) \\ = A \cdot g^{-1} \cdot A$$

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$$

$$g^{ab}(x) \rightarrow \tilde{g}^{ab}(\tilde{x}) = g^{cd}(x) \frac{\partial \tilde{x}^a}{\partial x^c} \frac{\partial \tilde{x}^b}{\partial x^d}$$

★ EX. Tangent Vector



$$x^a \rightarrow \tilde{x}^a(x)$$

$$T^a \rightarrow \tilde{T}^a = \frac{d\tilde{x}^a}{d\lambda} \stackrel{\text{Chain rule}}{=} \frac{\partial \tilde{x}^a}{\partial x^b} \frac{dx^b}{d\lambda} = T^b \frac{\partial \tilde{x}^a}{\partial x^b}$$

The tan. vec. $T^a = \dot{x}^a$ is a $(1, 0)$ -tensor
 $\equiv$ contravar. vec.

(EX). If T^a is a contravar. vec. (or a $(1, 0)$ -tensor),

then $T_a = g_{ab} T^b$ is a covar. vec.
 (or a $(0, 1)$ -tensor).

$$\text{Indeed, we can check: } T^a \rightarrow \tilde{T}^a = T^b \frac{\partial \tilde{x}^a}{\partial x^b}$$

$$T_c = g_{ca} T^a \longrightarrow \tilde{T}_c = \tilde{g}_{ca} \tilde{T}^a$$

(4)

$$\begin{aligned}\tilde{g}_{cd} \tilde{T}^d &= g_{ef} T^w \frac{\partial \tilde{x}^d}{\partial x^w} \frac{\partial x^e}{\partial \tilde{x}^c} \frac{\partial x^f}{\partial \tilde{x}^d} \\ &= \frac{\partial x^e}{\partial \tilde{x}^c} \underbrace{\left(\frac{\partial x^f}{\partial \tilde{x}^d} \frac{\partial \tilde{x}^d}{\partial x^w} \right)}_{= \delta^f_w} \\ &= \delta^f_w\end{aligned}$$

$$\tilde{T}_c = \tilde{g}_{cd} \tilde{T}^d = \underbrace{g_{ef} T^f}_{= T_e} \frac{\partial x^e}{\partial \tilde{x}^c} = T_e \frac{\partial x^e}{\partial \tilde{x}^c}$$



T_c is a $(0,1)$ -tnsr or a covar. vec.

★ EX. Raising & Lowering indices

If $T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}$ is (m, l) -tnsr, then

$$T^{\mu_1 \dots \mu_{m-1}}_{\alpha \nu_1 \dots \nu_l} = g_{\alpha \mu_m} T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l} \quad \text{is } (m-1, l+1)\text{-tnsr.}$$

$$T^{\mu_1 \dots \mu_{m-1} \beta}_{\nu_1 \dots \nu_l} = g^{\beta \nu_{l+1}} T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_{l+1}} \quad \text{is } (m+1, l-1)\text{-tnsr.}$$

GR ch. 1.

Transformation Laws

A rank (m, l) -tensor is an objt T that transforms as follows under a change of coord.s

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x) \quad \text{or} \quad x^a = x^a(\tilde{x})$$

(invertible)

$$T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}(x) \rightarrow \tilde{T}^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}(\tilde{x})$$

A scalar fld. $\varphi(x) \rightarrow$ invariant under arbitrary coord. transf.

$$\varphi(x) \rightarrow \tilde{\varphi}(\tilde{x}) = \varphi(x) = \varphi(x(\tilde{x}))$$

e.g. of scalar quantity $|\underline{E}|^2 - |\underline{B}|^2 = F_{\mu\nu}^2$

$$= g^{\mu\mu'} g^{\nu\nu'} F_{\mu\nu} F_{\mu'\nu'}$$

(EX) Higgs fld $\rightarrow$ also a scalar fld.

(c) Gradient of a scalar fld.

If $\varphi = \varphi(x)$ is a scalar fld.

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$$

$$\varphi(x) \rightarrow \tilde{\varphi}(\tilde{x}) = \varphi(x)$$

$$V_a = \partial_a \varphi(x) = \frac{\partial \varphi}{\partial x^a} = \varphi_{,a}$$

$$V_a \rightarrow \tilde{V}_a(\tilde{x}) = \frac{\partial}{\partial \tilde{x}^a} \tilde{\varphi}(\tilde{x})$$

$$\tilde{V}_a(\tilde{x}) = \frac{\partial}{\partial \tilde{x}^a} \varphi(x) = \frac{\partial x^b}{\partial \tilde{x}^a} \underbrace{\frac{\partial \varphi(x)}{\partial x^b}}_{= V^b(x)}$$

$$\tilde{V}_a(\tilde{x}) = \frac{\partial x^b}{\partial \tilde{x}^a} V_b \longrightarrow V_a \text{ is a vector}$$

or a rank- $(0,1)$ tensor

$$V_a V^a = \partial_a \varphi \partial_b \varphi g^{ab} \text{ is a scalar.}$$

②

(d) Derivs of tensors or vectors

Let V_a be a vec. $\tilde{V}^a(\tilde{x}) = \frac{\partial \tilde{x}^a}{\partial x^d} V^d(x)$.

$$Y_b^a(x) = \partial_b V^a = V^a_{,b} = \frac{\partial V^a(x)}{\partial x^b}$$

$$Y_b^a(x) \longrightarrow \tilde{Y}_b^a(\tilde{x}) = \frac{\partial}{\partial \tilde{x}^b} \tilde{V}^a(\tilde{x})$$

$$= \frac{\partial}{\partial \tilde{x}^b} \left(\frac{\partial \tilde{x}^a}{\partial x^d} V^d(x) \right)$$

$$\tilde{Y}_b^a(\tilde{x}) = \frac{\partial x^c}{\partial \tilde{x}^b} \frac{\partial}{\partial x^c} \left[\frac{\partial \tilde{x}^a}{\partial x^d} V^d(x) \right]$$

$$= \frac{\partial x^c}{\partial \tilde{x}^b} \frac{\partial \tilde{x}^a}{\partial x^d} \underbrace{\partial_c V^d(x)}_{= Y_c^d(x)} + \frac{\partial x^c}{\partial \tilde{x}^b} \frac{\partial^2 \tilde{x}^a}{\partial x^c \partial x^d} V^d$$

precisely the part that

tells us how a (1,1)-tensor transforms,

is typically
non-zero.

$Y_b^a = \partial_b V^a$ is NOT a tensor.

Define: $\psi(x) = Y_a^a(x) = \frac{\partial}{\partial x^a} V^a$

$$x \longrightarrow \tilde{x}(x)$$

$$\psi(x) \longrightarrow \tilde{\psi}(\tilde{x}) = \tilde{Y}_a^a(\tilde{x})$$

$$= \underbrace{\frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial \tilde{x}^a}{\partial x^d}}_{= \frac{\partial x^c}{\partial x^d} = \delta^c_d} \partial_c V^d + \underbrace{\frac{\partial x^c}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^c \partial x^d} V^d}_{\text{typically is } \neq 0.}$$

$$= \psi(x)$$

(3)

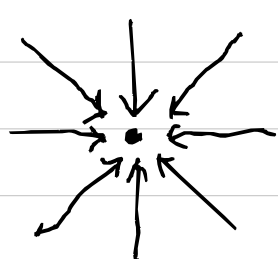
Under $x \rightarrow \tilde{x}(x)$,

$$\psi(x) = \partial_a V^a \longrightarrow \tilde{\psi}(\tilde{x}) = \psi(x) + \underbrace{\text{Mess}}_{\neq 0 \text{ typically}}$$

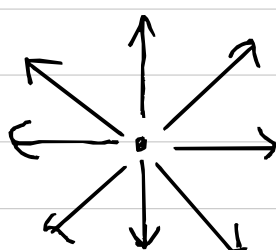
 ψ is NOT a scalar.

★ Div. of a vec.

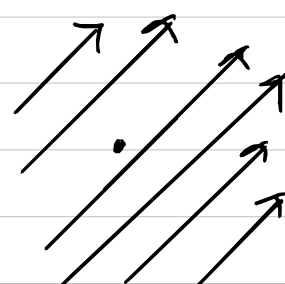
$$\underline{\nabla} \cdot \underline{v} < 0$$



$$\underline{\nabla} \cdot \underline{v} > 0$$



$$\underline{\nabla} \cdot \underline{v} = 0$$



(e) Covar. Deriv.

$$\begin{array}{ccc} \partial_a & \xrightarrow{\text{promoted}} & D_a = \frac{D}{dx^a} \quad \left(\text{or } \nabla_a \right) \\ \text{partial} & & \text{Covar. deriv.} \\ \text{deriv.} & & \end{array}$$

(comma)

Sometimes a semicolon (;) is used.

($\cdot$, a)($\cdot$; a)

$$D_a T^{\mu_1 \dots \mu_m}_{\nu_1 \dots \nu_l}$$

should be a $(m, l+1)$ -tnsr. (m, l) -tnsr★ For a scalar fld. ϕ ,

$$D_a \phi = \partial_a \phi(x)$$

→ already a $(0, 1)$ -tnsr.

★ Linearity

$$D_a (T^m_m + S^m_m) = (D_a T^m_m) + (D_a S^m_m)$$

★ Leibniz Rule

$$\begin{aligned} \partial_A (T^{ab}_{cde} S^{fgk}_{blm}) &= \left(\partial_A [T^{ab}_{cde}] \right) S^{fgk}_{blm} \\ &\quad + T^{ab}_{cde} \partial_A (S^{fgk}_{blm}) \end{aligned}$$

★ The same w/ $\partial_A \rightarrow D_A$

$$D_a (S T) = (D_a S) T + S (D_a T)$$

(4)

* Covar. Deriv. should be linearly related to the partial deriv.

$$D_a T^m_m = \partial_a T^m_m + \# T^m_m$$

For ex., if U^a is a vec.:

$$D_a U^a = \partial_a U^a + \underbrace{\gamma^a_{bc}}_{\text{linearity coefficients}} U^c$$

linearity coefficients

should transform as a tensor.

* Geodesic Eqn. : Tangent vec. $U^a = \dot{x}^a(\tau)$
(τ : affine parameter)

$$\underbrace{\ddot{x}^a(\tau)}_{= \frac{d^2 x^a}{d\tau^2}} + \Gamma^a_{bc} \dot{x}^b \dot{x}^c = 0 \equiv \frac{DU^a}{D\tau}$$

$$= \frac{d^2 x^a}{d\tau^2}$$

$$= \frac{d}{d\tau}(U^a)$$

$$\frac{DU^a}{D\tau} = \frac{dx^b}{d\tau} \frac{DU^a}{Dx^b}$$

Defn. of covar. deriv.

$$\frac{DU^a}{Dx^b} = \frac{\partial U^a}{\partial x^b} + \gamma^a_{bc} U^b U^c$$

$\frac{DU^a}{D\tau} = 0$ is the geodesic eqn. for $U^a = \dot{x}^a$.

If these coeff.s $\gamma^a_{bc} \equiv \Gamma^a_{bc} \equiv$ Christoffel symbols

$$\frac{DU^a}{Dx^b} = D_b U^a = U^a_{;b} = \partial_b U^a + \Gamma^a_{bc} U^c$$

Ch. I. V. Transformation Laws

Coord. Transfrm. $x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$

(1) Ex.s

* Derivative of a tensor① Partial deriv. $\rightarrow \partial_b V^a$ is NOT a tensor. $\partial_b V^a \equiv 0$ in 1 frame or coord. sys. DOES NOT imply $\partial_a V^b = 0$ in another frm. $\psi = \partial_a V^a$ is NOT a scalar,

$$\psi \rightarrow \tilde{\psi}(\tilde{x}) = \psi(x) + \frac{\partial x^b}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^b \partial x^c} V^c$$

② Covariant Deriv. D_a Christoffel symbols

$$D_b V^a \equiv \partial_b V^a + \Gamma_{bc}^a V^c$$

Satisfy:

- linearity
- Leibniz rule
- Does it transform as a tensor?

$\rightarrow$ Ans.: YES!

 Γ_{bc}^a are NOT tensors. $D_a V^a$ is a scalar.

$$\Gamma_{ac}^a = \frac{1}{2} g^{ae} \left(\cancel{\partial_a g_{ec}} + \cancel{\partial_c g_{ae}} - \cancel{\partial_e g_{ac}} \right)$$

$$= \frac{1}{2} g^{ac} \partial_b g_{ac}$$

 $x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$ invertible

$$D_a V^a \rightarrow \tilde{D}_a \tilde{V}^a$$

$$\tilde{V}^a = \frac{\partial \tilde{x}^a}{\partial x^b} V^b$$

$$\tilde{D}_a \tilde{V}^a = \tilde{\partial}_a \tilde{V}^a + \tilde{\Gamma}_{ab}^a \tilde{V}^b$$

$$= \partial_a V^a + \frac{\partial x^b}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^b \partial x^c} V^c$$

$$+ \frac{1}{2} \frac{\partial \tilde{x}^a}{\partial x^a} \frac{\partial \tilde{x}^c}{\partial x^a} \frac{\partial x^b}{\partial \tilde{x}^c} \frac{\partial x^d}{\partial \tilde{x}^b} g^{Ad} \frac{\partial}{\partial x^b} \left[\frac{\partial x^a}{\partial \tilde{x}^a} \frac{\partial x^b}{\partial \tilde{x}^c} g_{a\beta} \right] V^\sigma$$

$= \delta_{\sigma}^{\beta}$ $\tilde{g}_{ac}$

$\rightarrow V^\beta$

$$= D_a V^a + \frac{\partial x^b}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^b \partial x^c} V^c$$

$$+ \frac{\partial \tilde{x}^a}{\partial x^a} \frac{\partial \tilde{x}^c}{\partial x^a} \frac{\partial x^b}{\partial \tilde{x}^c} g^{Ab} g_{a\beta} \frac{\partial^2 x^a}{\partial x^b \partial \tilde{x}^a} V^\beta$$

③ $\frac{\partial \tilde{x}^a}{\partial x^a} \frac{\partial \tilde{x}^c}{\partial x^a} \frac{\partial x^b}{\partial \tilde{x}^c} g^{Ab} g_{a\beta} = \delta_{\beta}^{\alpha} = g^{A\alpha} g_{\alpha\beta}$

$$\frac{\partial x^b}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^b \partial x^c} V^c + \frac{\partial \tilde{x}^a}{\partial x^a} \frac{\partial^2 x^b}{\partial x^b \partial \tilde{x}^a} V^\beta$$

$$= \left[\frac{\partial x^b}{\partial \tilde{x}^a} \frac{\partial^2 \tilde{x}^a}{\partial x^b \partial x^c} + \frac{\partial \tilde{x}^a}{\partial x^a} \frac{\partial^2 x^b}{\partial x^b \partial \tilde{x}^a} \right] V^c$$

$$= \frac{\partial}{\partial x^c} \left[\frac{\partial \tilde{x}^a}{\partial x^b} \frac{\partial x^b}{\partial \tilde{x}^a} \right] = \frac{\partial}{\partial x^c} \left[\delta^a_a \right] = \frac{\partial(n)}{\partial x^c}$$

$$= 0$$

$$(n = \text{const.})$$

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$$

$$D_a V^a \rightarrow \tilde{D}_a \tilde{V}^a = D_a V^a$$

 $\rightarrow D_a V^a$ is a scalar!(d) Covariant deriv. on an arbitrary tensor. ϕ : scalar $D_a \phi = \partial_a \phi$ V^a is a vector. $D_b V^a = \partial_b V^a + \Gamma_{bc}^a V^c$

(2)

 V^a : contravariant vec. W_a : covariant vec. $\rightarrow \varphi = V^a W_a = \text{scalar}$

$$D_b [V^a W_a] = \partial_b [V^a W_a]$$

$$\begin{array}{ccc} \text{LHS} & & \text{RHS} \\ \downarrow & & \downarrow \\ [D_b V^a] W_a + V^a D_b W_a & = & [\partial_b V^a] W_a + V^a \partial_b W_a \\ \downarrow & & \downarrow \\ \cancel{\partial_b V^a} W_a + \Gamma_{bc}^a V^c W_a & = & \cancel{\partial_b V^a} W_a + V^a \partial_b W_a \end{array}$$

$$V^a (D_b W_a + \Gamma_{ba}^c W_c - \partial_b W_a) = 0$$

Lo $\forall$ any covariant vector V^a

$$D_b W_a = \partial_b W_a - \Gamma_{ab}^c W_c$$

for $\forall$ rank- (m, l) tensor $T_{b_1 \dots b_l}^{a_1 \dots a_m}$

$$D_c T_{b_1 \dots b_l}^{a_1 \dots a_m} = \partial_c T_{b_1 \dots b_l}^{a_1 \dots a_m} + \sum_{i=1}^m \Gamma_{c k}^{a_i} T_{b_1 \dots b_l}^{a_1 \dots a_{i-1} a_{i+1} \dots a_m} - \sum_{k=1}^l \Gamma_{b_k c}^{\beta} T_{b_1 \dots b_{k-1} \beta b_{k+1} \dots b_l}^{a_1 \dots a_m}$$

at position k

$$\Gamma_{bc}^a = \Gamma_{cb}^a \implies D_a D_b \varphi = D_b D_a \varphi$$

$$[D_a, D_b] \varphi = 0$$

$$(D_a D_b - D_b D_a) \varphi = \cancel{\partial_a \partial_b \varphi} - \cancel{\partial_b \partial_a \varphi} + (\cancel{\Gamma_{ab}^c} - \cancel{\Gamma_{ba}^c}) \varphi = 0$$

$$\begin{aligned} \text{Faraday Tensor } F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu \\ &\equiv D_\mu A_\nu - D_\nu A_\mu + \Gamma_{\mu\nu}^\alpha A_\alpha - \Gamma_{\nu\mu}^\alpha A_\alpha \end{aligned}$$

More generally,

 $[D_\mu, D_\nu] W_c \neq 0 \rightarrow$ is, in fact, related to curvature.

(3)

Our defn. of covariant deriv. is

metric-compatible:

$$D_a g_{bc} \equiv 0 \equiv D_a g^{bc} \\ \equiv 0 \equiv D_a \delta^b_c$$

Prove this in P53.

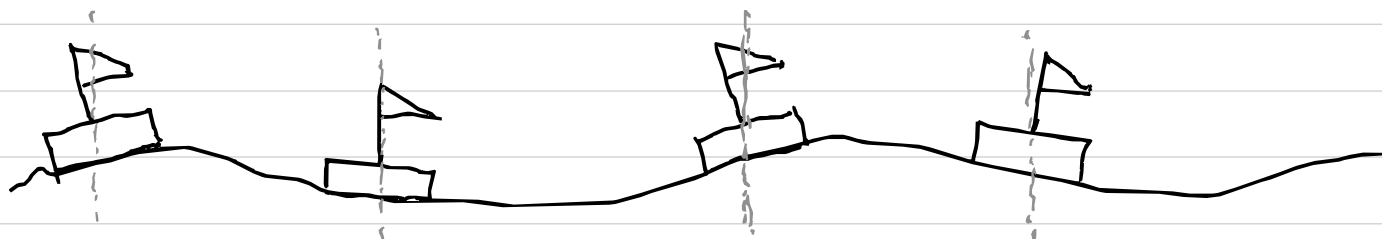
$$\text{If } D_b V^c = 0, \quad D_b V_c = 0.$$

$$D^b T = g^{ba} D_a T = D_a (g^{ab} T)$$

↑
Tensor

Ch. 2. Curvature

I. Parallel Transport



Vector V^a is parallel-transported along a curve $\mathcal{C}$ w/ $\dot{x}^a = \frac{dx^a(\lambda)}{d\lambda}$ if $D_\lambda V^a = 0$,

$\mathcal{C}: x^a = x^a(\lambda)$ λ : affine parameter

$$T^a = \frac{dx^a(\lambda)}{d\lambda} = \dot{x}^a(\lambda)$$

V^a is // transported along $\mathcal{C}$



$$D_\lambda U^a \equiv 0 \equiv \frac{dx^b}{d\lambda} \frac{D}{Dx^b} U^a = \dot{x}^b D_b U^a$$

$$D_\lambda U^a = \frac{\partial U^a}{\partial x^b} \dot{x}^b + \Gamma_{bc}^a \dot{x}^b U^c$$

Geodesic.

Curve for which the tan. vec. itself is parallel transported.

$$D_\lambda T^a \equiv \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

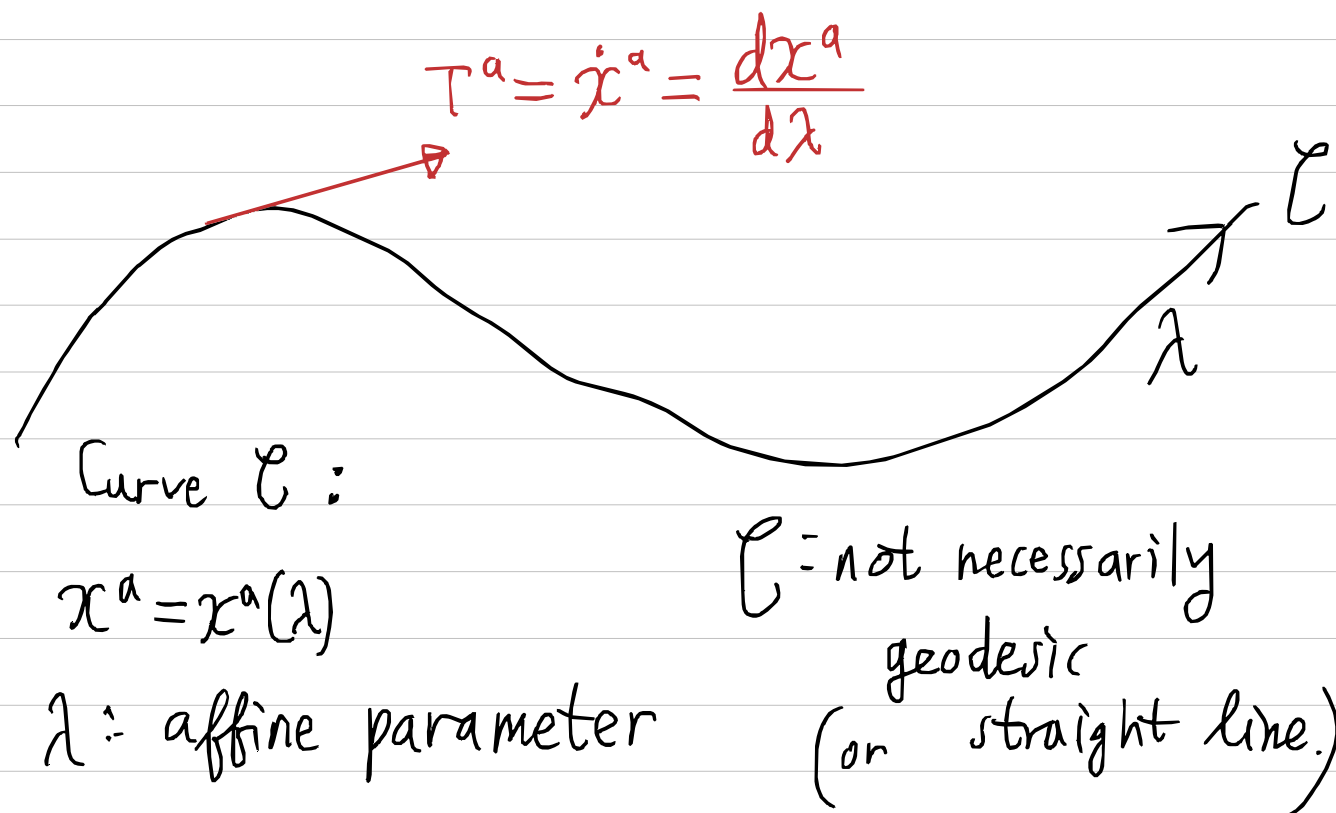
Ch. 2. Curvature.

This is where we'll understand how to identify the relevant curvature of a manifold.

In the next Chap., we'll see how "stuff" living on a manifold generates this curvature.

§2.1. Parallel Transport

(a) Parallel transport ($\parallel$) of a tensor along a curve



Vector V^a is $\parallel$ transported along C



$$D_\lambda V^a = 0$$

$$D_\lambda V^a \equiv \frac{DV^a}{d\lambda} \equiv T^b D_b V^a$$

$$= \dot{x}^b \frac{\partial}{\partial x^b} V^a + \Gamma_{bc}^a \dot{x}^b V^c$$

$$\underbrace{\hspace{1cm}}_{= \frac{dV^a}{d\lambda}} \equiv \dot{V}^a$$

Parallel transport ($\parallel$ transport) $\iff \dot{V}^a(\lambda) + \Gamma_{bc}^a(x(\lambda)) \dot{x}^b(\lambda) V^c(\lambda) = 0$

* If V^a is a vec., then $D_\lambda V^a$ is also a vec.

* If V^a is a vec., & U_a is a vec.

$$D_\lambda U_a = D_\lambda V^a = 0 \iff D_\lambda (U_a V^a) = 0$$

$\downarrow$
scalar

$$D_\lambda (U, V) = \frac{d}{d\lambda} (U \cdot V) = 0$$

$$= (U_a D_\lambda V^a) + V^a D_\lambda U_a$$

$$D_\lambda U_a = \dot{U}_a - \Gamma_{ab}^c \dot{x}^b U_c$$

U_a is $\parallel$ transported along C if $D_\lambda U_a = 0$,

more generically, if $\begin{matrix} \{a_j\} \\ \{b_k\} \end{matrix}$ is a

rank (m, l) -tensor :

$$D_\lambda \begin{matrix} \{a_j\} \\ \{b_k\} \end{matrix} = \underbrace{\frac{d}{d\lambda} \begin{matrix} \{a_j\} \\ \{b_k\} \end{matrix}}_{= \dot{\begin{matrix} \{a_j\} \\ \{b_k\} \end{matrix}}} + \sum_{\bar{j}=1}^m \Gamma_{ac}^{\bar{a}k} \begin{matrix} \{a_j\} \\ \{b_{\bar{j}}\} \end{matrix} \dot{x}^c - \sum_{k=1}^l \Gamma_{bc}^{\bar{b}\rho} \begin{matrix} \{a_j\} \\ \{b_{\bar{k}}, \dots, \bar{b}, \dots, \bar{b}_e\} \end{matrix} \dot{x}^c$$

A rank (m, l) -tensor Y is $\parallel$ transported

along curve w/ $\dot{x}^a \iff D_\lambda \begin{matrix} \{a_j\} \\ \{b_k\} \end{matrix} \equiv 0$.

if Y is a rank (m, l) -tensor, the $D_\lambda Y$ is also a rank (m, l) -tensor.

* If U^a, V^a are 2 vecs, // transported (2)
 along a curve C (not necessarily a straight line)

$\Rightarrow$ The I.P. $\underline{U \cdot V}$ is conserved along that curve.
 (hence, the angle)

* If the tan. vec. $T^a = \dot{x}^a = \frac{dx^a}{d\lambda}$ is

// transported along C

$\Leftrightarrow C$ is a straight line or a geodesic.

Straight line $D_\lambda T^a \equiv 0 \equiv \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$

* If V^a is // transported along a straight line, then

$$\underline{V \cdot \dot{x}} = g_{ab} V^a T^b \\ = g_{ab} V^a \dot{x}^b$$

is conserved :

$$\frac{d}{d\lambda} (\underline{V \cdot \dot{x}}) = 0$$

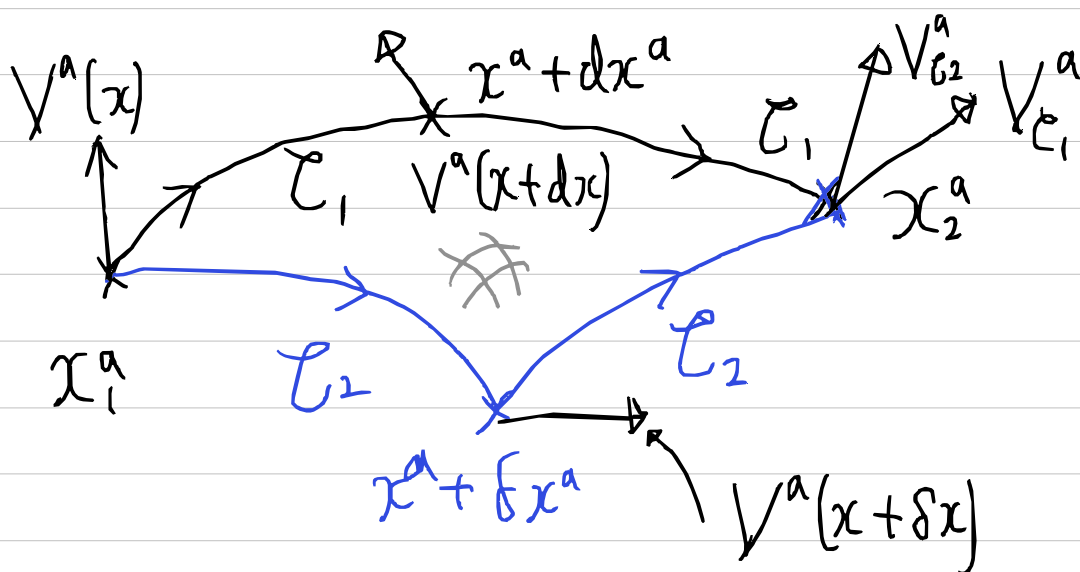
$\hookrightarrow$ angle b/t the vec. V^a & tan. vec.
 is const..

(b) // transport along a closed curve

V^a is // transported $x_1^a = x^a \rightarrow x_2^a$

$$= x^a + dx^a + \delta x^a$$

"
 2 infinitesimal
 separations



Infinitesimal displacements

$$D_\lambda V^a = \dot{x}^b \partial_b V^a + \Gamma_{bc}^a V^c \dot{x}^b = 0$$

$$\rightarrow dx^b \partial_b V^a + \Gamma_{bc}^a V^c dx^b = 0$$

Also, // transparent from $x^a \rightarrow x^a + \delta x^a$ ③

$$\delta x^b \partial_c V^a + \Gamma_{bc}^a \delta x^b V^c = 0$$

$x \longrightarrow x$

$$V^a(x+dx) = V^a(x) + dx^b \partial_b V^a + \mathcal{O}(dx^2)$$

(Taylor exp.)

$$= -\Gamma_{bc}^a dx^b V^c$$

$$V_{\mathcal{L}_1}^a = V^a(x+dx+\delta x) = V^a(x+dx) + \delta x^b \partial_b V^a(x+dx)$$

$$\begin{aligned} V_{\mathcal{L}_1}^a &= V^a + dx^b \partial_b V^a + \delta x^b \partial_b \left(V^a - \Gamma_{dc}^a dx^d V^c \right) \\ &= V^a + (dx^b + \delta x^b) \partial_b V^a - \partial_b \Gamma_{dc}^a \delta x^b dx^d V^c \\ &\quad - \Gamma_{dc}^a \partial_b V^{\mathcal{L}} \delta x^b \delta x^d \\ &\quad = -\Gamma_{bc}^{\mathcal{L}} \delta x^b V^{\mathcal{L}c} \end{aligned}$$

$$\begin{aligned} V_{\mathcal{L}_1}^a &= V^a + (dx^b + \delta x^b) \partial_b V^a \\ &\quad - \left[\partial_b \Gamma_{dc}^a - \Gamma_{df}^a \Gamma_{bc}^f \right] (\delta x^b dx^d V^c) \end{aligned}$$

$$\begin{aligned} V_{\mathcal{L}_2}^a &= \text{---//---} \text{ replacing } dx \leftrightarrow \delta x \\ &\equiv \text{replacing } b \leftrightarrow d \text{ inside the square brackets} \end{aligned}$$

$$\begin{aligned} V_{\mathcal{L}_1}^a - V_{\mathcal{L}_2}^a &= \left[\partial_b \Gamma_{dc}^a - \Gamma_{df}^a \Gamma_{bc}^f - \partial_d \Gamma_{bc}^a \right. \\ &\quad \left. + \Gamma_{bf}^a \Gamma_{dc}^f \right] dx^b \delta x^d V^c \end{aligned}$$

(4)

Riemann Curvature $R^a{}_{bcd}$

is a $(1,3)$ -tngr.

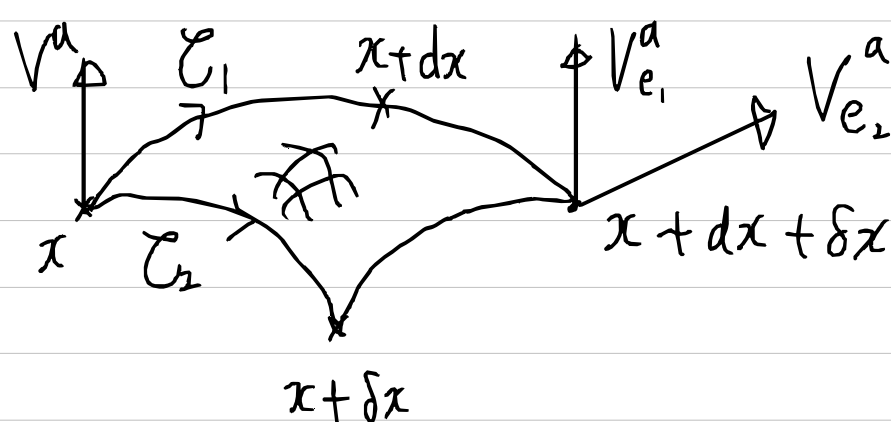
$$R^a{}_{cbd} = \partial_b \Gamma^a_{dc} - \partial_d \Gamma^a_{bc} + \Gamma^a_{be} \Gamma^e_{dc} - \Gamma^a_{de} \Gamma^e_{bc}$$

$$R^a{}_{cbd} = -R^a{}_{cdb}$$

$$\begin{aligned} [D_a, D_b] V^c &= D_a D_b V^c - D_b D_a V^c \\ &\equiv R^c{}_{dab} V^d \end{aligned}$$

Ch2: Curvature

§2.1. Parallel Transport



Vector V^a parallel ($\parallel$) transported along 2 pt.s
 $\neq$ Curves C_1 & C_2

$$C_1: x \rightarrow x+dx \rightarrow x+dx+\delta x$$

$$C_2: x \rightarrow x+\delta x \rightarrow x+\delta x+dx$$

§2.2. Riemann Curvature

(a) R tensor

Manifold
flat



$$R^a_{bcd} \equiv 0 \quad \forall x \in M$$

for all comp.s

(this is a covariant statement)

for Euclidean space
in Cartesian coord.s

$$g_{ab} = \delta_{ab}$$

for S.T. in Minkowski coord.s

$$g_{\mu\nu} = \eta_{\mu\nu}$$

(b) Commutation of covariant deriv.

If φ is a scalar $\longrightarrow D_a D_b \varphi = D_b D_a \varphi$

$$[D_a, D_b] \varphi = 0$$

If V^c is a contravariant vec.:

$$[D_a, D_b] V^c \neq 0.$$

$$[D_a, D_b] V^c = R^c_{dab} V^d \longrightarrow \text{Proof in the LN.}$$

$\longrightarrow$ Try it!

U_c is a covariant vec.

$V^c U_c$ is a scalar

$$[D_a, D_b] (V^c U_c) = 0$$

$$([D_a, D_b] V^c) U_c + V^c ([D_a, D_b] U_c) = 0$$

$$\hookrightarrow = + R^c_{dab} V^d U_c$$

$$\Rightarrow [D_a, D_b] U_c = -R^d_{cab} U_d$$

(2)

If $Y_{\{d_k\}}^{\{c_j\}}$ is a rank- (m, l) -tensor:

$$[D_a, D_b] Y_{\{d_k\}}^{\{c_j\}} = \sum_{j=1}^m R_{dab}^{c_j} Y_{\{d_k\}}^{c_1 \dots \overset{\text{position at } j}{c_j} \dots c_m} - \sum_{k=1}^l R_{d_k ab}^p Y_{d_1 \dots \underset{\text{position at } k}{p} \dots d_l}^{\{c_j\}}$$

Possible Exam Q.s

• Derive the non-zero comp.s of Riemann Curva. Tensors. on the sphr.

→ Practice the 3D version. predominantly.

→ Practice solving the Prb Book's Ch. 9 prob.s (Particularly, Q. 9.20)

§2.3. LIF. (Locally Inertial Frame)

(1) LIF. A coord. sys. at a pt. P fr.m.
where $\Gamma_{bc}^a \Big|_P = 0$

"=" means in the LIF at P

* Check: we can put ourselves always in an LIF at any pt. $P \in M$.

To simplify, we can match the origin of $\{x^a\}_{a=1}^n$ w/ P , $x_P^a = 0$.

Let's imagine that, in $\{x^a\}$, $\Gamma_{bc}^a \Big|_P \neq 0$.

$$x^a \rightarrow \tilde{x}^a = x^a + Q_{bc}^a x^b x^c + O(x^3)$$

const.
coeff.

$$x^a = x^a(\tilde{x}) = \tilde{x}^a - \frac{1}{2} Q_{bc}^a \tilde{x}^b \tilde{x}^c + O(\tilde{x}^3)$$

Under this change of coords,

$$\Gamma_{bc}^a \rightarrow \tilde{\Gamma}_{bc}^a = \Gamma_{\alpha\beta}^{\kappa} \frac{\partial x^{\alpha}}{\partial \tilde{x}^b} \frac{\partial x^{\beta}}{\partial \tilde{x}^c} \frac{\partial \tilde{x}^{\kappa}}{\partial x^{\alpha}} + \frac{\partial^2 x^{\sigma}}{\partial \tilde{x}^b \partial \tilde{x}^c} \frac{\partial \tilde{x}^{\sigma}}{\partial x^{\sigma}}$$

$$\left. \frac{\partial x^a}{\partial \tilde{x}^b} \right|_P = \delta_b^a + O(x)$$

$$\left. \frac{\partial \tilde{x}^a}{\partial x^{\kappa}} \right|_P = \delta_{\kappa}^a + O(\tilde{x})$$

$$\left. \frac{\partial x^{\sigma}}{\partial \tilde{x}^b \partial \tilde{x}^c} \right|_P = -Q_{bc}^{\sigma} + O(x)$$

$$\Gamma \Big|_P \rightarrow \tilde{\Gamma}_{bc}^a \Big|_P = \Gamma_{bc}^a \Big|_P - Q_{bc}^a$$

(3)

By choosing coeff.s of coord. transf.

$$Q_{bc}^a = \Gamma_{bc}^a \Big|_P, \text{ we end up in the LIF}$$

$$\text{at } P: \tilde{\Gamma}_{bc}^a \Big|_P = 0.$$

In LIF,

$$R^a_{bcd} \equiv \partial_c \Gamma_{bd}^a - \partial_d \Gamma_{bc}^a$$

(2) Geodesic Deviation

2 neighboring free massive particles

follow geodesics $\begin{cases} x_1^a(\tau) \\ x_2^a(\tau) = x_1^a(\tau) + \xi^a(\tau) \end{cases}$ Infinitesimal displacement $|\xi| \ll 1$

① $\left\{ \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\} \xrightarrow{\xi^a(\tau)} \left\{ \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\}$ $\dot{\xi}^a = 0$ if the 2 trajectories start // to each other,

$$x^a = x_1^a(\tau) \quad x^a = x_2^a(\tau) = x_1^a(\tau) + \xi^a(\tau)$$

Let's put ourselves in the LIF at P.

$$\text{Geodesic : } ① \ddot{x}_1(\tau) \equiv 0$$

$$② \ddot{x}_2(\tau) + \Gamma_{bc}^a \Big|_P \dot{x}_1^b \dot{x}_1^c = 0$$

$$\Gamma_{bc}^a \Big|_Q \equiv \Gamma_{bc}^a \Big|_P + \partial_d \Gamma_{bc}^a \xi^d + \mathcal{O}(\xi^2)$$

$$+\ddot{\xi}^c = \ddot{x}_2^c(\tau) = -\partial_d \Gamma_{be}^c \xi^d \dot{x}_1^b \dot{x}_1^e + \mathcal{O}(\xi^2)$$

$$\frac{D^2}{D\tau^2} \xi^a = \frac{d}{d\tau} (D_\tau \xi^a) + \underbrace{\Gamma_{cd}^a}_{\equiv 0} (D_c \xi^c) \dot{x}^d$$

$$= \frac{d}{d\tau} \xi^a + \Gamma_{bc}^a \xi^b \dot{x}^c$$

$$D_\tau^2 \xi^a \equiv \frac{d}{d\tau} \left(\dot{\xi}^a + \Gamma_{bc}^a \xi^b \dot{x}^c \right)$$

$$D_\tau^2 \xi^a \equiv \ddot{\xi}^a + \partial_d \Gamma_{bc}^a \dot{x}^d \dot{x}^c \xi^b$$

$$+ \Gamma_{bc}^a (\dot{\xi}^b \dot{x}^c + \xi^b \ddot{x}^c)$$

 $\hookrightarrow \equiv 0$

$$\equiv -\partial_d \Gamma_{bc}^a \dot{x}^b \xi^d \dot{x}^c$$

 $\equiv$: def'd in the LIF.

$$D_\tau^2 \xi^a \equiv \left[\partial_d \Gamma_{bc}^a - \partial_b \Gamma_{cd}^a \right] \dot{x}^b \dot{x}^c \xi^d$$

$$D_\tau^2 \xi^a = -R^a_{bcd} \dot{x}^b \dot{x}^c \xi^d$$

 $\rightarrow$ Geod. deviation

GR ch.2 : Curvature

§2.1. // transport

§2.2. Riemann Curvature

$$R^a{}_{bcd} = \partial_c \Gamma^a{}_{bd} - \Gamma^a{}_{ce} \Gamma^e{}_{bd} - (c \leftrightarrow d)$$

$$[D_a, D_b] V^c = +R^c{}_{dab} V^d$$

$$[D_a, D_b] V_c = -R^d{}_{cab} V_d$$

§2.3. Locally Inertial Frame (LIF)

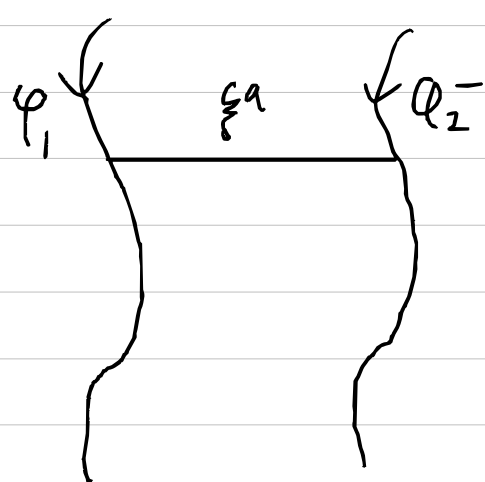
$$\text{LIF @ pt. } P \rightarrow \Gamma^a{}_{bc} \Big|_P = 0$$

$\forall a, b, c = 1, \dots, n$

(1) $\forall$ manifold M , $\forall$ pt. $P \in M$,
we can always put ourselves in LIF @ P
by appropriate coord. transformation.

(2) Geodesic Deviation

(a) neighboring geodesics



$$\frac{D^2 \xi^a(\tau)}{D\tau^2} = -R^a{}_{bcd} \dot{x}^b \dot{x}^d \xi^c$$

derived in LIF, now since this is a covariant eqn., it has to be true in every coord. sys.

(b) Newtonian Limit

Newtonian approximation $\tau \sim t$

$$x^\mu(\tau) = (t, x^i(t))$$

$$\dot{x}^\mu \sim (1, \mathcal{O}(v)) \quad \left(v \ll 1 \text{ in Newtonian limit} \right)$$

$\uparrow$
order of vel.

$$\underline{a} = -\underline{\nabla} \varphi, \quad \varphi : \text{Newtonian potl.}$$

acceleration
in Newtonian approx.

$$\ddot{x}^i = -\partial^i \varphi$$

$$= -\delta^{ij} \partial_j \varphi$$

$$\begin{cases} g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1) \\ g_{ij} = \delta_{ij} \end{cases}$$

$$\ddot{x}^i + \partial^i \varphi = 0$$

analogue to a "Newtonian" geodesic eqn.

$$\ddot{x}^\mu + \Gamma^\mu{}_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0$$

could be written : $\ddot{x}^i + \partial^i \varphi \dot{t} \dot{t} = 0$

In the Newtonian approx.,

$$\Gamma^i{}_{00} = \partial^i \varphi = \delta^{ij} \partial_j \varphi$$

all other $\Gamma^\mu{}_{\alpha\beta} = 0$, e.g., $\Gamma^0{}_{i0} = 0$ in Newtonian analogy.

$$R^i{}_{0j0} = \partial_j \Gamma^i{}_{00} + \cancel{\Gamma^i{}_{0\alpha} \Gamma^\alpha{}_{00}} - \cancel{\partial_0 \Gamma^i{}_{j0}} - \cancel{\Gamma^i{}_{0\alpha} \Gamma^\alpha{}_{j0}}$$

$$R^i{}_{0j0} = \partial_j \partial^i \varphi$$

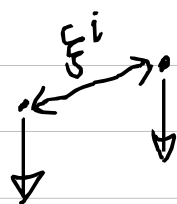
(2)

* Check what $R^0{}_{i0j}$ is

& see if Newtonian limit is consistent w/ properties of $R^a{}_{bcd}$ (P54)

* Geodesic Deviation in Newtonian approx.

$$\xi^\mu = (0, \xi^i)$$



$$\varphi(x^i)$$

$$\frac{D^2 \xi^i}{D\tau^2} = \ddot{\xi}^i = -R^i{}_{0j0} \underbrace{\dot{x}^0}_{=} \underbrace{\dot{x}^0}_{=} \xi^j = -\partial^i \partial_j \varphi$$

"Space-dependence of Newtonian potl."

So, already in Newtonian gravity, we have

tidal forces : $\ddot{\xi}^i = \delta^{ik} \partial_j \partial_k \varphi \xi^k$

It's in separation b/t 2 pt.s that we see tidal forces.

In GR, there's also time dependence " $\partial_t \varphi$ " really of $g_{\mu\nu}$ entering.

(EX 1)

• Tidal forces around the horizon of BHs

→ "spaghettification."

(EX 2)

"GWs" → Time dependence of the metric is important.

↓
GWs would not be described there

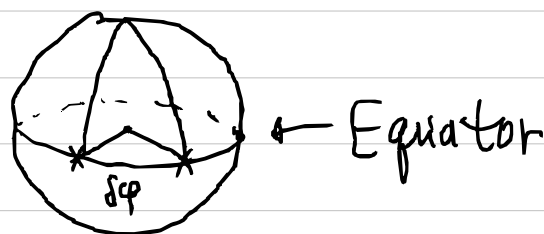
$\partial_t g_{\mu\nu}$ is relevant for GWs.

More generically, (not in Newtonian approx.)

$$\frac{D^2 \xi^a}{D\tau^2} = -\underbrace{R^a{}_{cbd} \dot{x}^c \dot{x}^d}_{S^a{}_b} \xi^b = -S^a{}_b \xi^b$$

$S^a{}_b$: Tidal Forces

(c) Geodesic deviation on sphere



$\xi^a = (0, \delta\varphi)$ → ξ^a decreases as you go straight North or South.

↑
 $\delta\varphi$ along the latitude dir.

0 along the longitudinal dir.

(3)

§2.5. Properties of the Riemann Tnsr.

(a) Riemann tnsr in LIF. $\Gamma \equiv 0$

$$R^a{}_{bcd} \equiv \partial_c \Gamma^a_{bd} - \partial_d \Gamma^a_{bc}$$

↳ when written in terms of metric:

$$\boxed{D_a g_{bc} = 0} \equiv \partial_a g_{bc}$$

↳ Need to know this.

$$\text{Eq. (2.67) of LN} \rightarrow D_a g_{bc} = 0.$$

(Correct ver.)

In LIF,

$$\partial_a \Gamma^b_{cd} \equiv \partial_a \left(\frac{1}{2} g^{be} (\partial_c g_{ed} + \partial_d g_{ce} - \partial_e g_{cd}) \right)$$

↳ can ignore $\partial_a g^{be}$ here, bc only the 2nd deriv. count.

$$\equiv \frac{1}{2} g^{be} (\partial_a \partial_c g_{ed} + \partial_a \partial_d g_{ce} - \partial_a \partial_e g_{cd})$$

$$R^a{}_{bcd}$$

$$\equiv \frac{1}{2} g^{ae} \left[\partial_c \partial_b g_{ed} + \partial_c \partial_d g_{eb} - \partial_c \partial_e g_{bd} - \partial_d \partial_b g_{ec} + \partial_d \partial_e g_{bd} \right]$$

when we interchange $c \leftrightarrow d$ it cancels.

symm. under $c \leftrightarrow d$

Define: covariant ver. of $R^a{}_{bcd}$

$$R_{abcd} = g_{ae} R^e{}_{bcd}$$

$$\equiv \frac{1}{2} \left[\partial_a \partial_c g_{db} + \partial_b \partial_d g_{ac} - \partial_a \partial_d g_{bc} - \partial_b \partial_c g_{ad} \right]$$

$= - (c \leftrightarrow d)$

Antisymm. under the $(c \leftrightarrow d)$ or $(a \leftrightarrow b)$

$$\underline{ac\ db + bd\ ac - ad\ bc - bc\ ad}$$

Symm. under $(a, b) \leftrightarrow (c, d)$

$$R_{abcd} = R_{cdab} = -R_{bacd} = -R_{abdc} = R_{badc}$$

These are covariant eqns, & they're true in any frm.

NEXT: Bianchi Identity

& Cyclic Iden.

$$\left\{ \begin{array}{l} \bigcirc_{abc} R_{abcd} = 0 \\ \bigcirc_{abc} D_a R_{bcde} = 0 \end{array} \right.$$

similar to $\bigcirc_{abc} D_a \Gamma_{bc} = 0.$

GR Ch.2. Curvature

RECAP

§2.1. // transport

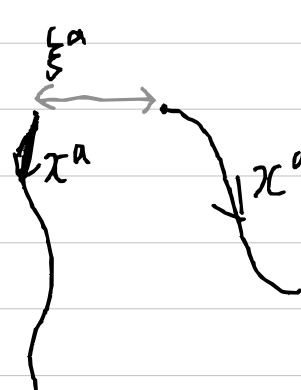
§2.2. Riemann Curvature Tnsr

$$R^a_{bcd} = \partial_c \Gamma^a_{bd} + \Gamma^a_{ce} \Gamma^e_{bd} - (c \leftrightarrow d)$$

§2.3. L.I.F. (a) $P : \Gamma^a_{bc} \Big|_p \stackrel{''}{=} 0$

§2.4. Geodesic Deviation

2 straight lines that start // to e.o. may deviate due to curvature



$$D_\tau^2 \xi^a = -R^a_{bcd} \dot{x}^b \dot{x}^c \xi^d$$

RECAP

§2.5. Properties of Riemann Curvature Tnsr

(1) (anti)-symmetric R_{abcd} * Anti-symm. under $a \leftrightarrow b$ or under $c \leftrightarrow d$ * Symmetric under $(a, b) \leftrightarrow (c, d)$

$$R^a_{bcd} \stackrel{''}{=} \partial_c \Gamma^a_{bd} - \partial_d \Gamma^a_{bc}$$

(2) Ricci Tensor, Scalar Curvature, Einstein Tensor

 $R_{bd} = R^a_{bad}$ is the Ricci tnsr rank-(0,2)-tnsr. $R = g^{ab} R_{ab}$ is a scalar Ricci scalar or scalar curvatureEinstein tnsr : $G_{ab} = R_{ab} - \frac{1}{2} R g_{ab}$
(in any dimensions)to invert this relation : $G^c_c = g^{cd} G_{cd}$

$$G^c_c = R - \frac{n}{2} R = -\frac{(n-2)}{2} R \rightarrow R = -\frac{2}{n-2} G^c_c$$

$$\text{For } n \neq 2: R_{ab} = G_{ab} - \frac{1}{n-2} G^c_c g_{ab}$$

$$R^a_{bcd} \stackrel{''}{=} \partial_c \Gamma^a_{bd} - \partial_d \Gamma^a_{bc}$$

$$g^{ab} g_{bc} = \delta^a_c = \begin{pmatrix} 1 & & 0 \\ & 1 & \\ 0 & & 1 \end{pmatrix}$$

$$g^{ab} g_{ba} = \text{Tr}[g] = n$$

$$R_{ab} = R^c_{acb} = g^{cd} R_{cadb} = g^{dc} R_{bdac} = R_{ba}$$

(Ricci Tensor & Einstein tnsr are symmetric.)

(3) Bianchi Id.

$$\text{EM: } F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -F_{\nu\mu}$$

Faraday & Ampère's Laws derived from the Cyclic Identity

$$\bigcirc_{abc} \partial_a F_{bc} \equiv 0$$

Identity.

$$\bigcirc_{abc} = abc + bca + cab$$

a direct analogue is the 1st Bianchi iden.

$$\bigcirc_{bcd} R^a_{bcd} = 0$$

(2)

$$\bigcup_{bcd} R^a_{bcd} = R^a_{bcd} + R^a_{cdb} + R^a_{dbc}$$

$$\begin{aligned} \text{"="} \quad & \cancel{\partial_c F^a_{bd}} - \cancel{\partial_d F^a_{bc}} + \cancel{\partial_d F^a_{cb}} - \cancel{\partial_c F^a_{db}} \\ & + \cancel{\partial_b F^a_{dc}} - \cancel{\partial_b F^a_{cd}} \end{aligned}$$

"=" 0 Since this is a tensor eq., it has to be true in any frame

$$\rightarrow \bigcup_{bcd} R^a_{bcd} = 0 \iff \bigcup_{abc} R_{abcd} = 0$$

$$\bigcup_{abc} \partial_a F_{bc} = 0 \quad \left(\text{really } \bigcup_{abc} D_a F_{bc} = 0 \right)$$

PS5

2nd Bianchi Identity $\bigcup_{abc} D_a R_{bcde} = 0$

Can prove it by writing

$$R_{abcd} = \frac{1}{2} [g_{ac, bd} + \dots]$$

$$g^{ad} g^{ce} [D_a R_{bcde} + D_b \underbrace{R_{cade}}_{=-R_{acde}} + D_c R_{abde}] = 0$$

$$D_a g_{bc} = 0$$

$$g^{ad} D_a R_{bd} - D_b R + g^{ce} D_c R_{bd} = 0$$

$$2 D_a R^a_b - \underbrace{D_b R}_{\text{"=" } \frac{1}{2} R} = 0 \rightarrow D_a G^a_b = 0$$

Identity

always true.

$$(\equiv D_\mu D_\nu F^{\mu\nu} = 0)$$

$$D_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

$$D_\mu D_\nu F^{\mu\nu} = 0 = -\mu_0 D_\nu J^\nu$$

4-current is conserved: $D_\mu J^\mu = 0$

$$J^\mu = (\rho, \vec{r})$$

$$\dot{\rho} + \nabla \cdot \underline{j} = 0 \quad \text{: continuity eqn.}$$

(3)

$D_\mu G^\mu_\nu = 0$: Einstein tensor is transversal.
(conservation laws associated to that)

(3) # of indep. compo.s

$$R_{abcd}, \quad a, b, c, d = 1, \dots, n$$

(= 0, ..., d in (d+1)-dim.)

A priori, $\exists$ n^4 components if A, B had N compo.s,
then Riemann

$$R_{\underbrace{ab}_A \underbrace{cd}_B} = R_{\underbrace{cd}_B \underbrace{ab}_A}$$

$I(N) = \frac{N(N+1)}{2}$
is # of indep. compo.s
of R a priori.

A, B are each $n \times n$ anti-symm. tensors

$$A = \{a, b\}, \quad a, b = 1, \dots, n$$

$$R_{abB} = -R_{baB} \quad \& \quad R_{Acd} = -R_{Adc}$$

$$R_{AB} = -R_{A^TB}$$

A & B have $N = \frac{n(n-1)}{2}$ indep. compo.s

The Bianchi Idem. carries info if all 4 compo.s are

$$\neq \bigcup_{abc} R_{abcd} = 0$$

of indep. eqn.s
in Bianchi identity

is

$$\binom{4}{n} = \frac{n!}{(n-4)!4!}$$

n choices of a
 $(n-1)$ choices of b
 $(n-2)$ " c
 $(n-3)$ " d

ordering irrelevant,
so divide by $4!$

In total: R^a_{bcd} carries

$$N(n) = I(N(n)) - \frac{n!}{(n-4)!4!} \quad \text{of indep. compo.}$$

$$= \frac{n^2(n^2-1)}{12}$$

§2.6. Isometries

(a) Killing Vectors

$$\nabla_a G^a_b = 0$$

§2.7. Weyl & Conformal Relations

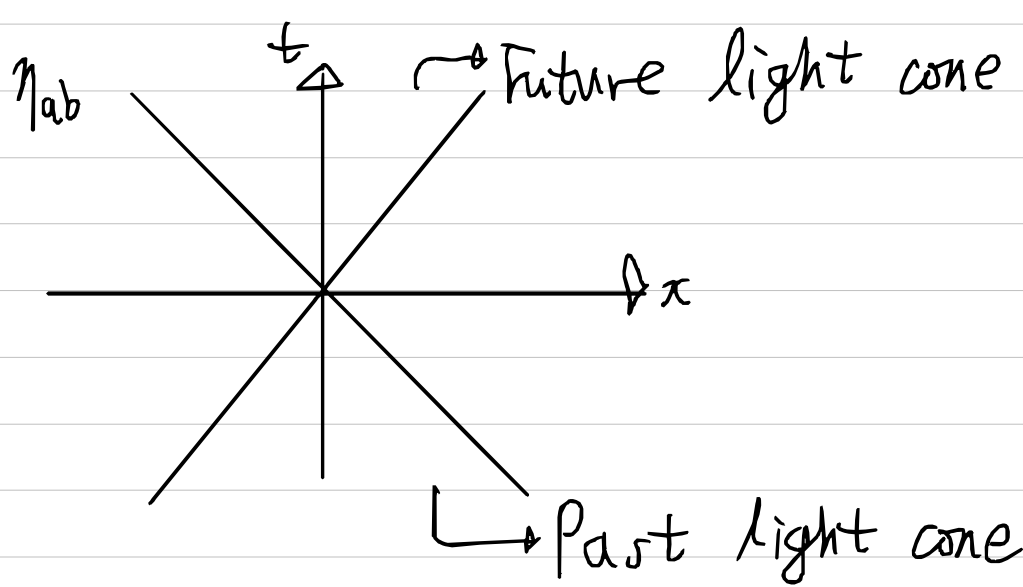
$$g_{ab} \rightarrow \bar{g}_{ab} = \Omega^2 g_{ab} \quad : \text{Weyl Relations}$$

Conformal Factor

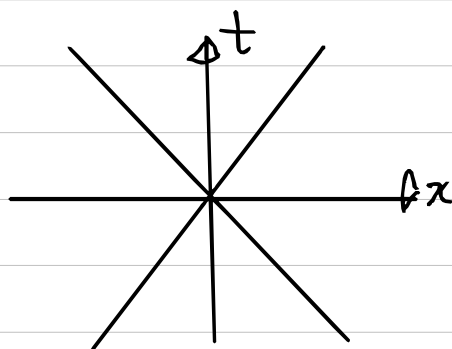
* Conformally Flat Space.

$$g_{ab} = \Omega^2(x) \eta_{ab}$$

conformally related to Minkowski space.



$$\eta_{ab} \rightarrow g_{ab}$$

keep the
L.C. struc.

* line element for light is

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu = 0$$

(Minkowski)

In conformally flat sp.-tm., $(\eta_{\mu\nu} \rightarrow g_{\mu\nu})$

$$d\tilde{s}^2 = g_{\mu\nu} dx^\mu dx^\nu = \Omega^2 \eta_{\mu\nu} dx^\mu dx^\nu$$

Causal struc.

$$= \Omega^2 ds^2$$

remain the same

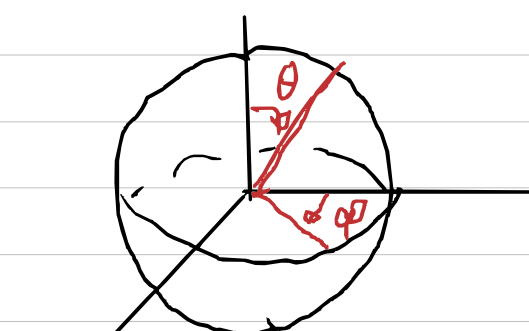
$$= 0$$

In 2D, any manifold is conformally flat:

(EX) 2-sphere

$$ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2$$

$$= \Omega^2 [dx^2 + d\varphi^2]$$

 $\{\theta, \varphi\}$
 $\downarrow$
 $\{x, \varphi\}$

$$\Omega^2 dx^2 = \sin^2 \theta(x) dx^2 = d\theta^2$$

$$\int dx = \int \frac{1}{\sin \theta} d\theta = \ln \left| \tan \left(\frac{\theta}{2} \right) \right|$$

$$x = x(\theta) = \ln \left| \tan \left(\frac{\theta}{2} \right) \right|$$

$$\theta = \theta(x) = 2 \tan^{-1}(e^x)$$

$$\sin \theta = \frac{1}{\cosh x}$$

$$ds^2 = d\theta^2 + \sin^2 \theta d\varphi^2 = \sin^2 \theta(x) [dx^2 + d\varphi^2]$$

$$= \frac{1}{\cosh^2 x} [dx^2 + d\varphi^2]$$

In PS4, generically any 2-D metric ②

$$ds^2 = g_{ab} dx^a dx^b = \tilde{g}_{ab} d\tilde{x}^a d\tilde{x}^b$$

$$x^a \rightarrow \tilde{x}^a = \tilde{x}^a(x)$$

$$w/ \quad g_{ab} = \Omega^2 \eta_{ab} \quad (\text{for } sp=tm.)$$

$$\underline{OR} \quad g_{ab} = \Omega^2 \delta_{ab} \quad (\text{for } space)$$

In 2D, Riemann tensor is uniquely determined by Ω & its deriv.s. $\partial_\mu(\Omega)$.

$$R^a_{bcd} \xrightarrow{\frac{(n^2(n^2-1))}{12} \text{ indep. compo.}} \downarrow \begin{cases} C^a_{bcd} \text{ [Weyl Tensor]} \\ \text{(Conformally Invariant Contribution)} \\ * \text{ Traceless \& Conformally } C^a_{bcd} \text{ Invar.} \end{cases} \begin{cases} \rightarrow G_{ab} \neq \text{Sourced by matter flds in our manifold. (E.T.)} \\ \rightarrow G_{ab} \text{ has } \left(\frac{n(n+1)}{2}\right) \text{ indep. compo.s.} \end{cases}$$

$$C_{abcd} = R_{abcd} + \frac{1}{n-2} \left(R_{ad} g_{bc} - R_{ac} b_{bd} - R_{bd} g_{ac} + R_{bc} g_{ad} \right) + \frac{R}{(n-1)(n-2)} (g_{ac} g_{bd} - g_{ad} g_{bc})$$

W.T. has the same properties as R.T. (R) (C)

$$* \text{ Traceless} \quad g^{ac} C_{abcd} = C^a_{bad} = 0$$

* Conformally Invariant

$$C^a_{bcd} [\Omega^2 g_{\mu\nu}] = C^a_{bcd} [g_{\mu\nu}]$$

C^a_{bcd} for a conformally flat sp-tm.

$$\hookrightarrow \equiv 0 = C^a_{bcd} [\Omega^2 \eta_{\mu\nu}]$$

$$0 = C^a_{bcd} [\eta_{\mu\nu}]$$

Weyl Tensor has the same # of additional constraints as [Riemann T - $\frac{n(n+1)}{2}$]

continuous from tracelessness, $\downarrow$

$$W_{\text{Weyl}}(n) = \frac{n^2(n^2-1)}{12} - \frac{n(n+1)}{2}$$

$$\downarrow$$

$$C^a_{bad} = 0$$

$$\text{Riemann T} \rightarrow \text{split into } \textcircled{0} \text{ (E.T. } G_{ab}) \left[\frac{n(n+1)}{2} \text{ indep. compo.s.} \right]$$

$$+ \textcircled{2} \text{ (Weyl T.)}$$

$\downarrow$ cares abt matter flds sourced by

$$C^a_{bcd} \left[\frac{n^2(n^2-1)}{2} - \frac{n(n+1)}{2} \right]$$

indep. compo.s

$$T_{ab} \text{ (S-E. T.)}$$

which can be $\neq 0$ even in the absence of sources.

W.T. tells us abt local distortions, like GWs, Tidal Forces

(Quantities that describe Sp.-Tm.)

$$= \text{const.} \times \left(\text{Quantities that describe sources living in the } S-T \right)$$

$G_{\mu\nu}$ $T_{\mu\nu}$

$8\pi G$ Φ

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

§3.1. Analogy w/ EM

4-vec. potl. A_μ (massless Spin-1 fld)

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi(x) \quad \text{photon}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \text{: Faraday Tensor}$$

$\mathcal{L}_{EM}$ $\xrightarrow{\text{Lagr. den.}}$ scalar invariant under L.T. & gauge Tr.

$$\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \rightarrow a = -\frac{1}{4}$$

$$\mathcal{L}_{EM} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \mu_0 A_\mu \boxed{J^\mu}$$

$\downarrow$ 4-current

$$S_{EM} = \int d^4x \mathcal{L}_{EM}[A_\mu]$$

$$\delta_{A_\mu} S_{EM} = 0 \rightarrow \partial_\mu F^{\mu\nu} = -\mu_0 J^\nu$$

In GR, $\rightarrow$ massless spin-2 fld.
(Graviton)

Invariant under generic coord. Transformations

$$x^\mu \rightarrow \tilde{x}^\mu = \tilde{x}^\mu(x)$$

4-vec. current $J^\mu = (-\rho, \vec{j})$

Contin. Eq. $-\dot{\rho} + \nabla \cdot \vec{j} = 0$

↓

En. Cons.

* Faraday Tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$
 $= -F_{\nu\mu}$

rank $-(0,2)$ tensor L.T.

All of this can be easily generalized to curved S-T with full covariance i.e., full coord. transform. (invariance).

Lagr. for EM (Maxwell Lagr.)

$$\mathcal{L}_{EM} = \mathcal{L}_{EM}(A_\mu, \partial_\alpha A_\mu)$$

$$= -\frac{1}{4} F_{\mu\nu}^2 + \mu_0 A_\mu J^\mu$$

$$F_{\mu\nu}^2 = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\mu\nu} F_{\alpha\beta}$$

$$S_{EM}[A_\mu] = \int d^n x \mathcal{L}(A, \partial A)$$

↳ invariant under Poincaré Transfrm.

$$x^\mu \rightarrow \tilde{x}^\mu = \Lambda^\mu_\alpha x^\alpha + \bar{x}^\mu$$

$A_\mu J^\mu$ & $F_{\mu\nu}$ ^② are scalars

$$F_{\mu\nu}^2 \rightarrow \tilde{F}_{\mu\nu}^2 = F_{\mu\nu}^2$$

$$A_\mu J^\mu \rightarrow \tilde{A}_\mu \tilde{J}^\mu = A_\mu J^\mu$$

$$\mathcal{L}_{EM} \rightarrow \tilde{\mathcal{L}}_{EM} = \mathcal{L}_{EM}$$

$$\eta_{\mu\nu} \rightarrow \tilde{\eta}_{\mu\nu} = \eta_{\mu\nu}$$

$$d^n x \rightarrow d^n \tilde{x}$$

Under Poincaré

↳ Putting all tog, → the action is invariant.

∴ $S_{EM} = \int d^n x \mathcal{L}_{EM} \Rightarrow \tilde{S}_{EM} \rightarrow S_{EM}$
 Invar. under Poincaré

* Gauge Invariance with gauge parameter χ :

$$A_\mu \rightarrow A'_\mu = A_\mu + \partial_\mu \chi$$

$$F_{\mu\nu} \rightarrow F'_{\mu\nu} = F_{\mu\nu}$$

$$A_\mu J^\mu \rightarrow A'_\mu J^\mu = A_\mu J^\mu + \partial_\mu \chi J^\mu$$

$$S_J = \mu_0 \int d^n x A_\mu J^\mu$$

↳ $S'_J = \mu_0 \int d^n x A_\mu J^\mu + \mu_0 \int d^n x \partial_\mu \chi J^\mu$ // S_J

$\mu_0 \int d^n x \partial_\mu \chi J^\mu$ (source part) $= -\mu_0 \int d^n x \chi \underbrace{\partial_\mu J^\mu}_{\text{should be 0}}$

Gauge invar imposes $\partial_\mu J^\mu = 0$.

$$S_J \rightarrow S'_J = S_J$$

∴ S_{EM} is invar. under Gauge Transfrm.

Guiding principle : S_{EM} invar. under G, T .
 $(A_\mu \rightarrow A_\mu + \partial_\mu \chi)$
 & P.T.

$$\& S_{EM} = \int d^n x \mathcal{L}_{EM}$$

Physics
 (usually $\mathcal{L}$
 2nd-order
 deriv.s

$$\mathcal{L}_{EM} \sim (\partial A)^2$$

$\hookrightarrow$ higher order
 corrections
 can be added.

quadratic
 in ∂ & A

From $S_{EM}[A_\mu]$, a semi-classical sol.

$$\delta A_\mu \delta S_{EM} = 0$$

(thru the E-L E)

$$\delta S_{EM} = \int d^n x \left[-\frac{1}{4} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) F^{\mu\nu} + \mu_0 (\delta A_\mu) J^\mu \right]$$

$$\delta S_{EM} = (\text{boundary terms})$$

$$+ \int d^n x [-\partial_\nu F^{\mu\nu} + \mu_0 J^\nu] \delta A_\mu$$

$$\delta S_{EM} = 0 \implies \partial_\mu F^{\mu\nu} = \mu_0 J^\nu$$

"Bianchi Id":

$$\partial_\mu G^{\mu\nu} = 0$$

$$G \sim R \sim \partial T \sim T^2$$

$$\downarrow$$

$$\approx \partial(g) \approx \partial^2(g)$$

as a consequence
 of Bian. Id.

$$\partial_\mu \partial_\nu F^{\mu\nu} = 0 \implies \text{4-current } J^\mu \text{ conserved}$$

$$= \mu_0 \partial_\mu J^\mu$$

* Gauss Eq

$$\text{For E.F.} : \nabla \cdot \vec{E} = 4\pi \rho_e$$

$$\text{Gauss for M.F.} : \nabla \cdot \vec{B} = 0$$

Identities

$$\text{Faraday} : \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\text{Ampere} : \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \vec{J}$$

$$\bigcirc_{\alpha\mu\nu} \partial_\alpha F_{\mu\nu} = 0 \text{ identity of}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

(3)

Gravity

 $n = (d+1)$ -dim. Sp-Tm. (no longer flat)

$$\{x^\mu\}_{\mu=0}^d = \{t, x^{\bar{i}}\}_{\bar{i}=1}^d$$

 $g_{\mu\nu}(x)$

built covariant formalism
under an essential symm:
Invariance under full coord.
Trans.

Non-linear diffeomorphism: $x^\mu \rightarrow \tilde{x}^\mu = \tilde{x}^\mu(x)$ ✓

(1) From non-lin. diffeo. $\xrightarrow{to}$ lin. diffeo.

To compare w/ the coord. trans. in EM:

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi(x)$$

✓ P55

→ Look at small fluctuations

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}(x) \quad (h \ll 1)$$

$$x^\mu \rightarrow \tilde{x}^\mu = x^\mu - \chi^\mu(x) + \dots$$

↳ to 1st order

$$x^\mu(\tilde{x})$$

(small change $|\chi| \ll 1$)

$$x^\mu = \tilde{x}^\mu + \chi^\mu(\tilde{x}) + \dots$$

$$g_{\mu\nu}(x) \rightarrow \tilde{g}_{\mu\nu}(\tilde{x}) = g_{\alpha\beta} \frac{\partial x^\alpha}{\partial \tilde{x}^\mu} \frac{\partial x^\beta}{\partial \tilde{x}^\nu}$$

$$\eta_{\mu\nu} + h_{\mu\nu}(x) \rightarrow \eta_{\mu\nu} + \tilde{h}_{\mu\nu}(\tilde{x})$$

$$= (\eta_{\alpha\beta} + h_{\alpha\beta}) \left(\delta_\mu^\alpha + \partial_\mu \chi^\alpha \right) \times \left(\delta_\nu^\beta + \partial_\nu \chi^\beta \right)$$

+ higher order corrections.

$$= \eta_{\alpha\beta} + h_{\alpha\beta} + \partial_\mu \chi_\nu + \partial_\nu \chi_\mu + \text{H.O.C.}$$

Under Linear Diffeomorph.

$$h_{\mu\nu}(x) \rightarrow \tilde{h}_{\mu\nu}(\tilde{x}) = h_{\alpha\beta} + \partial_\mu \chi_\nu + \partial_\nu \chi_\mu$$

G.T. w/ gauge parameter χ_μ

⬆
L.T. might be included.

Isometries

Local symmetry / Gauge Symmetry

$$\bar{x}^\mu = x^\mu + \epsilon^\mu(x)$$

like in particle
phys.

In EM $\rightarrow A_\mu(x) \rightarrow A_\mu + \partial_\mu X(x)$

Physics doesn't care whether which coord.s
we choose.

Redundancy rather than symmetry.

Global symmetries (or Actual symmetries
or Real symm.)

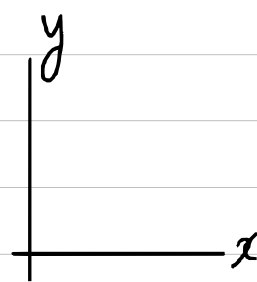
$$ds^2 = g_{\mu\nu}(x) dx^\mu dx^\nu = \bar{g}_{\alpha\beta}(\bar{x}) d\bar{x}^\alpha d\bar{x}^\beta$$

$$g_{\mu\nu}(x) = \bar{g}_{\alpha\beta}(\bar{x}) \frac{\partial \bar{x}^\alpha}{\partial x^\mu} \frac{\partial \bar{x}^\beta}{\partial x^\nu}$$

Scalars: $\phi(x) = \bar{\phi}(\bar{x})$

EX

$$ds^2 = dx^2 + dy^2$$



$$g_{\text{ct}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad g_{xx}=1, \quad g_{xy}=0, \quad g_{yy}=1$$

NOT

A
SYMMETRY

$$ds^2 = dr^2 + r^2 d\theta^2$$

$$g_{\text{sp}} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

$$g_{\text{ct}} \neq g_{\text{sp}}$$

ISOMETRY
= SYMMETRY in GR

$$\left. \begin{aligned} \bar{x} &= x \cos \theta - y \sin \theta \\ \bar{y} &= y \cos \theta + x \sin \theta \end{aligned} \right\} \rightarrow ds^2 = dx^2 + dy^2 = d\bar{x}^2 + d\bar{y}^2$$

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \& \quad \bar{g} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

ISOMETRY $\equiv$ Global Symmetry

$$x^\mu \rightarrow \bar{x}^\mu \Rightarrow \bar{g}_{\mu\nu}(x) = g_{\mu\nu}(x)$$

↑ ↑
same argument x

Focusing on infinitesimal rotations:

$$\bar{x}^\mu = x^\mu + \epsilon E^\mu(x)$$

$E = \text{infinitesimal const.}$

$$g_{\mu\nu}(x) = \bar{g}_{\alpha\beta}(\bar{x}) = \bar{g}_{\alpha\beta}(x + \epsilon E) \frac{\partial(x^\alpha + \epsilon E^\alpha)}{\partial x^\mu} \frac{\partial(x^\beta + \epsilon E^\beta)}{\partial x^\nu}$$

$$= \left[\bar{g}_{\alpha\beta}(x) + \epsilon E^\omega \partial_\omega \bar{g}_{\alpha\beta} + O(\epsilon^2) \right] \left[\delta_\mu^\alpha + \epsilon \partial_\mu E^\alpha \right]$$

(Taylor Expansion)

$$\times \left[\delta_\nu^\beta + \epsilon \partial_\nu E^\beta \right]$$

$$= \bar{g}_{\alpha\beta}(x) \delta_\mu^\alpha \delta_\nu^\beta + \epsilon \left(\bar{g}_{\alpha\beta} \delta_\mu^\alpha \partial_\nu E^\beta + \bar{g}_{\alpha\beta} \partial_\mu E^\alpha \delta_\nu^\beta + E^\omega \partial_\omega \bar{g}_{\alpha\beta} \delta_\mu^\alpha \delta_\nu^\beta \right) + O(\epsilon^2)$$

$$= \bar{g}_{\mu\nu}(x) + \epsilon \left(\bar{g}_{\mu\beta} \partial_\nu E^\beta + \bar{g}_{\alpha\nu} \partial_\mu E^\alpha + E^\omega \partial_\omega \bar{g}_{\mu\nu} \right) + O(\epsilon^2)$$

Isometry: $g_{\mu\nu}(x) = \bar{g}_{\mu\nu}(x)$

$$g_{\mu\beta} \partial_\nu E^\beta + g_{\alpha\nu} \partial_\mu E^\alpha + E^\omega \partial_\omega g_{\mu\nu} = 0$$

Lie derivative

$$\mathcal{L}_E g_{\mu\nu} = g_{\mu\beta} \partial_\nu E^\beta + g_{\alpha\nu} \partial_\mu E^\alpha + E^\omega \partial_\omega g_{\mu\nu} = 0$$

$$\mathcal{L}_E \varphi = E^\mu \partial_\mu \varphi$$

$$\partial_\omega g_{\mu\nu} = \Gamma_{\mu\omega}^\alpha g_{\alpha\nu} + \Gamma_{\omega\nu}^\alpha g_{\mu\alpha}$$

$$(D_\omega g_{\mu\nu} = 0)$$

Killing Eq.

$$(D_\mu E^\omega) g_{\omega\nu} + (D_\nu E^\omega) g_{\mu\omega} = 0$$

Killing vec. satisfies (E_ν)

$$D_\mu E_\nu + D_\nu E_\mu = 0$$

$$E^\omega g_{\mu\omega} = E_\mu$$

In 4D, 10 eq.s for 4 quantities. (In 4D)
No generic sol.s

EX D=2

$$ds^2 = dx^2 + dy^2, \quad g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \eta$$

$$\Gamma_{\nu\beta}^\mu = 0$$

$$\partial_\mu E_\nu + \partial_\nu E_\mu = 0$$

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 0$$

$$\begin{aligned} x, x: \partial_x \xi_x + \partial_x \xi_x &= 2 \partial_x \xi_x = 0 \\ y, y: \partial_y \xi_y + \partial_y \xi_y &= 2 \partial_y \xi_y = 0 \\ x, y: \partial_x \xi_y + \partial_y \xi_x &= 0 \end{aligned} \rightarrow \text{sol.: } \begin{aligned} \xi_x &= y \\ \xi_y &= -x \\ \partial_x \xi_y &= -1 \\ \partial_y \xi_x &= 1 \end{aligned}$$

$$\begin{aligned} \bar{x} &= x \cos \theta + y \sin \theta \\ \bar{y} &= y \cos \theta - x \sin \theta \end{aligned}$$

$$\xi^\mu = (y, -x)$$

$$\begin{aligned} \bar{x} &= x + y \delta \\ \bar{y} &= y - x \delta \end{aligned} \rightarrow \bar{x}^\mu = x^\mu + \xi^\mu$$

Minkowski sp.-tm.

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & & \\ & 1 & \\ & & \ddots & \\ & & & 1 \end{pmatrix}$$

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = 0$$

$$\xi_\mu = C_\mu + \omega_{\mu\nu} x^\nu$$

4 (transl) 6 (LT)

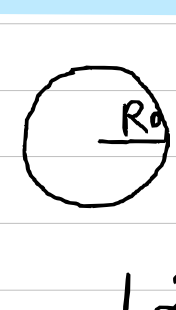
$$\partial_\nu \xi_\mu = \omega_{\mu\nu}$$

$$\omega_{\mu\nu} = \omega_{[\mu\nu]} = -\omega_{\nu\mu} : \text{anti-symm.}$$

10 symmetries
4 translations
6 Lorentz grp
(3 rotations)
(3 boosts)
Lorentz

EX.

2-sphr. S_2



$$R_0^2 = x_1^2 + x_2^2 + x_3^2$$

$$\begin{aligned} x_1 &= R_0 \cos \theta_1 \\ x_2 &= R_0 \sin \theta_1 \cos \theta_2 \\ x_3 &= R_0 \sin \theta_1 \sin \theta_2 \end{aligned}$$

$$\begin{aligned} ds^2 &= dx_1^2 + dx_2^2 + dx_3^2 \\ &= R_0^2 (d\theta_1^2 + \sin^2 \theta_1 d\theta_2^2) \end{aligned}$$

Eq. (2.96)

$$g = \begin{pmatrix} R_0^2 & 0 \\ 0 & R_0^2 \sin^2 \theta_1 \end{pmatrix}$$

$$\xi^\omega \partial_\omega g_{\mu\nu} + g_{\mu\nu} \partial_\mu \xi^\omega + g_{\mu\omega} \partial_\nu \xi^\omega = 0$$

$$\theta_1, \theta_1 : \partial_{\theta_1} \xi^{\theta_1} = 0$$

$$\theta_1, \theta_2 : (\sin^2 \theta_1) \partial_{\theta_1} \xi^{\theta_2} + \partial_{\theta_2} \xi^{\theta_1} = 0$$

$$\theta_2, \theta_2 : \xi^{\theta_1} \partial_{\theta_1} (\sin^2 \theta_1) + 2 \sin^2 \theta_1 \partial_{\theta_2} \xi^{\theta_2} = 0$$

$$\xi^\mu = (\xi^{\theta_1}, \xi^{\theta_2})$$

$\sin \theta_1 \stackrel{?}{=} 1$ in LN

$$\xi_1^\mu = (0, 1)$$

$$\xi_2^\mu = (\cos \theta_2, -\cos \theta_1 \sin \theta_2)$$

$$\xi_3^\mu = (\sin \theta_2, \cos \theta_1 \cos \theta_2)$$

$$\begin{aligned} \mathcal{L} \varphi &= \xi^\mu \partial_\mu \varphi \\ &= \frac{\partial \varphi}{\partial \theta_2} \end{aligned}$$

The same thing

$$\mathcal{L} = \frac{\partial}{\partial \theta_2} : \text{op.} \leftrightarrow L_z = -i\hbar \frac{\partial}{\partial \theta_2}$$

Lie in GR

z-compo. of the ang mtn in QM.

§3.2. Einstein-Hilbert Action

$$S_{GR} = \int \underbrace{d^4x \sqrt{-g}}_{\text{Vol. elem.}} \left[\underbrace{\frac{1}{2K}}_{\text{Const.}} \underbrace{R}_{\text{scalar curvature}} + \underbrace{\mathcal{L}_m(g_{\mu\nu}, \psi_i)}_{\text{Matter Lagr.}} \right]$$

$r^2 \sin \theta dr d\theta d\phi$

$\mathcal{L}_m(g_{\mu\nu}, \psi_i)$ incl.s all fld.s ψ_i living on the spacetime.

In PS5, a & $\rightarrow \mathcal{L}_m \supset \mathcal{L}_{EM}[g, A_\mu]$

$$\mathcal{L}_{EM}[g, A_\mu] = -\frac{1}{4} F_{\mu\nu}^2 = -\frac{1}{4} g^{\alpha\mu} g^{\beta\nu} F_{\mu\nu} F_{\alpha\beta}$$

K : const. that can be recovered by comparing with Newton's law using weak fld approximation.

EOMs come from $\delta_g S_{GR} = 0 \quad \forall \delta g_{\mu\nu}$

The Soln extremizes E-H action.

Using the Minimum

$$\delta_g S_{GR} = \int d^4x \left\{ \frac{\delta \sqrt{-g}}{\delta g_{\mu\nu}} \left[\delta g_{\mu\nu} \left[\frac{1}{2K} R + \mathcal{L}_m \right] \right] \right\}$$

$$R = g^{\alpha\beta} R_{\alpha\beta}$$

$$= \int d^4x \left\{ \underbrace{\frac{\delta \sqrt{-g}}{\delta g_{\mu\nu}} \delta g_{\mu\nu}}_{\textcircled{1}} \frac{1}{2K} R + \frac{\sqrt{-g}}{2K} \left[\underbrace{\left(\frac{\delta g^{\alpha\beta}}{\delta g_{\mu\nu}} \delta g_{\mu\nu} \right)}_{\textcircled{2}} R_{\alpha\beta} + \underbrace{\left(\frac{\delta R_{\alpha\beta}}{\delta g_{\mu\nu}} \right)}_{\textcircled{3}} \underbrace{\delta g_{\mu\nu} g^{\alpha\beta}}_{\textcircled{4}} \right] + \frac{\delta \sqrt{-g} \mathcal{L}_m}{\delta g_{\mu\nu}} \delta g_{\mu\nu} \right\}$$

*① Variation of $\det g_{\mu\nu} = g$

* Jacobi iden : A_{ij}

$$\frac{\partial(\det A)}{\partial A_{ij}} = (\text{adj } A)^{ij}$$

$$(\text{adj } A) \underline{A} = (\det A) \underline{1}$$

$$\text{adj } A = (\det A) A^{-1}$$

$$\frac{\delta g}{\delta g_{\mu\nu}} = g g^{\mu\nu} \quad \& \quad \frac{\delta \sqrt{-g}}{\delta g_{\mu\nu}} = \frac{1}{2} \sqrt{-g} g^{\mu\nu}$$

$$\frac{\delta \sqrt{-g}}{\delta g^{\mu\nu}} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu}$$

*② Variation of inverse metric

$$\frac{\delta g_{\alpha\beta}}{\delta g_{\mu\nu}} = \delta^\mu_\alpha \delta^\nu_\beta = \frac{1}{2} \left[\delta^\mu_\alpha \delta^\nu_\beta + \delta^\mu_\beta \delta^\nu_\alpha \right]$$

$$g^{\alpha\sigma} g_{\sigma\tau} = \delta^\alpha_\tau \quad \& \quad \delta(g^{\alpha\sigma} g_{\sigma\tau})$$

$$\frac{d}{dx} (x^{-1}) = -x^{-2} = 0$$

$$\delta(g^{\alpha\sigma} g_{\sigma\tau}) = \delta(g^{\alpha\sigma}) g_{\sigma\tau} + g^{\alpha\sigma} \delta(g_{\sigma\tau})$$

$$\delta g^{\alpha\sigma} = -g^{\alpha\beta} g^{\sigma\tau} \delta g_{\beta\tau}$$

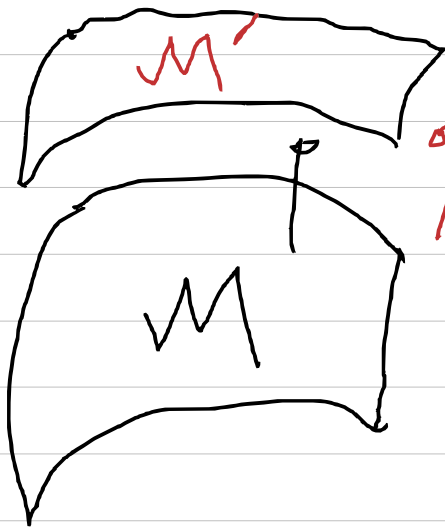
$$\frac{\delta g^{\alpha\beta}}{\delta g_{\mu\nu}} = -g^{\alpha(\mu} g^{\nu)\beta}$$

(2)

* (3) Variation of Ricci Tensor

In RFS $\rightarrow$ Q (Last)

$$\delta R_{\alpha\beta} = D_\sigma \Gamma_{\alpha\beta}^\sigma - D_\alpha \Gamma_{\sigma\beta}^\sigma$$



$$R'_{\alpha\beta} = R_{\alpha\beta} + \delta R_{\alpha\beta}$$

derivation $\delta R_{\alpha\beta}$

$R_{\alpha\beta}$

* Covariant Deriv. ^{obeys} $\rightarrow$ Leibniz rule

$$D_\mu [S \cdots T \cdots] = D_\mu [S \cdots] T \cdots + S \cdots D_\mu [T \cdots]$$

We can do

IbPs w/ covar. deriv. D_μ rather than ∂_μ :

$$\begin{aligned} & \int d^4x S \cdots D[T \cdots] \\ &= \int d^4x D_\mu [S \cdots T \cdots] - \int d^4x (D_\mu S \cdots) T \cdots \end{aligned}$$

(boundary terms)

PS 6 (#2(c))

(3)

$$D_\sigma g^{\mu\nu} = 0 = D_\sigma g_{\mu\nu} \quad \& \quad D_\sigma \sqrt{-g} = 0$$

Part (3):

$$\int d^4x \sqrt{-g} g^{\alpha\beta} \delta R_{\alpha\beta}$$

$$= \int d^4x \sqrt{-g} g^{\alpha\beta} \left[D_\alpha (\delta T^\sigma_{\beta\sigma}) - D_\beta (\delta T^\sigma_{\alpha\sigma}) \right]$$

↳ "IBP-ing"

$$= - \int d^4x \left[\cancel{D_\sigma (\sqrt{-g} g^{\alpha\beta})} \delta T^\sigma_{\alpha\beta} - (\alpha \leftrightarrow \sigma) \right]$$

$$= 0 - 0 = 0$$

$$\textcircled{3} = 0$$

$$\textcircled{4} \rightarrow \frac{\int d^4x (\delta \sqrt{-g}) \mathcal{L}_m(g, \psi_i)}{\delta g_{\mu\nu}} \equiv \frac{1}{2} \sqrt{-g} T^{\mu\nu}$$

For EM,

$$T^{\mu\nu} = \frac{1}{4\pi} \left(F^{\mu\alpha} F^\nu_{\alpha} - \frac{1}{4} F^2 g^{\mu\nu} \right)$$

||
J-E Tensor

for the metric
flds living on
the S-T,

$$\delta g S_{GR} = 0 = \int d^4x \sqrt{-g} \left[\frac{1}{2K} \left(\frac{1}{2} R g^{\mu\nu} - \underbrace{R^{\mu\nu}}_{-G^{\mu\nu}} \right) + \frac{1}{2} T^{\mu\nu} \right] \delta g_{\mu\nu}$$

$$= \int d^4x \sqrt{-g} \left(- \frac{G^{\mu\nu}}{2K} + \frac{1}{2} T^{\mu\nu} \right) \delta g_{\mu\nu}$$

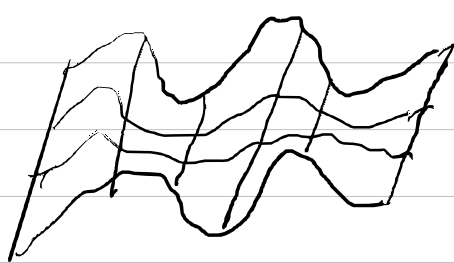
$$\forall \delta g_{\mu\nu}, \quad - \frac{G^{\mu\nu}}{2K} + \frac{1}{2} T^{\mu\nu} = 0$$

$$\frac{G^{\mu\nu}}{2K} = \frac{1}{2} T^{\mu\nu}$$

$$G^{\mu\nu} = K T^{\mu\nu} = 8\pi G T^{\mu\nu} \quad : \text{Einstein fld Eq.}$$

$$K = 8\pi G$$

$$T^{\mu\nu} \equiv \frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} \mathcal{L}_m(g, \psi_i)}{\delta g_{\mu\nu}}$$



Manifold
w/ $g_{\mu\nu}$



manifold w/
 $g_{\mu\nu} + \delta g_{\mu\nu}$

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \quad : \text{EE}$$

★ In the "vacuum" absence ψ_i :

$$T_{\mu\nu} = 0$$

$$G_{\mu\nu} = 0 = R_{\mu\nu}$$

defined flat S-T. as Minkowski :

$$R_{abcd} = 0$$

Conservation of $G_{\mu\nu}$:

$$\text{Bianchi Id. : } D_\mu G^\mu{}_\nu = 0$$

$$D_\mu D_\nu F^{\mu\nu} = 0 = -\mu_0 D_\mu J^\mu$$

Conservation of $T^{\mu\nu}$:

$$D_\mu T^\mu{}_\nu = 0 \quad \text{in most standard cases.}$$

$\equiv$ EOM for that matter fld.

★ Cosmological const. Λ

We think of the C.C. Λ as being included in $\mathcal{L}_m$

$$\mathcal{L}_m = -\Lambda \quad : \text{just a const.}$$

$$\frac{\delta \sqrt{-g}}{\delta g_{\mu\nu}} = \frac{1}{2} \sqrt{-g} g^{\mu\nu}$$

$$\begin{aligned} T_\Lambda^{\mu\nu} &= -\frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} \Lambda}{\delta g_{\mu\nu}} = -\frac{2}{2} \Lambda g^{\mu\nu} \\ &= -\Lambda g^{\mu\nu} \end{aligned}$$

$$\therefore D_\mu T_\Lambda^{\mu\nu} = \partial_\nu \Lambda = 0$$

$\hookrightarrow$ a const.

EE including the C.C. are:

$$G_{\mu\nu} + 8\pi G \Lambda = 0$$

Soln. : "de Sitter space"

* EM

$$\mathcal{L}_{EM} = -\frac{1}{4} F_{\mu\nu}^2$$

$$= -\frac{1}{4} g^{\mu\nu} g^{\alpha\beta} F_{\mu\alpha} F_{\nu\beta}$$

$$T^{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} \mathcal{L}_m}{\delta g_{\mu\nu}}$$

Equivalently,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} \mathcal{L}_m}{\delta g^{\mu\nu}}$$

$$\frac{\delta g^{\alpha\beta}}{\delta g_{\mu\nu}} = -g^{\alpha(\mu} g^{\nu)\beta}$$

$$\frac{\delta \sqrt{-g}}{\delta g^{\mu\nu}} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu}$$

$$T_{\mu\nu}^{EM} = -\frac{2}{\sqrt{-g}} \frac{\delta \sqrt{-g} \left(-\frac{1}{4} F^2\right)}{\delta g^{\mu\nu}}$$

$$= -\frac{2}{\sqrt{-g}} \sqrt{-g} \left(-\frac{1}{4} \frac{\delta g^{\alpha\alpha'} g^{\beta\beta'}}{\delta g^{\mu\nu}} F_{\alpha\beta} F_{\alpha'\beta'} \right)$$

$$= -\frac{1}{4} F^2 g_{\mu\nu} + F_{\mu\beta} F_{\nu}{}^{\beta}$$

$$\downarrow$$

Traceless: $T_{EM\mu}{}^{\mu} = g^{\mu\nu} T_{\mu\nu}^{EM}$

$$= F^2 - \frac{1}{4} F^2 \cdot 4 = 0$$

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} \longrightarrow G^{\mu}{}_{\mu} = 0$$

$$R = 0$$

(No scalar curvature if dominated by "radiation")

$$\Lambda = 0$$

$$D_{\mu} T_{EM\nu}{}^{\mu} = D_{\mu} F^{\mu}{}_{\alpha} F_{\nu}{}^{\alpha} - \frac{1}{2} (D_{\nu} F_{\alpha\beta}) F^{\alpha\beta} + F^{\mu\alpha} D_{\mu} F_{\nu\alpha}$$

Cyclic id. for $D_{\nu} F_{\alpha\beta}$:

$$\bigcup_{\nu\alpha\beta} D_{\nu} F_{\alpha\beta} = 0 = D_{\nu} F_{\alpha\beta} + D_{\alpha} F_{\beta\nu} + D_{\beta} F_{\nu\alpha}$$

$$D_{\nu} F_{\alpha\beta} = -D_{\alpha} F_{\beta\nu} - D_{\beta} F_{\nu\alpha} \\ = D_{\alpha} F_{\nu\beta} - D_{\beta} F_{\nu\alpha}$$

$$(D_{\nu} F_{\alpha\beta}) F^{\alpha\beta} = (D_{\alpha} F_{\nu\beta} - D_{\beta} F_{\nu\alpha}) F^{\alpha\beta}$$

$$D_{\mu} T_{EM\nu}{}^{\mu} = (D_{\mu} F^{\mu}{}_{\alpha}) F^{\alpha}{}_{\nu} + \cancel{F^{\mu\alpha} D_{\mu} F_{\nu\alpha}} - \cancel{F^{\alpha\beta} D_{\alpha} F_{\nu\beta}}$$

$$\begin{array}{l} \mu\alpha \rightarrow \alpha\beta \\ \mu \rightarrow \alpha \\ \nu\alpha \rightarrow \nu\beta \end{array}$$

For EM:

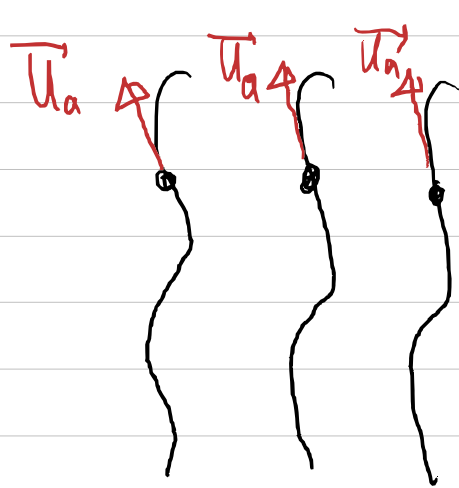
$$(D_{\mu} F^{\mu}{}_{\alpha}) F^{\alpha}{}_{\nu} = 0 \\ \longrightarrow D_{\mu} F^{\mu}{}_{\nu} = 0$$

$$F_{\mu\nu} = \partial_{\mu} A_{\nu} - \partial_{\nu} A_{\mu}$$

Perfect Fluid

P=0, pressureless

Ex. "Dust" (Non-interactive flow of massive particles)



world lines w/
4-vec.
fld. flow.

$$U^a = \frac{dx^a}{d\tau} = U^a(x)$$

Tangent Vec.

τ = proper time along worldlines of a dust particle.

Scalar fld $\rho_0(x)$ = proper density of the flow
= density measured by a co-moving observer
(an obs. moving along w/ the flow)

(3)

$$T^{\mu\nu} = \rho_0 U^\mu U^\nu = \text{EM Tensor for pressureless fluid}$$

$$(P=0)$$

$$(\rho = \rho_0 = \text{const.})$$

Minkowski
SR

$$U^a = \gamma(1, \vec{v})$$

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \quad : \text{Lorentz factor}$$

$$\rho = \gamma^2 \rho_0 = \text{energy density}$$

$$T^{\mu\nu} = \rho \begin{pmatrix} 1 & v^i \\ v^i & v^i v^j \end{pmatrix}$$

In SR, $\partial_\mu T^\mu_\nu = 0$

If $\nu = 0$,
(tm) $\frac{\partial \rho}{\partial t} + \frac{\partial (\rho v^i)}{\partial x^i} = 0$

$$\dot{\rho} + \vec{\nabla} \cdot (\rho \vec{v}) = 0$$

continuity Eq.

If $\nu = i$,
(space)

$$\frac{\partial}{\partial t} (\rho v^i) + \frac{\partial}{\partial x^i} (\rho v^j v^j) = 0$$

$$\Rightarrow \rho [\dot{\vec{v}} + (\vec{v} \cdot \vec{\nabla}) \vec{v}] = 0$$

Navier - Stokes Eq.

4

More generically.

If there's "pressure" P :

$$T_{\mu\nu}^{\text{perfect fluid}} = (\rho + P) u^\mu u^\nu + P g^{\mu\nu}$$

$$T^{\mu\nu} = \begin{pmatrix} T^{00} & T^{0j} \\ T^{\bar{i}0} & T^{\bar{i}j} \end{pmatrix}$$

T^{00} : Energy density

T^{0j} : Energy flux

$T^{\bar{i}0}$: mom density

$T^{\bar{i}j} = \begin{cases} \text{shear} \\ \text{(spatial stress)} \\ \text{(\bar{i} \neq \bar{j})} \\ \text{pressure} \\ \text{(\bar{i} = \bar{j})} \end{cases}$

$$P = \omega \rho$$

ω : EOS parameter

For superconductor $\omega = -1$

For dust $\omega = 0$

Perfect Fluid:

$$T^{\mu\nu} = (\rho + P) u^\mu u^\nu + P g^{\mu\nu}$$

 u^μ : 4-vec. flow P : Pressure ρ : energy density

Sometimes we use

$$M_{\text{Planck}}^2 = \frac{1}{8\pi G_N}$$

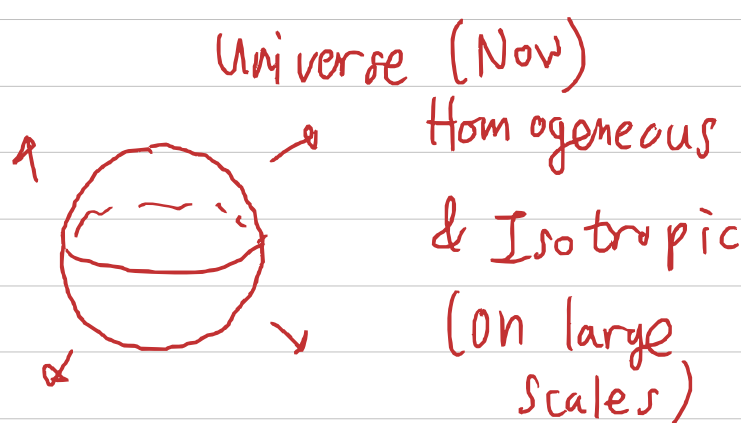
$$S_{\text{GR}}[g, \psi_i] = \int d^4x \sqrt{-g} \left\{ \frac{1}{2} M_{\text{Pl}}^2 R + \mathcal{L}_m(g, \psi_i) \right\}$$

$$G_{\mu\nu} = \frac{1}{2 M_{\text{Planck}}^2} T_{\mu\nu}$$

§3. Cosmology

Navier-Stokes Eq. $\rightarrow$ result of E cons.

Rely on symmetries.



Homogeneity & Isotropy

 $\rightarrow$ Friedmann-Lemaître-Robertson-Walker's (FLRW)Metric $\rightarrow$ The most general metric exhibiting these symmetries.

$$g_{\mu\nu} = (t, \cancel{x^i}) \rightarrow g_{0i} = 0, \quad "g_{ij} \sim \delta_{ij}"$$

spatial

$$ds_{\text{FLRW}}^2 = -N^2(t) dt^2 + a(t) \underbrace{\sigma_{ij} dx^i dx^j}_{\text{3-line element}}$$

lapse function $= \frac{d\tau}{dt}$

scale factor

$$ds_3^2 = \sigma_{ij} dx^i dx^j$$

using solid angle

(3-spatial metric)

$$d\Omega_2^2 = d\theta^2 + \sin^2\theta d\varphi^2$$

$$ds_3^2 = \begin{cases} dx^2 + x^2 d\Omega_2^2 & \rightarrow \text{Flat} \end{cases}$$

$$dx^2 + \sin^2 x d\Omega_2^2$$

3-sphere (closed)

$$dx^2 + \sinh^2 x d\Omega_2^2$$

3-hyperbolic open 3-space

Change of variables

$$\{x, \theta, \varphi\} \longrightarrow \{r, \theta, \varphi\}$$

$$r = r(x) = \begin{cases} x & , \text{ flat} \\ \sin x & , \text{ closed} \\ \sinh x & , \text{ open} \end{cases}$$

$$r' = \frac{dr}{dx}$$

(2)

$$dr = r' dx$$

$$= \begin{cases} 1 dx, & \text{flat} \\ \cos x dx, & \text{closed} \\ -\cosh x dx, & \text{open} \end{cases}$$

$$\cos x = \sqrt{1 - r^2}$$

$$\cosh x = \sqrt{1 + r^2}$$

$$ds^2 = \frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 = \sigma_{ij} dx^i dx^j$$

$$k = \begin{cases} 0 & \text{flat} \\ +1 & \text{closed} \rightarrow \text{posi. curvature} \\ -1 & \text{open} \rightarrow \text{neg. curvature} \end{cases}$$

* General $\{g_{\mu\nu}(x^\alpha)\}$: "mini-superspace"
 $\delta_g S_{GR}$ {scale factor, lapse}
 $= \{a(t), N(t)\}$

$\delta_N S_{GR} \rightarrow$ Friedmann Eq.

$\delta_a S_{GR} \rightarrow$ Raychaudhuri Eq.

(redundant consistency relation)

In "mini-superspace"

$$S_{GR}[a, N, p] = \int d^4x \sqrt{-g_{FLRW}} \left\{ \frac{1}{8\pi G} R - p \right\}$$

$\uparrow$ $\uparrow$ $\uparrow$
 $g_{\mu\nu}$ ψ_i For FLRW

FLRW $\rightarrow$ conformally flat
 can easily compute 4D Ricci tensors

$$R = 6 \left[\frac{k}{a^2} + \frac{\dot{a}^2}{a^2 N^2} - \frac{\dot{a} \dot{N}}{a N^3} + \frac{\ddot{a}}{a N^2} \right]$$

In S_{GR} , boundary parts need attention.

(3)

$$\sqrt{-g} = \frac{a^3 N}{\sqrt{1 - kr^2}}$$

$$\int dt \cancel{a^3 N} \frac{\ddot{a}}{aN^2} = \int dt \frac{a^2 \ddot{a}}{N}$$

Cancelling
some
terms

$$= \int dt \left[\cancel{\frac{2a\dot{a}^2}{N}} + \cancel{\frac{N\dot{a}a^2}{N^2}} \right]$$

$$R = 6 \left[\cancel{\frac{k}{a^2}} + \frac{\ddot{a}^2 a}{a^2 N^2} - \cancel{\frac{\dot{a}N}{aN^3}} + \frac{\ddot{a}a^2}{aN^2} \right]$$

$$\int_{GR}^{Mini} = \int d^4x \frac{1}{\sqrt{1 - kr^2}} \left\{ \frac{3}{8\pi G} \left(aNk - \frac{a\dot{a}^2}{N} \right) - a^3 N \rho \right\}$$

* Friedmann Eq. $S_N S_{GR}$

$$\left(\frac{3}{8\pi G} \left(\cancel{\frac{k}{a^2}} + \frac{a\dot{a}^2}{Na^2} \right) \right) = \cancel{a^3} \rho$$

$$H = \frac{\dot{a}}{aN} : \text{Hubble parameter (expansion rate)}$$

$$\therefore H^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} \quad \text{: Friedmann Eq.}$$

(EOM w.r.t. N)
Lapse

Set $N=1 \rightarrow$ Gauge choice

$$\text{time slicing } d\tilde{t} = N dt$$

$$\tilde{t} = \int^t N dt$$

$$\left(N = \frac{d\tilde{t}}{dt} \right)$$

$\tilde{t}$: cosmic time or physical time

Now, we work w/ $\tilde{t}$ (setting $N=1$)

w/o tilde.
 $t \leftrightarrow \tilde{t}$

Conservation of $T^{\mu\nu}$:

$$T^{\mu\nu} = (\rho + P) u^\mu u^\nu + P g^{\mu\nu} \quad (\text{Perfect fluid})$$

$$\rho = \rho(t), \quad P = P(t) : \text{only depend on } t$$

$$\rho(a(t)) \rightarrow \dot{\rho} = \dot{a} \partial_a \rho$$

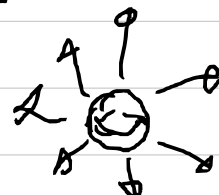
$$P = \omega \rho \rightarrow \omega : \text{EOS parameter}$$

$$u^\mu = (\#, i)$$

$\bar{i}=0 \rightarrow$ Isotropic (no preferred dir.)

$$\# = 1 \rightarrow N=1$$

$$\therefore \underline{u^\mu = (1, 0)}$$



$$g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & a^{-2} \gamma^{ij} \end{pmatrix}$$

$$(\gamma^{ij} = (\gamma_{ij})^{-1})$$

$$\text{So, } T^{\mu\nu} = \begin{pmatrix} \rho & 0 \\ 0 & P a^{-2} \gamma^{ij} \end{pmatrix}$$

$$T^\mu{}_\nu = \begin{pmatrix} \rho & 0 \\ 0 & -P a^{-2} \delta^i{}_j \end{pmatrix}$$

$$0 = D_\mu T^\mu{}_\nu = \partial_\mu T^\mu{}_\nu - \Gamma_{\mu\alpha}^\mu T^\alpha{}_\nu + \Gamma_{\mu\nu}^\alpha T^\mu{}_\alpha$$

$$\underline{\nu=0} \rightarrow \partial_t T^0{}_0 - \Gamma_{\mu 0}^\mu T^0{}_0 + \Gamma_{0\nu}^\alpha T^0{}_\alpha = 0$$

$$\rightarrow -\dot{\rho} + \cancel{\Gamma_{00}^0 T^0{}_0} + \Gamma_{i0}^i T^0{}_0 - \cancel{\Gamma_{00}^0 T^0{}_0} - \Gamma_{i0}^i \rho = 0$$

$$\rightarrow \dot{\rho} + 2\Gamma_{i0}^i \rho = 0$$

$$\dot{\rho} + 3H(\rho + p) = 0 \quad \leftarrow D_\mu T^\mu{}_\nu = 0$$

$$\dot{\rho} = -3H\rho(1+w)$$

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} \quad : \text{Friedmann Eq.}$$

$$H = \frac{\dot{a}}{a}$$

$w = \text{EOS parameter}$

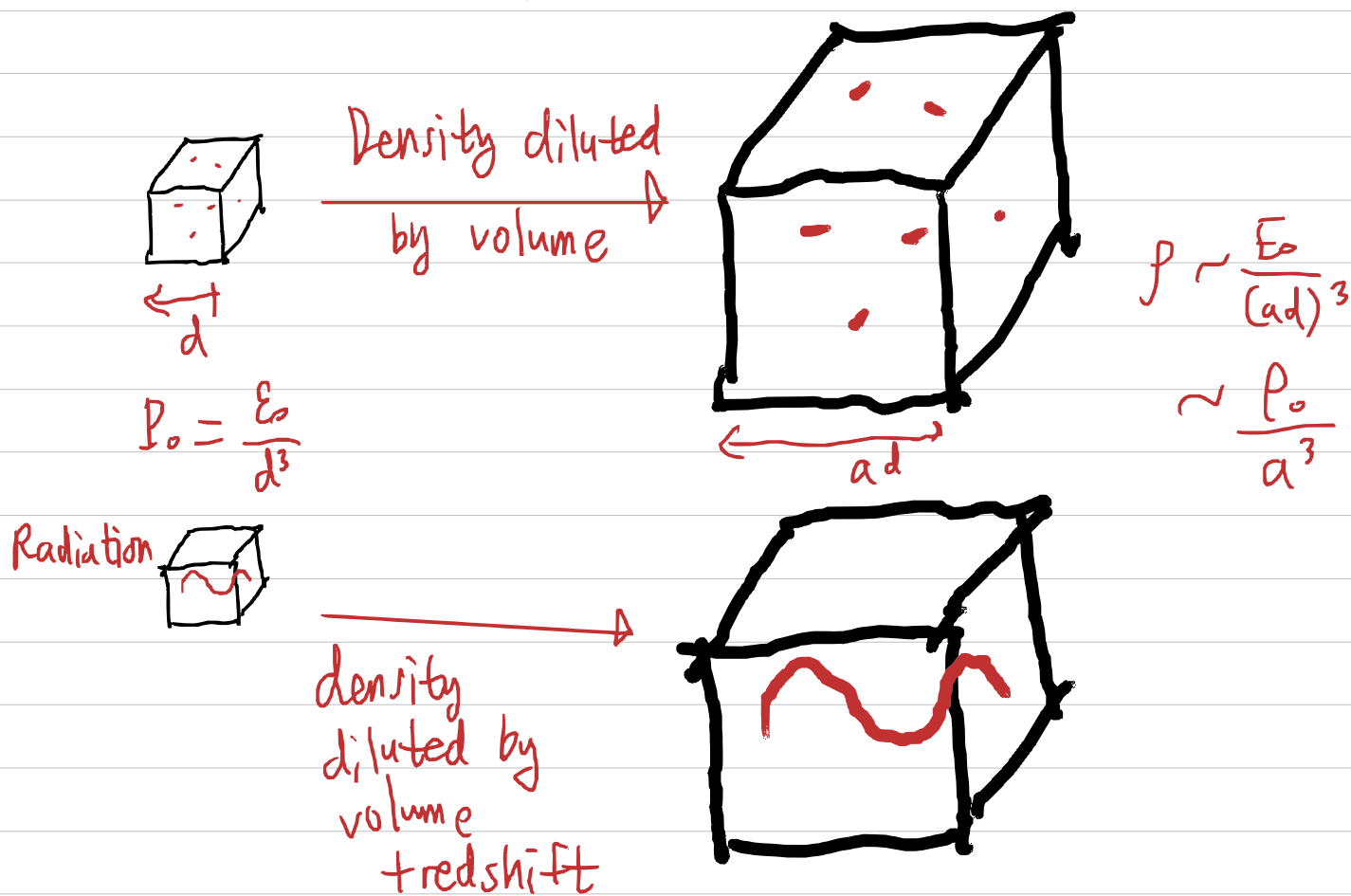
$$\dot{\rho} = -3 \frac{\dot{a}}{a} (1+w)\rho \rightarrow \partial_a \rho = -3(1+w)\rho$$

If w is const., $\rho \sim a^{-3(1+w)}$

e.g., "dust" pressureless matter $p=0$

EOS para. $w=0$

$\rho \sim a^{-3}$ dilutes like the volume.



Radiation $w = +\frac{1}{3}$

$$\rho = a^{-4}$$

$\rightarrow$ diluted by Volume + Redshift $\frac{1}{a}$

* 3-curvature (k)

$$\rho \sim -\frac{k}{a^2}$$

$$-3(1+w) = -2 \rightarrow w = -\frac{1}{3}$$

* Cosmological Const. Λ

$$p = \text{const.} = \frac{\Lambda}{8\pi G}$$

$$H^2 = \frac{\Lambda}{3} \rightarrow H = \text{const.} = H_0 = \frac{\dot{a}}{a}$$

$$\rightarrow a(t) = a_0 e^{+H_0 t} \rightarrow \text{exponential expansion of the Universe}$$

$$\dot{a} = H_0 a$$

$$\frac{\ddot{a}}{a} = H_0^2 > 0 \rightarrow \text{accelerated expansion}$$

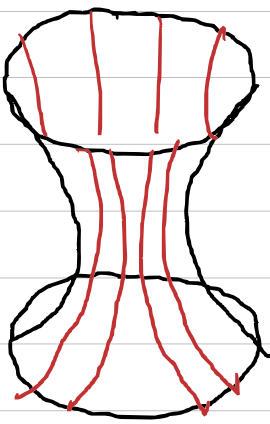
$$p = \text{const.} \rightarrow \dot{p} = 0 \rightarrow \omega = -1$$

$\rightarrow$ resulting metric is d-S space

$$ds^2 = -dt^2 + e^{2H_0 t} dS_{(3)}^2$$

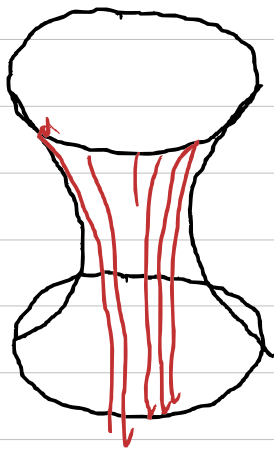
de Sitter

$$R_{\mu\nu} = 3H_0^2 g_{\mu\nu}$$



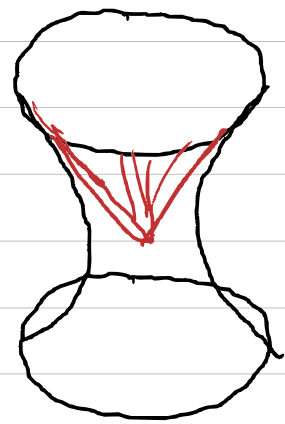
closed

(contraction followed by expansion)



flat

(exponential expansion)



open

(hyperbolic)

$$(p_{\text{rad}} \sim \frac{1}{a^4})$$

Radiation-dominated

$$(p_{\text{matter}} \sim \frac{1}{a^3})$$

Matter-dominated

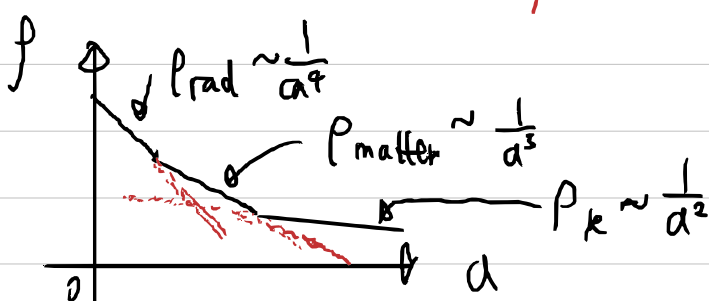
$$(p_k \sim \frac{1}{a^2})$$

Dark Energy

(Λ)-dominated

Inflation

Big Bang



The Schwarzschild Metric

$$dS_{\text{Schwar}}^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2$$

$$(d\Omega_2^2 = r^2 d\theta^2 + \sin^2\theta d\phi^2)$$

§4.1. The metric

Vacuum $\rightarrow T_{\mu\nu} = 0 \rightarrow R_{\mu\nu} = 0$

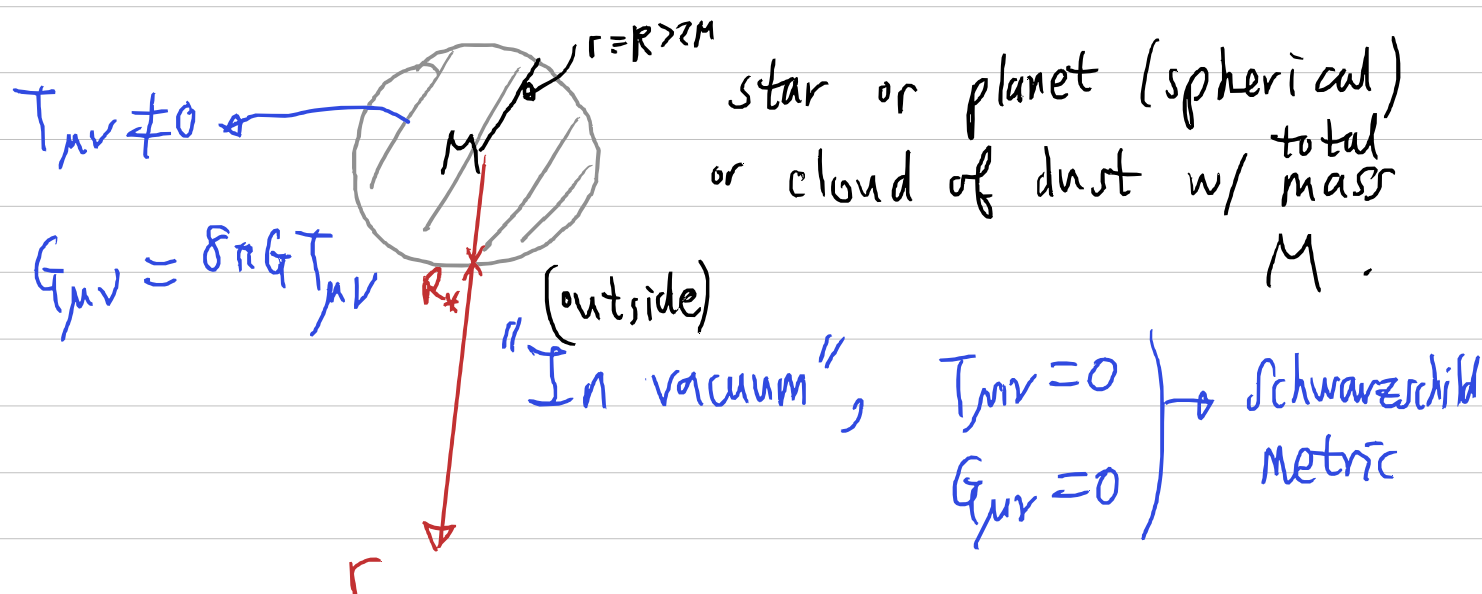
In the vacuum, the most general spherically symm. soln. to Einstein's Eq. is the Schwarzschild metric.

$$dS_{\text{Schwar}}^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2$$

$$(-+++)$$

$$(d\Omega_2^2 = r^2 d\theta^2 + \sin^2\theta d\phi^2)$$

$r_s = 2GM$, $M = \text{mass at the center.}$



For a star of mass M , size R^* , $r_s = 2GM$,

* $r \rightarrow \infty$, $ds^2 = -dt^2 + dr^2 + \dots$
 Minkowski metric
 (asymptotically flat)

* $r > R^*$, Schwarzschild metric r_s

* $r < R^*$, no longer in vacuum

assuming $R^* > r_s$.

$r_s \approx 8\text{mm}$
 (Earth)

$(r_{s(\text{sun})} \approx 3\text{km})$

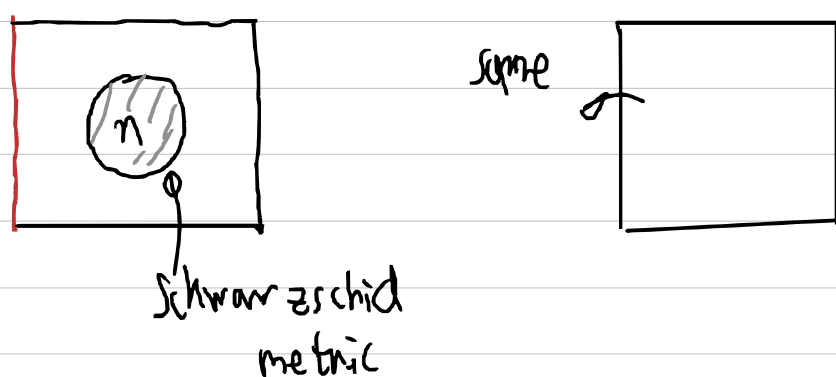
If $r_s > "R^*" \rightarrow \text{BH}$

$\partial_t g_{\mu\nu} = 0$
 $g_{t\theta} = 0$
 $g_{t\phi} = 0$

$(t \leftrightarrow -t)$

} metric is static

* all that matter is the tot. mass (not size)



II. Birkhoff's Thm

Sch M. :

$$ds^2 = - \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2$$

$$d\Omega_2^2 = r^2 d\theta^2 + \sin^2\theta d\phi^2$$

* asymptotically flat as $r \rightarrow \infty$ * only depends on Mass M .

Steps to proving Birkhoff's Thm:

* Spherically symmetric

$$ds^2 = -g_{tt}(r,t) dt^2 + 2g_{rt}(r,t) dr dt + g_{rr}(r,t) dr^2 + g_{\theta\theta}(r,t) d\Omega_2^2$$

$$d\Omega_2^2 = r^2 d\theta^2 + \sin^2\theta d\phi^2$$

by spherical symm., $g_{\mu\nu} = g_{\mu\nu}(t, r)$
(aside from $\sin^2\theta$)

* Change coords (use our gauge freedom)

$$\begin{array}{c} r \\ t \end{array} \longrightarrow \begin{array}{c} \tilde{r} = \tilde{r}(r, t) \\ \tilde{t} = \tilde{t}(r, t) \end{array}$$

$$g_{\theta\theta}(r, t) \longrightarrow \tilde{r}^2$$

Set 1 func. (1 gauge freedom) by demanding

$$g_{\theta\theta}(r, t) = \tilde{r}^2 \xrightarrow{s.t.} g_{\theta\theta}(r, t) d\Omega_2^2 = \tilde{r}^2 d\Omega_2^2$$

We still 1 free func. in the change of coords

 $\tilde{t}(r, t)$ worth of freedom
(1-func.)

$$g_{rt}(r, t) \longrightarrow \tilde{g}_{\tilde{r}\tilde{t}}(\tilde{r}, \tilde{t}) = g_{rr} \frac{\partial r}{\partial \tilde{r}} \frac{\partial r}{\partial \tilde{t}}$$

freedom of func.

 $\tilde{t}$ or its inverse

$$t = t(\tilde{r}, \tilde{t})$$

 $\longrightarrow$ We solve in principle for $t = t(\tilde{r}, \tilde{t})$
so that

$$\tilde{g}_{\tilde{t}\tilde{r}} = 0 \quad (1 \text{ eq. for 1-func.})$$

removing $\sim$ tildes, in this new gauge,

$$ds^2 = -A(r, t) dt^2 + C(r, t) dr^2 + r^2 d\Omega_2^2$$

(spher. symm. metric)

(2)

2nd assumption: $T_{\mu\nu} = 0$ In the vacuum, $G_{\mu\nu} = 0$

$$G_{00} \sim -1 + c + r \frac{c'}{c} \quad \left\{ \begin{array}{l} \frac{\partial}{\partial r} = ' \\ \frac{\partial}{\partial t} = \cdot \end{array} \right.$$

$$G_{rr} \sim 1 - c - r \frac{A'}{A}$$

$$G_{0r} \sim \dot{c} \quad \dot{c} = 0$$

$$c = \text{const. in time} \rightarrow \underline{c(r) dr^2}$$

$$(C(r,t) dr^2) \rightarrow$$

$$\left. \begin{array}{l} G_{00} = 0 = -1 + c + r \frac{c'}{c} \\ G_{rr} = 0 = 1 - c - r \frac{A'}{A} \end{array} \right\} \rightarrow \frac{A'}{A} = -\frac{c'}{c}$$

$$G_{0r} \sim \dot{c}$$

$$A(r,t) = \frac{f(t)}{C(r)}$$

$$A(r,t) dt^2 = \frac{f(t)}{C(r)} dt^2 = \frac{1}{C(r)} d\hat{t}^2$$

Let

$$(f(t) dt^2 = d\hat{t}^2)$$

$$\frac{d\hat{t}^2}{dt^2} = f(t) \quad (N)$$

$\hookrightarrow$ similar to setting lapse to unity ($N=1$)

In terms of the rescaled time $\hat{t}$,

$$\hat{A}(r, \hat{t}) = \frac{1}{C(r)} \quad (\text{residual gauge freedom})$$

omit " $\hat{}$ " from now on.

$\rightarrow$ resultant metric is STATIC.

$$\text{Gauss's Law} \quad \left(\begin{array}{l} \vec{\nabla} \cdot \vec{E} = 4\pi \rho_e \\ \vec{\nabla} \cdot \vec{B} = 0 \end{array} \right)$$

& corresponding cons. laws are built into the Einstein Eq.

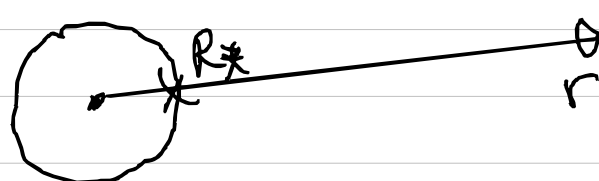
$$G_{00} \rightarrow -1 + c + r \frac{C'(r)}{C(r)} = 0$$

$$C(r) = \frac{1}{1 - \frac{\text{const.}}{r}} \quad (\text{PS7, \#3})$$

$$\therefore ds^2 = - \overset{A(r,t)}{\left(1 - \frac{\text{const.}}{r}\right)} dt^2 + \overset{C(r)}{\frac{1}{1 - \frac{\text{const.}}{r}}} dr^2 + r^2 d\Omega_2^2$$

Const. can be set by comparing w/ Newtonian approx.

$r \rightarrow \infty$ (far away)



(PS5, 61) (RF)

Weak fld approx.

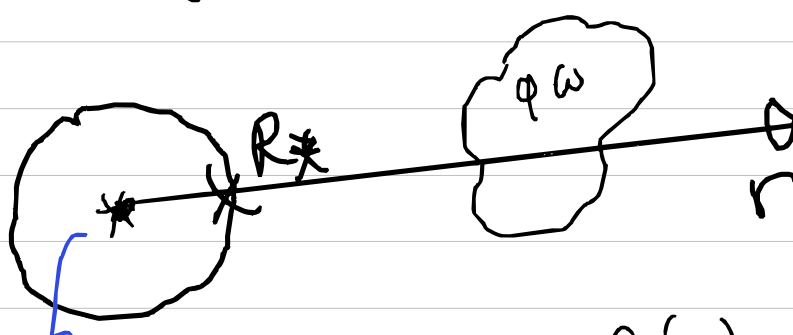
$$ds_{\text{Newtonian approx.}}^2 = -(1+2\phi) dt^2 + (1-2\phi) dr^2 + r^2 d\Omega_2^2$$

$$= -4\pi G M \delta^{(3)}(\underline{x}) \quad \text{(for mass } M \text{ at origin)} \quad (3)$$

$$\nabla^2 \phi = 4\pi G \rho \quad \text{Poisson's Eq.}$$

ρ = mass density ($\rho = -M \delta^{(3)}(\underline{x})$)
 ϕ = Newton's potential

G = grav. const.



$$\phi(r) = -\frac{MG}{r}$$

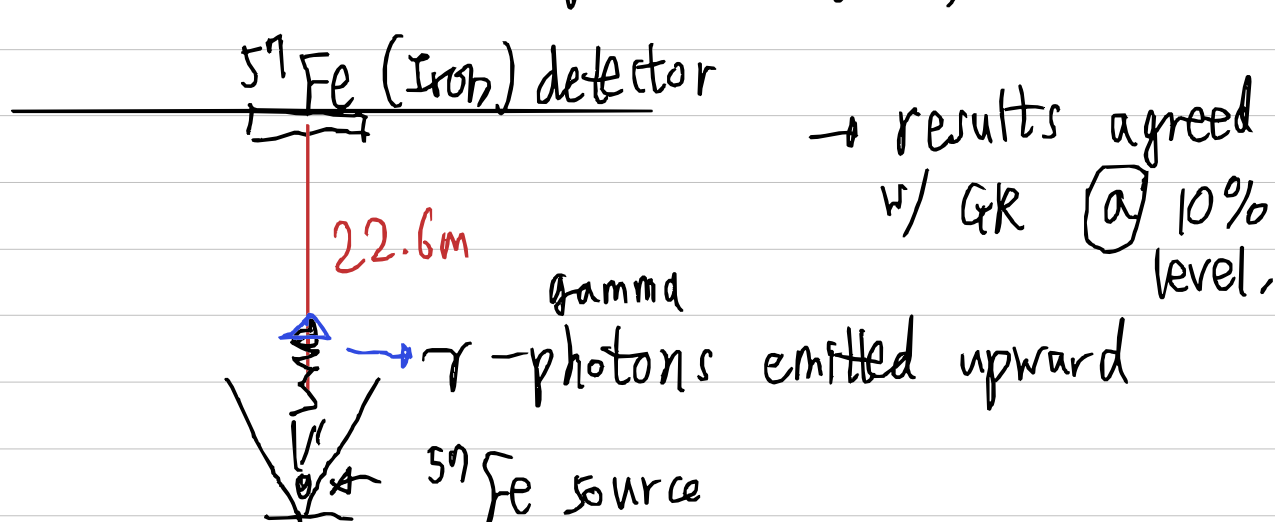
$$M \delta^{(3)}(\underline{x}) = M \frac{\delta(r)}{4\pi r^2}$$

$$\therefore ds^2_{\text{Newton app.}} = -\left(1 - \frac{2MG}{r}\right) dt^2 + \left(1 + \frac{2MG}{r}\right) dr^2 + r^2 d\Omega^2$$

const. = $2MG = r_s$
 $\hookrightarrow$ Schwarzschild metric

* EX. Jupiter
 photons get redshifted when they escape the grav. well
 $f \downarrow, \lambda \uparrow$
 ROYGBIV

* Pound - Rebka Experiment (1960)



$$\left(1 - \frac{2MG}{r}\right) \rightarrow \text{redshift}$$

Moon
 100 yrs

Earth
 100yrs - 3sec.

Jupiter
 100 yrs

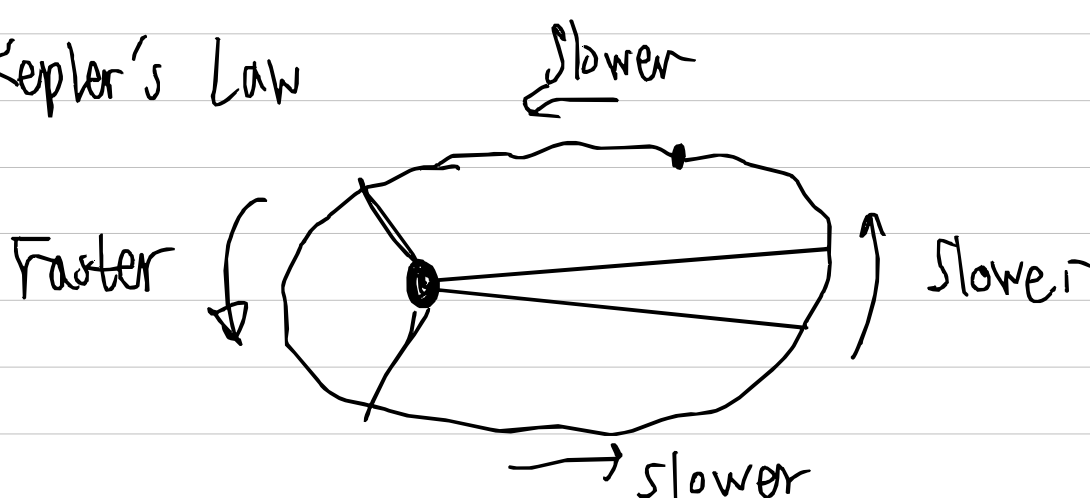
100yrs + 1min.

* GPS

Effects from SR: $+7 \mu\text{sec/day}$
 $\parallel$ GR: $-45 \mu\text{sec/day}$

$$38 \mu\text{sec} = 38000 \text{ nsec}$$

* Kepler's Law



* Precession of Mercury



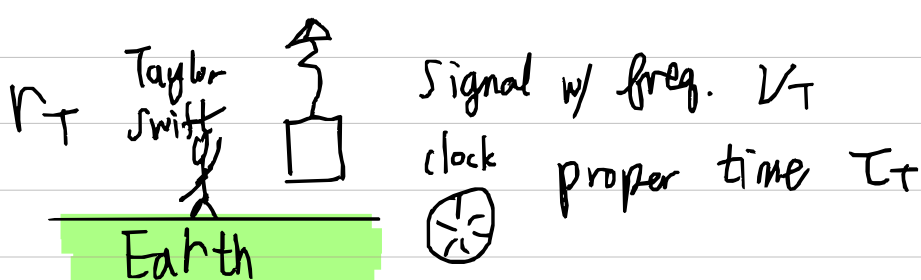
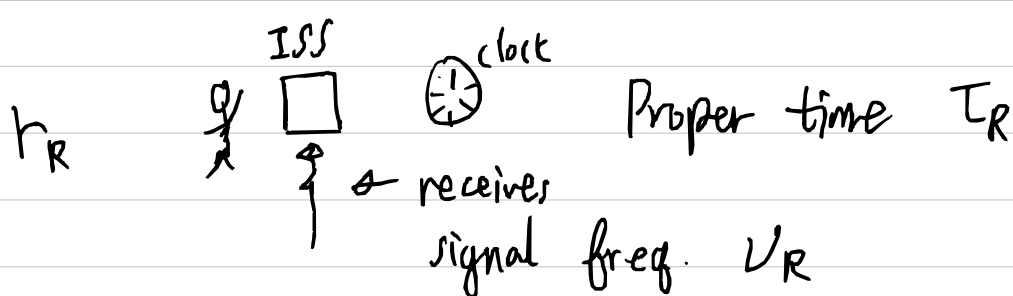
$\rightarrow$ also GR effect

Advancement of perihelion of Mercury:

$$GR: 0.15 \dots$$

GR L21

Gravitational Redshift



Taylor position r_T , τ_T
Receiver (astronaut) position r_R , τ_R } Both static

$$-d\tau_T^2 = -\left(1 - \frac{r_s}{r_T}\right) dt^2 + \dots$$

$$d\tau_T = \sqrt{\left(1 - \frac{r_s}{r_T}\right)} dt$$

t = coordinate time

Rob: $d\tau_R^2 = \left(1 - \frac{r_s}{r_R}\right) dt^2$

$$\frac{d\tau_T}{d\tau_R} = \sqrt{\frac{1 - \frac{r_s}{r_T}}{1 - \frac{r_s}{r_R}}}$$

Schwarzschild
radius of Earth
 $r_s = 2M_\odot G$
 $\approx 8\text{mm}$

a lapse of time

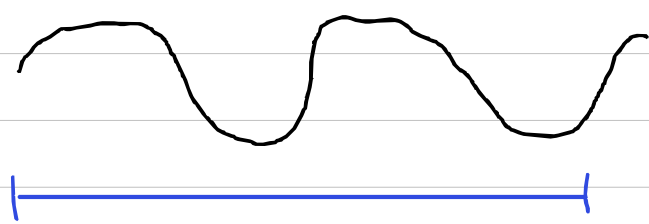
$$\Delta\tau_T = \sqrt{\frac{1 - \frac{r_s}{r_T}}{1 - \frac{r_s}{r_R}}} \Delta\tau_R$$

T sends a signal w/ freq. ν_T :



In $\Delta\tau_T$, n oscillations

for R, they will see the signal:



n oscillations in $\Delta\tau_R$

$$\nu_T = \frac{n}{\Delta\tau_T} \quad \& \quad \nu_R = \frac{n}{\Delta\tau_R}$$

$$\frac{\nu_T}{\nu_R} = \frac{\Delta\tau_R}{\Delta\tau_T} = \sqrt{\frac{1 - \frac{r_s}{r_R}}{1 - \frac{r_s}{r_T}}}$$

$$r_R = r_T + h$$

altitude of R

③

If the particle is moving along the plane $\theta = \frac{\pi}{2}$,

$$\dot{\theta} = 0,$$

$$\mathcal{L} = \mathcal{L}[t, \dot{t}, r, \dot{r}, \theta, \dot{\theta}, \phi, \dot{\phi}]$$

$$= - \left(1 - \frac{r_s}{r} \right) \dot{t}^2 + \left(1 - \frac{r_s}{r} \right)^{-1} \dot{r}^2$$

~~$$+ r^2 \dot{\theta}^2 + r^2 \dot{\phi}^2$$~~

GR L22

$$\mathcal{L} = -g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = \mathcal{L}[x^\mu]$$

considers plane $\theta \equiv \frac{\pi}{2} \rightarrow \dot{\theta} = 0$

doesn't explicitly depend on

$$\mathcal{L} = \mathcal{L}(\cancel{t}, \dot{t}, r, \dot{r}, \cancel{\phi}, \dot{\phi}, \cancel{\theta})$$

$$= -\left(1 - \frac{r_s}{r}\right) \dot{t}^2 + \left(1 - \frac{r_s}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2$$

$\rightarrow$ 3 conservation laws

$$\frac{\partial \mathcal{L}}{\partial t} = 0 = \frac{d\mathcal{L}}{dt}$$

$$\mathcal{L} \equiv \text{conserved} = \begin{cases} -1 & \text{if massive } (\tau: \text{proper time}) \\ 0 & \rightarrow \text{if massless (affine parameter)} \end{cases}$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0 \rightarrow \frac{\partial \mathcal{L}}{\partial \dot{t}} = \text{const.} = -2E$$

$$\text{Energy } E \rightarrow \text{conserved} : -2E = -2\left(1 - \frac{r_s}{r}\right) \dot{t}$$

$$\dot{t} = \frac{E}{1 - \frac{r_s}{r}} \quad \left(\text{asymptotically } \dot{t} \sim E \right)$$

$\dot{t} \rightarrow E$

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0 \rightarrow \text{ang. mom. conservation}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}} \right) = 0 \rightarrow \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = 2H = 2r^2 \dot{\phi} = \text{const.}$$

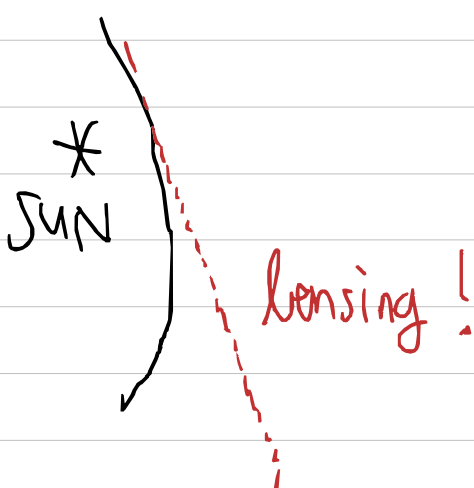
$$H = r^2 \dot{\phi} \rightarrow \dot{\phi} = \frac{H}{r^2}$$

Plug in $\dot{t}$ & $\dot{\phi}$:

$$\mathcal{L} = \frac{\dot{r}^2 - E^2}{1 - \frac{r_s}{r}} + \frac{H^2}{r^2} = \text{const.} = \begin{cases} -1 & \text{massive} \\ 0 & \text{massless} \end{cases}$$

when dealing w/ deflection of light

$$\mathcal{L} = \frac{\dot{r}^2 - E^2}{1 - \frac{r_s}{r}} + \frac{H^2}{r^2} = 0$$

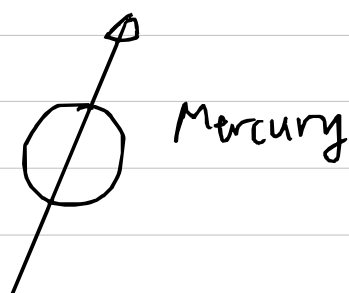


$$r_{\text{sun}} \sim 10^6 \text{ km}$$

$$r_{\text{sch (sun)}} \sim 3 \text{ km}$$

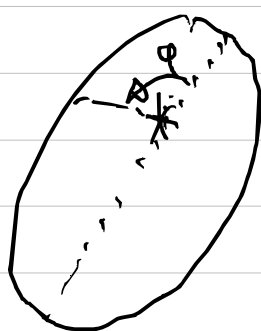
(2)

V. Orbits of Mercury

Commit to a massive
"particle"

$$\tilde{r}^2 = E^2 - \frac{H^2}{r^2} \left(1 - \frac{r_s}{r} \right)$$

$$r = r(\varphi(t))$$

change variable
to

$$u(\varphi) = \frac{1}{r(\varphi)}$$

$$\dot{r} = \dot{\varphi} r'(\varphi) = -\frac{\dot{\varphi}}{u^2} u'(\varphi) = -H u'(\varphi)$$

$$\cdot = \frac{d}{dt}$$

$$' = \frac{d}{d\varphi}$$

$$\left(\dot{\varphi} = \frac{H}{r^2} = H u^2 \right)$$

$$H^2 (u')^2 = -(1 - u r_s) (1 + u^2 H^2) + E^2$$

$$H^2 (u'^2 + u^2) = E^2 - 1 + u r_s (1 + u^2 H^2)$$

$$u'^2 + u^2 = \frac{E^2 - 1 + u r_s (1 + u^2 H^2)}{H^2}$$

primes!

$$2(u' u'' + u u') = \frac{u' r_s}{H^2} (1 + 3u^2 H^2)$$

$$u'' + u = \frac{r_s}{2H^2} (1 + 3u^2 H^2)$$

Exact

$$r_s = 2GM$$

Binet Eq. describing orbits in solar sys.
accounting for gravitational corrections.

$$u'' + u = \frac{GM}{H^2} (1 + 3u^2 H^2)$$

Exact

Can start from

(Classical Poisson Eq.)

$$\nabla^2 \varphi = 4\pi G \rho$$

$$r \rightarrow \infty \quad \& \quad u \rightarrow 0$$

(3)

"Newtonian approx." Weak fld limit

$$\begin{pmatrix} r \rightarrow \infty \\ u \rightarrow 0 \end{pmatrix}$$

↓ We get

Original Binet Eq.

$$u'' + u = \frac{GM}{H^2} \quad \text{// const.} \quad \left(\text{Newtonian laws of motion} \right)$$

$$\nabla^2 \varphi = -M G \delta^{(3)}(r)$$

φ : Gravitational potential

$$= 4\pi G \rho$$

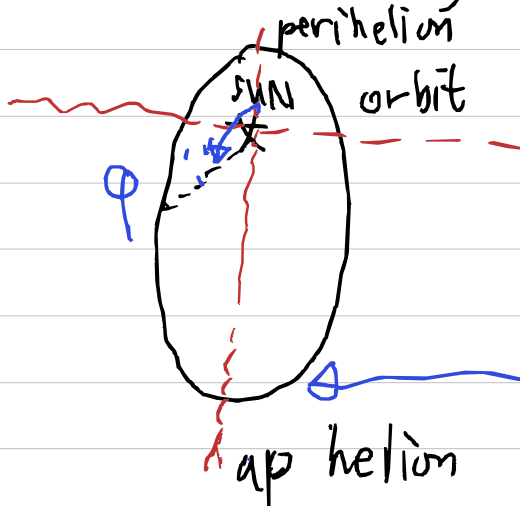
$$u(\varphi) = \frac{GM}{H^2} (1 + \varepsilon \cos \varphi)$$

↓

ε : eccentricity

$$\frac{1}{u(\varphi)}$$

In the Newtonian limit,



$$r(\varphi) = \frac{H^2}{GM} \frac{1}{1 + \varepsilon \cos \varphi}$$

Semi major axis

$$\begin{cases} r_{\min} = \frac{H^2}{GM} \frac{1}{1 + \varepsilon} \\ r_{\max} = \frac{H^2}{GM} \frac{1}{1 - \varepsilon} \end{cases}$$

$$\begin{aligned} \bar{r} &= \frac{1}{2} (r_{\min} + r_{\max}) = \frac{H^2}{2GM} \left(\frac{1}{1 + \varepsilon} + \frac{1}{1 - \varepsilon} \right) \\ &= \frac{H^2}{GM} \frac{1}{1 - \varepsilon^2} \quad \text{Semi-major} \end{aligned}$$

closed orbits → $r(\varphi) = r(\varphi + 2\pi)$ periodic

In GR,

$$u'' + u = \frac{GM}{H^2} \left(1 + 3 \frac{u^2 H^2}{c^2} \right)$$

// const.

↓

the origin of the advance of perihelion

(4)

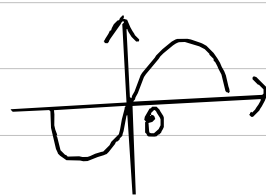
* 1st order in grav. corrections $u^2 H^2$,

$3u^2 H^2 \rightarrow$ can take it to be coming from Newtonian approx.

* 1st order in ϵ $\left(\begin{array}{l} \epsilon \cos \varphi \text{ on RHS} \\ \text{but not } \epsilon^2 \cos^2 \varphi \\ \text{"} \\ \mathcal{O}(\epsilon^2) \end{array} \right)$

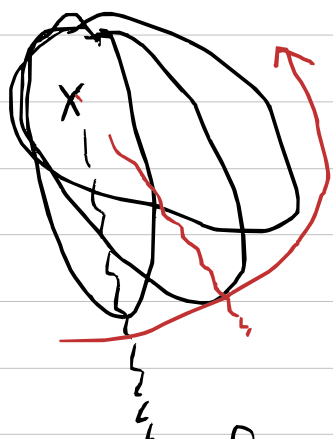
$$u(\varphi) = \frac{GM}{H^2} \left[\epsilon(1 + \cos \varphi) + \frac{3GM^2}{H^2} (1 + \epsilon \varphi \sin \varphi) + \dots \right]$$

$\varphi \sin \varphi \rightarrow$ NOT periodic
 not periodic $\uparrow$ periodic



$$u(\varphi) = u(\varphi + 2\pi)$$

In GR



Shift in semi-major axis,
from 1 orbit to another.

$\delta \varphi$ coming from $\varphi \sin \varphi$

$$u(\varphi) = \frac{GM}{H^2} \left[1 + \frac{3GM^2}{H^2} + \epsilon \cos \left(\varphi - \frac{3GM^2}{H^2} \varphi \right) + \dots \right]$$

$$\cos(\varphi + \Delta \varphi) = \cos \varphi - \Delta \varphi \sin \varphi$$

$$\Delta \varphi = \frac{3GM^2}{H^2}$$

"new notion" of effective periodicity

$$T = \frac{2\pi}{1 - \frac{3GM^2}{H^2}} + \dots \approx 2\pi \left(1 + \frac{3GM^2}{H^2} \right) + \dots$$

φ $\delta \varphi$

$$\left(1 - \frac{3GM^2}{H^2} \right)^{-1} \approx 1 + \frac{3GM^2}{H^2}$$

The advance of the perihelion

$$\delta \varphi = \frac{6\pi M^2 G^2}{H^2}$$

$$H^2 = \frac{r}{\bar{r}} (GM) (1 - \epsilon^2) \quad \text{from semimajor ax.}$$

* For Mercury

$$\delta \varphi \approx 0.01 \text{ deg per century} \quad (100 \text{ yr})$$

$$\epsilon_{\text{Mercury}} \approx 0.2$$

C4. Schwarzschild Metric

VI. Schwarzschild BHs (non-rotating BHs)

(1) Gravitational Collapse

Massive particle in free-fall

Pressureless gas

Massive Particle (e.g. Hydrogen atom)

at the edge of star

Starts at $\tau=0$ at rest at $r=R_0$ $\theta = \text{const.}, \phi = \text{const.} \longrightarrow H=0$ w.r.t.
the particle's own proper time

$$\dot{r}^2 = -\left(1 - \frac{r_s}{r}\right) + E^2$$

$$\dot{r}^2 \Big|_{r=R_0} = 0 \implies E^2 = 1 - \frac{r_s}{R_0}$$

$$\mathcal{L} = \frac{\dot{r}^2 - E^2}{1 - \frac{r_s}{r}} + \frac{H^2}{r^2} = \begin{cases} -1, & \text{massive} \\ 0, & \text{massless} \end{cases}$$

$$\dot{r}^2 = r_s \left(\frac{1}{r} - \frac{1}{R_0} \right) = \frac{r_s}{R_0} \left(\frac{R_0 - r}{r} \right)$$

Inwardly falling massive particle

$$\frac{dr}{d\tau} = \dot{r} = - \sqrt{\frac{r_s}{R_0}} \sqrt{\frac{R_0 - r}{r}}$$

$$d\tau = \frac{1}{\dot{r}} dr = - \sqrt{\frac{r_s \cdot r}{R_0(R_0 - r)}} dr$$

 $\Delta\tau$ = proper time taken to reach r_s

$$\Delta\tau = \int dt = - \int_{R_0}^{r_s} \sqrt{\frac{R_0}{r_s}} \sqrt{\frac{r}{R_0-r}} dr \quad (2)$$

USE

$$\int \frac{\sqrt{r}}{\sqrt{R_0-r}} dr = -\sqrt{r(R_0-r)} + R_0 \arcsin\left(\sqrt{\frac{r}{R_0}}\right)$$

$$\Delta\tau = \sqrt{R_0(R_0-r_s)} - \frac{R_0^{3/2}}{\sqrt{r_s}} \arcsin\left(\sqrt{\frac{r_s}{R_0}}\right) + \frac{R_0^{3/2}}{\sqrt{r_s}} \frac{\pi}{2}$$

$$= R_0 \left[\sqrt{1 - \frac{r_s}{R_0}} - \left(\frac{R_0}{r_s}\right)^{\frac{1}{2}} \arccos\left(\sqrt{\frac{r_s}{R_0}}\right) \right]$$

for Schw, $r_s = 3\text{km}$
 $R_0 \approx 7 \times 10^8 \text{m}$

$$\Delta\tau \sim \frac{1}{2} \text{ hour (Finite amount)}$$

Upon reaching r_s ,

$$\dot{r} \Big|_{r=r_s} = -\sqrt{\frac{r_s}{R_0}} \sqrt{\frac{R_0-r_s}{r_s}} \sim -\sqrt{1 - \frac{r_s}{R_0}}$$

Nothing special happens
upon reaching r_s

(event horizon
or Schwarzschild horizon)

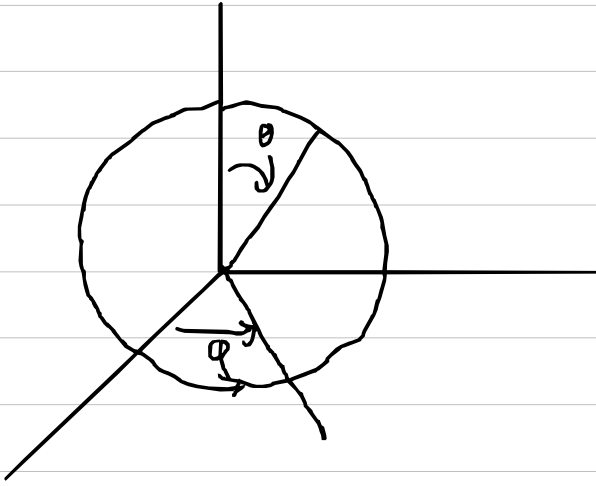
(aside from the fact that you can no longer
escape)

Very close to the spd
of lgt.

Rewatch her lec.

(3)

(2) Coordinate vs. Physical Singularities



Metric on S^2 radius R_0

$$ds^2 = R_0^2 (d\theta^2 + \sin^2\theta d\phi^2)$$

Metric has singularities

at $\theta = 0, \pi$.

$$\det g = R_0^4 \sin^2\theta = 0 \text{ for } \theta = 0, \pi.$$

Non-invertible at these pt.s.

$$R_{abcd} = \frac{1}{R_0^2} (g_{ac}g_{bd} - g_{ad}g_{bc})$$

Scalar Curvature

$$R = \frac{2}{R_0^2}$$

$$\text{"Ricci Tensor Squared"} \quad R_{ab} R^{ab} = \frac{2}{R_0^2}$$

$$R_{abcd} R^{abcd} = \frac{4}{R_0^2} \rightarrow \text{finite at } \theta = 0, \pi.$$

$$D_\mu R_{abcd} D^\mu R^{abcd} = 0$$

→ As a result, we have a coordinate singularity. ($\theta = 0, \pi$)

as opposed to a physical singularity.

(Take a divergence ↓)

(4)

(b) Schwarzschild Metric

↳ 3 kinds of singul.s.

$$ds^2 = - \left(1 - \frac{r_s}{r}\right) dt^2 + \frac{1}{1 - \frac{r_s}{r}} dr^2 + r^2 (d\theta^2 + \sin^2\theta d\phi^2)$$

* $\theta = 0, \pi$ (coord. singul.) $r=0$ $\rightarrow$ Something physical there

$$G_{\mu\nu} = R_{\mu\nu} = 0 \rightarrow R=0$$

$$R_{\mu\nu} R^{\mu\nu} = 0 \rightarrow \text{Scalar quantity } R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} = 12 \frac{r_s^2}{r^6}$$

(PS 7)
(#3)

Same in any coord. sys.

* $r=r_s$ (event horizon)

$$R=0, \quad R_{\mu\nu}^2=0,$$

$$R_{\mu\nu\alpha\beta}^2 \Big|_{r_s} = 12 \frac{1}{r_s^6} \rightarrow \text{finite}$$

↳ always diverges at $r=0$.
 ↳ i.e., physical singul. at $r=0$.

To confirm that Schwarzschild radius is a coord. singul. ($r=r_s$)
 NOT a phys. singul.,

we use a new coord. sys. $\rightarrow$

(3) Eddington-Finkelstein Coord. Sys.

$$(t, r, \theta, \phi) \rightarrow (\tilde{t}, \tilde{r}, \tilde{\theta}, \tilde{\phi})$$

$$t \rightarrow \tilde{t} = \tilde{t}(t, r) = t + r_s \log\left(\frac{r}{r_s} - 1\right)$$

$$\rightarrow ds^2 = - \left(1 - \frac{r_s}{r}\right) d\tilde{t}^2 + \frac{2r_s}{r} dr d\tilde{t} + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

$$\det g \neq 0 \quad \text{at } r=r_s.$$

When $R^2 \sim$ When $r \rightarrow 0$

(physical singul.)

$$\left(\frac{1}{l_{\text{Planck}}^4} \right)$$

Planck length.

Scalar $R_{abcd} R^{abcd} = 12 \frac{r_s^2}{r^6}$ ($\rightarrow \infty$)
diverges at $r=0$

Schwarzschild Metric

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

 $\theta = 0, \pi$: coord. singularity (most of the time, we work in* Coordinate Singularity $r = r_s$ ($\theta = \frac{\pi}{2}$ plane.)

* Escape Velocity $v_{\text{esc}} = \sqrt{\frac{2GM}{r}}$

$$KE = E_k = \frac{1}{2} M v^2 - \frac{GM}{r}$$

$$E = \frac{1}{2} M \left(v^2 - \frac{r_s}{r} \right)$$

If a particle of mass M reaches $r \rightarrow \infty$,
(escapes)

$$E \Big|_{r \rightarrow \infty} = \frac{1}{2} m v^2 \geq 0$$

 v : non-rel. velocity

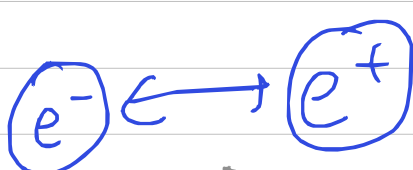
$$E \geq 0 \quad \text{at all } r \quad \left(\text{for } r > r_s \right) \\ \text{or } r < r_s$$

$$v^2 \geq \frac{r_s}{r} > 1$$

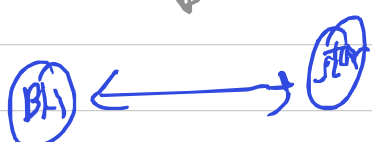
for a particle to escape grav. pull of an objt
(e.g. BH)



(a particle needs to have $v > 1$ to escape a distance $r < r_s$)

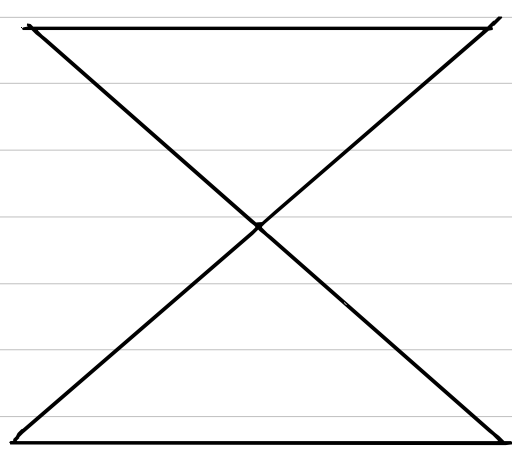


Difference

Equivalence Principle

(2)

* Light-cone structure "Escape region in light"

 $(n+1)$ dim.Eddington-Finkelstein
coord. syr.

Schw. Coords

Tortoise Coords

$$(t, r, \theta, \varphi) \longrightarrow (\tilde{t}, r, \theta, \varphi)$$

$$t \rightarrow \tilde{t} = \tilde{t}(t, r) = r_s \log\left(\frac{r}{r_s} - 1\right) + t$$

$$dt = d\tilde{t} - \frac{r_s}{1 - \frac{r_s}{r}} \frac{dr}{r_s} = d\tilde{t} - \frac{r_s dr}{r - r_s}$$

$$ds^2 = - \left(1 - \left(\frac{r_s}{r}\right)\right) \left(d\tilde{t} - \frac{r_s dr}{r - r_s}\right)^2$$

$$+ \left[1 - \frac{r_s}{r}\right]^{-1} dr^2 + r^2 d\Omega_2^2$$

$$= - \left(1 - \frac{r_s}{r}\right) d\tilde{t}^2 + 2 \frac{r_s}{r} d\tilde{t} dr$$

$$+ \left[1 - \frac{r_s}{r}\right]^{-1} dr^2 + r^2 d\Omega_2^2$$

$$\tilde{g}_{\tilde{t}\tilde{t}} = 0 \text{ at } r = r_s$$

$$\left(\begin{array}{cc|c} -\left(1 - \frac{r_s}{r}\right) & \frac{r_s}{r} & 0 \\ \frac{r_s}{r} & \left(1 + \frac{r_s}{r}\right) & 0 \\ \hline & 0 & r^2 \gamma_{ij} \end{array} \right)$$

Null coords

$$\mu = \tilde{t} - r$$

$$\nu = \tilde{t} + r$$

 $\downarrow$

$$d\tilde{t} = d\nu - dr$$

 $\downarrow$

$$ds^2 = - \left(1 - \frac{r_s}{r}\right) (d\nu - dr)^2 + r^2 d\Omega_2^2$$

$$+ \left(1 + \frac{r_s}{r}\right) dr^2 + 2 \frac{r_s}{r} (d\nu - dr) dr$$

$$= - \left(1 - \frac{r_s}{r}\right) d\nu^2 + 2 \left(1 - \frac{r_s}{r}\right) (d\nu dr)$$

$$+ \left(1 - \frac{r_s}{r}\right) dr^2 + \left(1 + \frac{r_s}{r}\right) dr^2 \quad \rightarrow 2 dr^2$$

$$+ r^2 d\Omega_2^2 + 2 \frac{r_s}{r} (d\nu - dr) dr$$

$$= - \left(1 - \frac{r_s}{r}\right) d\nu^2 + 2 (d\nu dr) + r^2 d\Omega_2^2$$

$$= ds_{E-F}^2$$

$$\tilde{g}_{\mu\nu}^{E-F} = \left(\begin{array}{cc|c} -\left(1 - \frac{r_s}{r}\right) & 1 & 0 \\ 1 & 0 & 0 \\ \hline 0 & 0 & r^2 \gamma_{ij} \end{array} \right)$$

In Flat space, the null coord.

$$g_{\mu\nu} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\det g = -1$$

$$\tilde{g}_{\mu\nu}^{E-F} = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & -\left(1 - \frac{r_s}{r}\right) & 0 \\ \hline 0 & 0 & r^{-2} \gamma_{ij} \end{array} \right)$$

Any o, t built out of $\tilde{g}_{\mu\nu}^{E-F}$ & $g_{\mu\nu}^{EF}$
(+ deriv.s)
are finite at $r = r_s$.

(3)

Null geodesics (for light cone structure)

$$\mathcal{L} = g_{\mu\nu}^{\text{EF}} \dot{x}^\mu \dot{x}^\nu = g_{\mu\nu}^{\text{EF}} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}$$

$$= 0 \quad (\text{for null geodesics})$$

For simplicity, we set

$\theta = \text{const.}$ & $\phi = \text{const.}$

$\lambda \equiv$ affine parameter

$$\dot{x}^\mu = (\dot{t}, \dot{r}, 0, 0)$$

with null $\dot{x}^\mu = (\dot{v}, \dot{r}, 0, 0)$

$$\mathcal{L} = -\left(1 - \frac{r_g}{r}\right) \dot{v}^2 + 2 \dot{v} \dot{r} = 0$$

$\rightarrow$ 2 solns

$$\left(1 - \frac{r_g}{r}\right) \dot{v}^2 = 2 \dot{v} \dot{r}$$

④

2 solns

$$\left(1 - \frac{r_s}{r}\right) \dot{v}^2 = 2 \dot{v} \dot{r}$$

(1)

• When $(\dot{v} = 0) \equiv (\dot{t} = -\dot{r})$

(2)

$$\dot{r} = \frac{1}{2} \left(1 - \frac{r_s}{r}\right) \dot{v}$$

$$= \frac{1}{2} \left(1 - \frac{r_s}{r}\right) (\dot{t} - \dot{r})$$

$$\left(1 - \frac{1}{2} + \frac{1}{2} \frac{r_s}{r}\right) \dot{r} = \frac{1}{2} \left(1 - \frac{r_s}{r}\right) \dot{t}$$

$$\dot{r} = \frac{1 - \frac{r_s}{r}}{1 + \frac{r_s}{r}} \dot{t}$$

$$= \frac{r - r_s}{r + r_s} \dot{t}$$

(1) $\frac{d\tilde{t}}{dr} = -1$ & (2) $\frac{d\tilde{t}}{dr} = \frac{r + r_s}{r - r_s}$

⑤

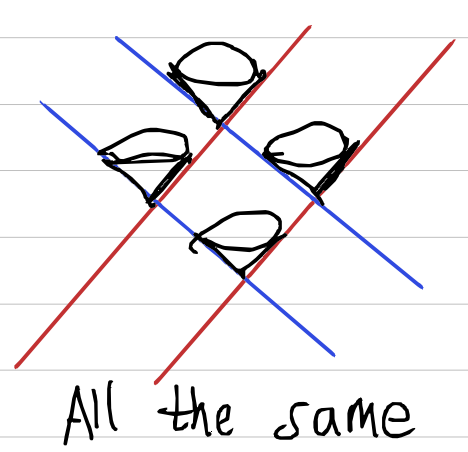
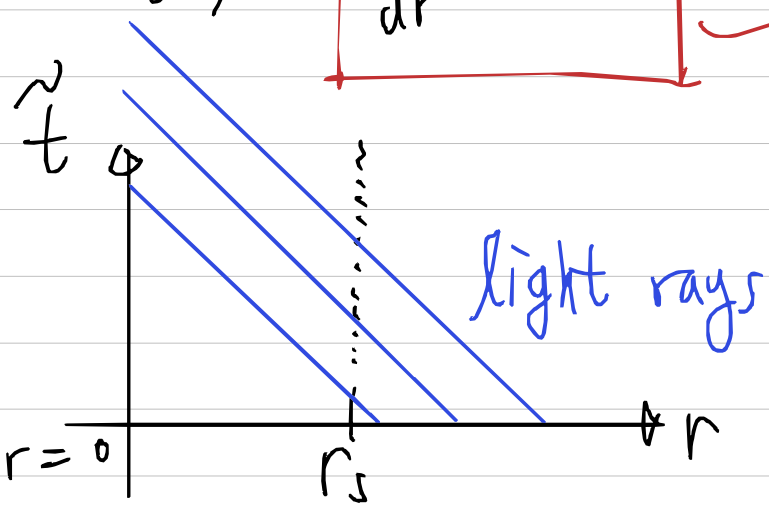
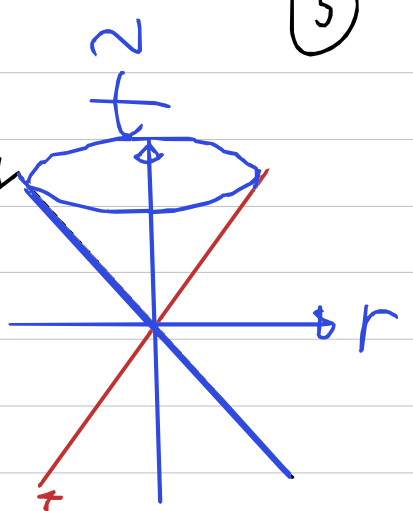
When $r \rightarrow \infty$

(1)

$$\frac{d\tilde{t}}{dr} = -1$$

(2)

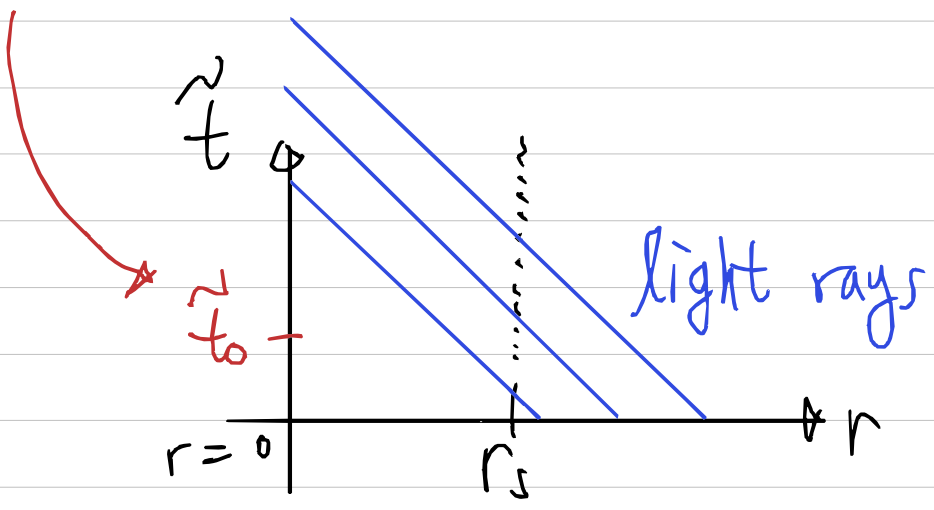
$$\frac{d\tilde{t}}{dr} = +1$$



All the same

light cone structure

(1) $\tilde{t} = \tilde{t}_0 - r$



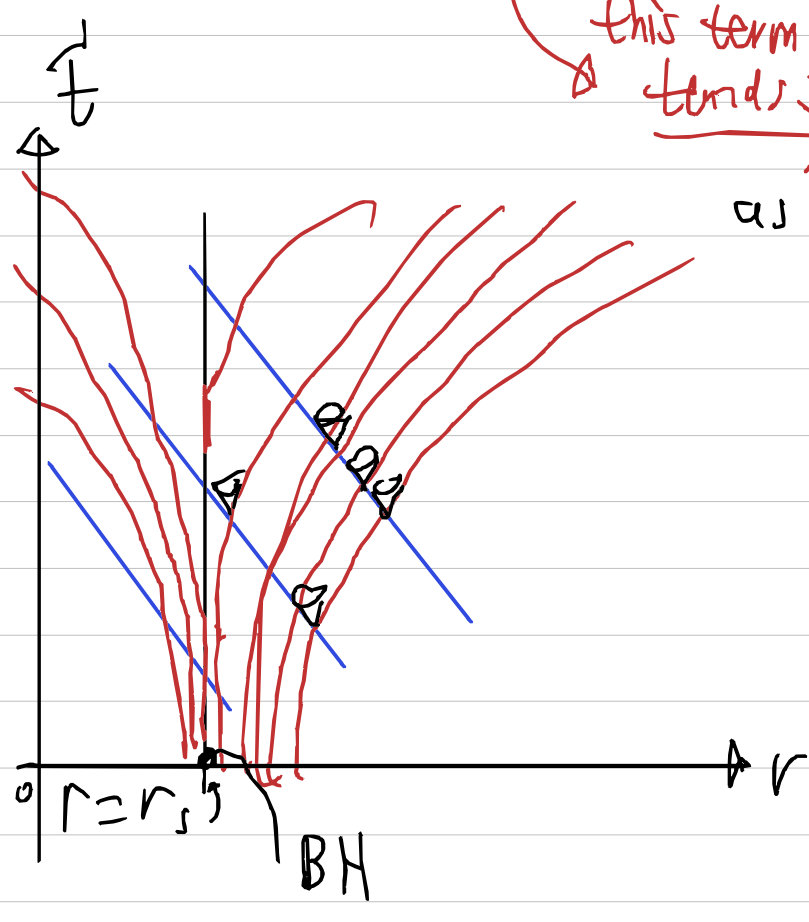
given

(2)
$$\int dr \frac{r+r_s}{r-r_s} = r + 2r_s \log \left| \frac{r-r_s}{r_s} \right| + \text{const.}$$

So,
$$\tilde{t} = \tilde{t}_0 + r + 2r_s \log \left| \frac{r-r_s}{r_s} \right|$$

$$= r \left(1 + \frac{2r_s}{r} \log \left| \frac{r-r_s}{r_s} \right| \right)$$

 this term tends to 0 as $r \rightarrow \infty$
 $\rightarrow 0$ (as $r \rightarrow \infty$)



GR L25

$$ds_{\text{Schwarzschild}}^2 = -\left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2$$

$$ds_{\text{EF}}^2 = -\left(1 - \frac{r_s}{r}\right) d\tilde{t}^2 + \frac{2r_s}{r} d\tilde{t}dr + \left(1 + \frac{r_s}{r}\right) dr^2 + r^2 d\Omega_2^2$$

$$ds_{\text{EF}}^2 \Big|_{r=r_s} = 2 d\tilde{t}dr + 2dr^2 + r^2 d\Omega_2^2$$

perfectly fine

(2)

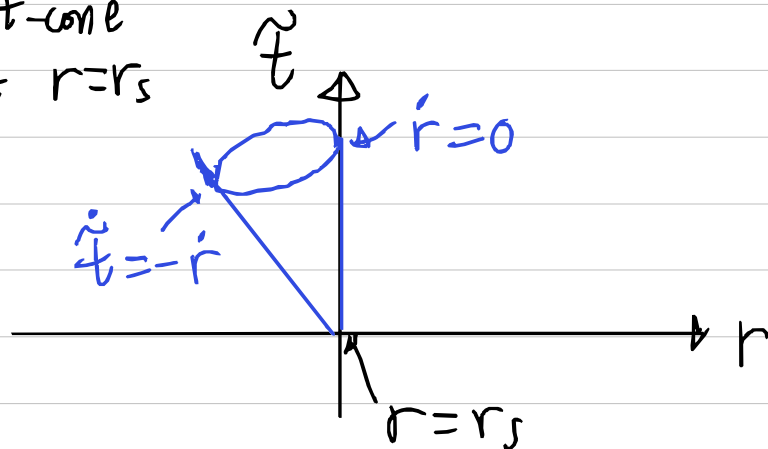
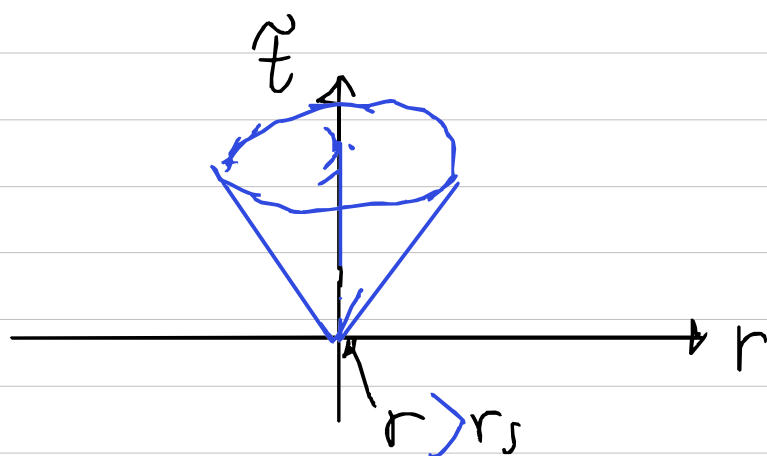
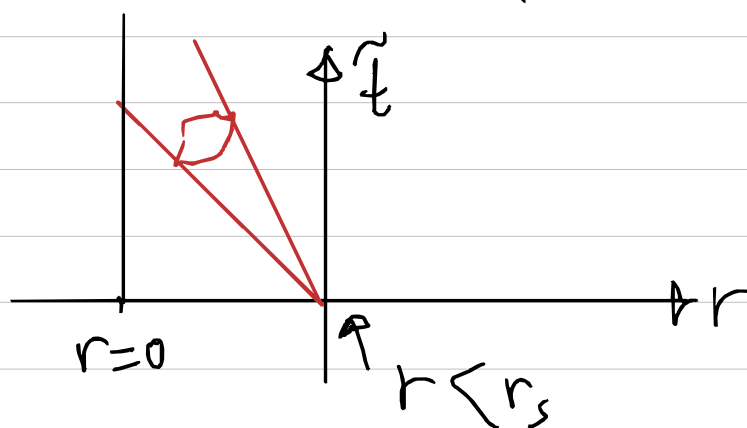
* Null geodesics

$$ds^2 = 0 = 2 d\tilde{t} dr + 2dr^2 + r^2 d\Omega^2$$

At $r = r_s$, $\dot{\tilde{t}} \dot{r} + \dot{r}^2 = 0$

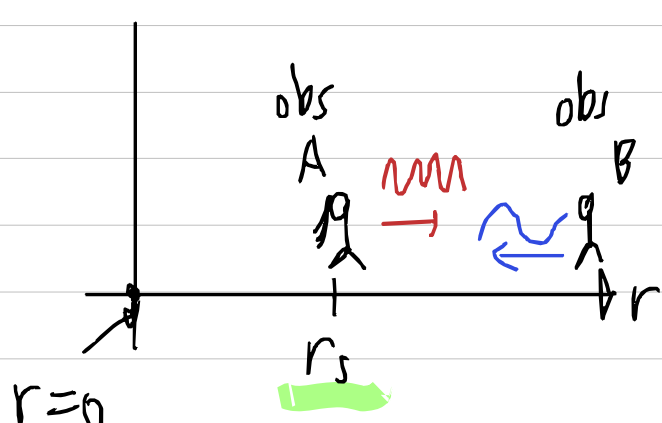
const. angles $\rightarrow \dot{\tilde{t}} = -\dot{r}$ OR

$$\dot{r} = 0$$

* light-cone
at $r = r_s$ * light-cone for $r > r_s$ * lightcone for $r < r_s$ 

* Gravitational Redshift

(3)



A sends signal to B

$$\text{freq } f_{AB} = \nu_A$$

B receives a signal w/

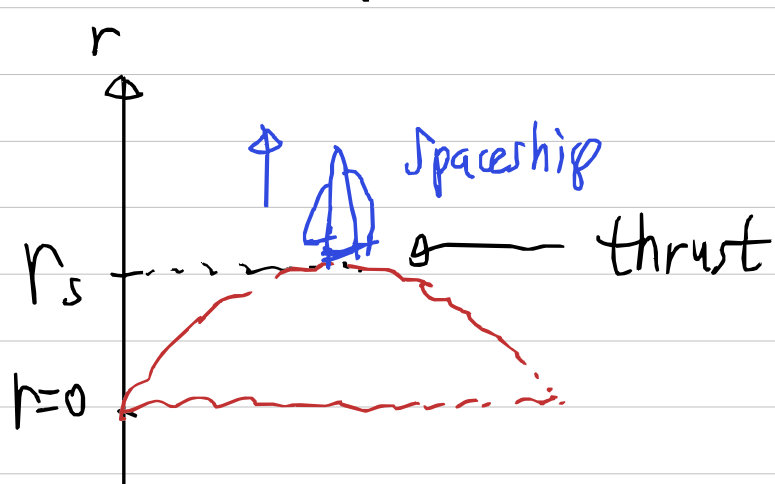
$$\text{freq } \nu_B.$$

$$\frac{\nu_B}{\nu_A} = \left(\frac{1 - \frac{r_s}{r_A}}{1 - \frac{r_s}{r_B}} \right)^{\frac{1}{2}}$$

As $r_A \rightarrow r_s$,
(w/ fixed r_B)
 $\nu_B \rightarrow 0$ ($\forall \nu_A$)

B (at fixed location) never actually sees A falling into the BH
(they see A frozen as $r_A \rightarrow r_s$)

(4) Hovering near horizon of a BH



$$r = R (> r_s)$$

$$R \rightarrow r_s$$

Const. angles

Const. radial position $r = R$

$$d\tau^2 = -\left(1 - \frac{r_s}{R}\right) dt^2 \rightarrow \dot{t} = \left(1 - \frac{r_s}{R}\right)^{-\frac{1}{2}}$$

4-vel. vec: $U^\mu = (\dot{t}, \dot{r}, \dot{\theta}, \dot{\phi})$
 $= \dot{t} \delta^\mu_0 = 0$

4-accel. vec: $a^\mu = D_\tau U^\mu$
 $= U^\alpha D_\alpha U^\mu$
 $= U^\alpha (\partial_\alpha U^\mu + \Gamma^\mu_{\alpha\beta} U^\beta)$

$$a^\mu = \dot{t} \delta^\mu_0 (\partial_\alpha \dot{t} \delta^\mu_0 + \Gamma^\mu_{\alpha 0} \dot{t})$$

$$= \dot{t} (\cancel{\delta^\mu_0 \partial_t \dot{t}} + \Gamma^\mu_{00} \dot{t})$$

$$\dot{t} = \left(1 - \frac{r_s}{R}\right)^{-\frac{1}{2}} \rightarrow \partial_t \dot{t} = 0$$

$$\Gamma^\mu_{00} = \delta^\mu_r \left(1 - \frac{r_s}{r}\right) \frac{r_s}{2r^2}$$

↑
p59

(4)

$$\Gamma_{rr}^{\mu} = -\delta_r^{\mu} \left(1 - \frac{r_s}{r}\right) \frac{r_s}{2r^2}$$

$$a^{\mu} = \delta_r^{\mu} \frac{1 - \frac{r_s}{R}}{1 - \frac{r_s}{R}} \frac{r_s}{2R^2} \quad (r=R)$$

$$= \delta_r^{\mu} \frac{r_s}{2R^2}$$

$$= \left(0, \frac{r_s}{2R^2}, 0, 0\right) \rightarrow 4\text{-vec.}$$

non-zero only in r -dir.

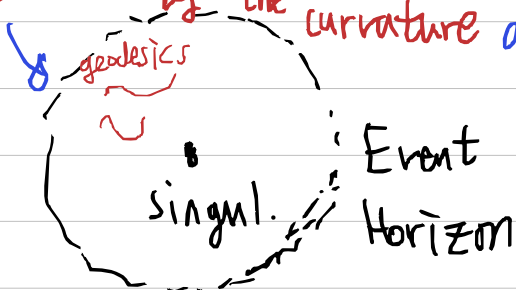
Scalar : $a^2 = g_{\alpha\beta} a^{\alpha} a^{\beta}$ $\rightarrow$ invariant under coord. transf.

$$\begin{aligned} r\text{-dir} : a^2 &= g_{rr} a^r a^r \\ &= \frac{1}{1 - \frac{r_s}{R}} \left(\frac{r_s}{2R^2} \right)^2 \\ &\quad \parallel \\ &\quad (a^r)^2 \end{aligned}$$

As $R \rightarrow r_s$, $a^2 \rightarrow \infty$

$\downarrow$ magnitude of acceleration.

2 geodesics $\rightarrow$ affected by the curvature as they go through the horizon.



(5)

(4) Life in a BH.

Inside the BH : $r < r_s$ 

proper time :

$$dt^2 = -\left(\frac{r_s}{r} - 1\right) dt^2 + \left(\frac{r_s}{r} - 1\right)^{-1} dr^2 - r^2 d\Omega^2$$

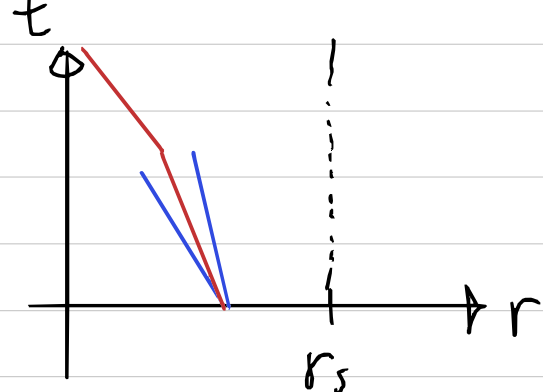
As it enters the BH's E.H.,

"radial" dim. = "timelike" dim.

"time" dim. = "spacelike" dim.

Center of the BH $\rightarrow$ located at $r=0$,

↳ is not "somewhere" in space, ("timelike" dir.)

but it's "sometime" in r (timelike).

To maximize the am. of proper time, you have left until hit the singul.

You need to have



Tidal forces extreme before hitting the singul.

$$\dot{t} = \dot{\theta} = \dot{\phi} = 0, \quad \dot{r} \neq 0$$

$$\dot{r}^2 = \left(\frac{r_s}{r} - 1\right)$$

$$\int d\tau = \int_{r_s}^0 \left(\frac{r_s}{r} - 1\right)^{-\frac{1}{2}} dr = \Delta\tau$$

$$= \frac{\pi}{2} r_s = \Delta\tau$$

: Maximum am. of proper time left in the BH.

$$3 \times 10^8 \text{ m/s}$$

$$10^3 \text{ m}$$

⑥

For a BH of $\sim 1 M_\odot$ (w/ $r_s \sim 3 \text{ km}$)

$$\begin{aligned} \text{MAX } \Delta\tau &= 15 \mu\text{sec} \\ &= 15 \times 10^{-6} \text{ s} \end{aligned}$$

* BH at center of galaxy $\rightarrow \text{Sgr A}^*$

$$M_{\text{Sgr A}^*} \approx 4.3 \times 10^6 M_\odot$$

$$r_{\text{(Sch)}} \text{Sgr A}^* = r_s \text{Sgr A}^* \approx 10^7 \text{ km}$$

$$\Delta\tau \approx 66 \text{ sec} \sim 1 \text{ min}$$

* GR breaks down at least when

$$R_{abcd} R^{abcd} = 12 \frac{r_s^2}{r^6} \sim \frac{1}{l_{\text{Planck}}^4}$$

$$l_p^2 \approx \frac{1}{M_{\text{Pl}}^2} = 8\pi G_N$$

$$\text{STOP : } r^6 \sim 12 r_s^2 l_p^4 \approx 12 r_s^2 (8\pi G_N)^2$$

$$\begin{aligned} \text{For Sgr A}^*, \rightarrow r &\approx 10^{-20} \text{ m} \\ &(\text{timescale} \sim 10^{-28} \text{ sec}) \end{aligned}$$

To maximize proper time in BH, we must have

$$U^\mu = (0, \dot{r}, 0, 0) \quad \Gamma_{rr}^\mu \dot{r}^2$$

$$a^\mu = U^\alpha \left[\partial_\alpha U^\mu + \Gamma_{\alpha\beta}^\mu U^\beta \right]$$

$$\hookrightarrow = \dot{r} \left(\partial_r \dot{r} \delta_r^\mu + \delta_r^\mu \left(\frac{r_s}{r} - 1 \right)^{-1} \dot{r} \frac{r_s}{2r^2} \right)$$

$$= \delta_r^\mu \dot{r} \left[\frac{1}{2} \frac{r_s}{r^2} \frac{1}{\sqrt{\frac{r_s}{r} - 1}} + \frac{r_s}{2r^2} \left(\frac{r_s}{r} - 1 \right)^{-1} \right]$$

$$\stackrel{\text{red}}{=} 0 = (0, 0, 0, 0)$$

Inwards

$$\dot{r} = - \left(\frac{r_s}{r} - 1 \right)^{\frac{1}{2}}$$

$$\Gamma_{rr}^\mu = \delta_r^\mu \left(\frac{r_s}{r} - 1 \right)^{-1} \frac{r_s}{2r^2}$$

Check

Lecture on GWs

EM

 A_μ

Gauge Invariance

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi$$

GR

↙ Minkowski

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

$$x^\mu \rightarrow \tilde{x}^\mu = \tilde{x}^\mu(x)$$

$$g_{\mu\nu} \rightarrow \tilde{g}_{\mu\nu} = g_{\alpha\beta} \frac{\partial x^\alpha \partial x^\beta}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu}$$

For linearized gravity,

$$x^\mu \rightarrow \tilde{x}^\mu = x^\mu + \xi^\mu$$

$$h_{\mu\nu} \rightarrow \tilde{h}_{\mu\nu} = h_{\mu\nu} + \underbrace{\partial_{(\mu} \xi_{\nu)}}_{\substack{\text{symm.} \\ \partial_\mu \xi_\nu + \partial_\nu \xi_\mu}}$$

$$\partial_\mu \tilde{F}^\mu{}_\nu = -\mu_0 J_\nu \quad \text{Maxwell's Eq.}$$

$$\square A_\nu - \partial_\nu \partial_\alpha A^\alpha = -\mu_0 J_\nu$$

$$\square \equiv \eta^{\mu\nu} \partial_\mu \partial_\nu = -\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2$$

In EM, we can choose "Lorenz" gauge on harmonic gauge

$$\partial_\mu A^\mu = 0 \quad \text{in that gauge}$$

$$\square A_\nu = -\mu_0 J_\nu \rightarrow \text{In vacuum, } J_\nu = 0$$

(Wave Eq.)

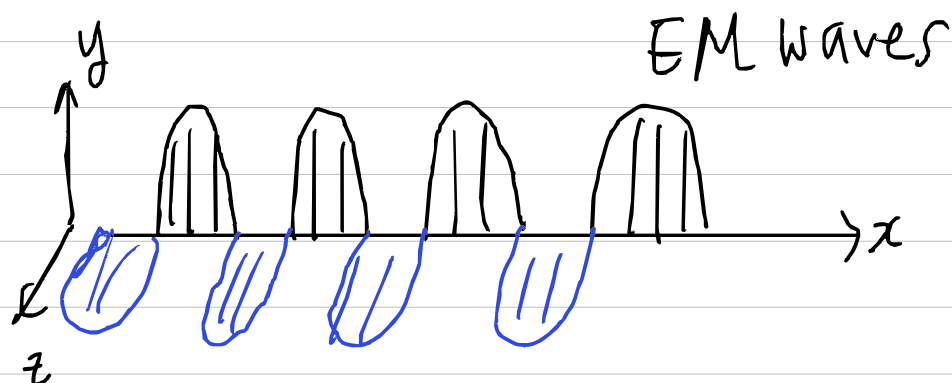
$$\square A_\nu = 0$$

(EX) Wave travelling in x-dir.

↪ we can use the residual gauge freedom

$$(\square \chi = 0) \quad \text{s.t.}$$

$$A_\mu(t, x) = \begin{pmatrix} 0 \\ 0 \\ A_y(t, x) \\ A_z(t, x) \end{pmatrix} \quad \left. \begin{matrix} A_0 = A_x = 0. \\ 2 \text{ indep. polarization} \\ \text{for EM waves} \\ (\text{E \& B polarizations of light}) \end{matrix} \right\}$$

transverse to
line of
propagation

$$-\partial_t^2 A_y + \partial_x^2 A_y = 0$$

$$A_y(t, x) = A_y^+(t-x) + A_y^-(t+x)$$

§4. Acceleration on Earth

Consider Massive particles $r = R > 2MG$ } const.
 $\theta = \theta_0, \varphi = \varphi_0$

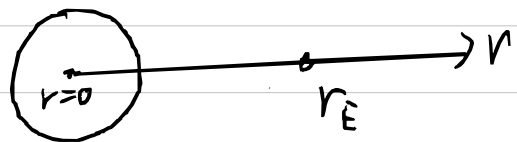
Parameter -

$$(a) \quad U^\mu = \frac{dx^\mu}{d\tau} = \left(\frac{dt}{d\tau}, 0, 0, 0 \right)$$

$$d\tau^2 = -ds^2 = \left(1 - \frac{2MG}{r} \right) dt^2 + 0$$

$$\rightarrow \frac{dt}{d\tau} = \left(1 - \frac{2MG}{r} \right)^{-\frac{1}{2}}$$

$$U^\mu = \left(\left(1 - \frac{2MG}{r} \right)^{-\frac{1}{2}}, 0, 0, 0 \right)$$



2 pol.s travelling at spd. of lgt. $c=1$ (2)

along $\pm x$ -dir. $\left(\underset{z}{A}_y^{\pm}(t-x) \right)$

- " $\left(\underset{z}{A}_y^{\mp}(t+x) \right)$

For Gravity, we can also use our gauge freedom

$\rightarrow$ left w/ only 2 indep. components. in 4D.

$\equiv$ 2 pol.s of GWs in 4D

$$\square J_\nu = 0 \quad (\text{in vacuum})$$

$$\square h_{\mu\nu} = 0 \quad \text{for GR (linearized)}$$

Choose de Donder gauge
(harmonic)

$$\partial_\mu h^\mu{}_\nu = \frac{\partial_\nu h^\alpha{}_\alpha}{2}$$

$\rightarrow$ Linearized Einstein's
Eqs in vacuum

$$\square h_{\mu\nu} = 0.$$

For a wave travelling in x -dir.,

we can fix the residual gauge freedom s.t.

$$h_{0\mu} = h_{x\mu} = 0 \quad (\mu = 0, 1, 2, 3)$$

$$h_{\mu\nu} = \left(\begin{array}{cc|cc} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & h_+ & h_x \\ 0 & 0 & h_x & -h_+ \end{array} \right)$$

$\rightarrow$ only 2-independent
compo.s in $h_{\mu\nu}$
(in 4D)

$$\square \underset{x}{h}_+(t, x) = 0 \quad \text{in Vacuum}$$

$$(-\partial_t^2 + \partial_x^2) \square \underset{x}{h}_+(t, x) = 0$$

2 pol.s $\perp$ ^(transverse) line of propagation

travelling at

$c=1$ along $\pm x$ -dir.s.

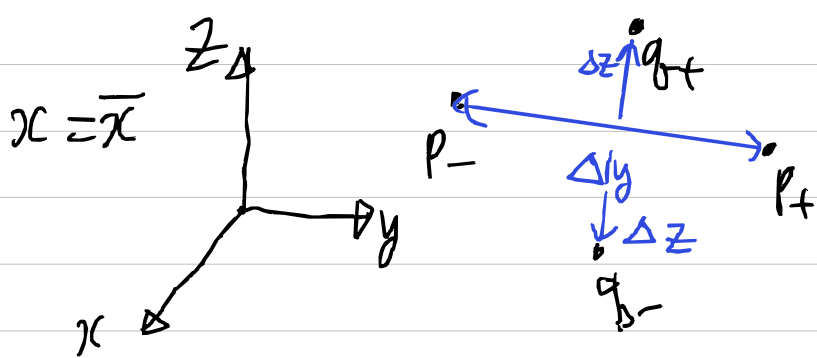
Effect of GWs on free-falling test particles (3)

$$ds^2 = (\eta_{\mu\nu} + h_{\mu\nu}) dx^\mu dx^\nu$$

$$= -dt^2 + dx^2 + (1 + h_+(t, x)) dy^2 + (1 - h_+(t, x)) dz^2 + 2h_x(t, x) dy dz$$

For a + -polarized GW (i.e., $h_x = 0$),

4 massive test particles



$$p_{\pm} : x_{p_{\pm}}^\mu = (t, \bar{x}, \pm \frac{1}{2} \Delta y, 0)$$

$$q_{\pm} : x_{q_{\pm}}^\mu = (t, \bar{x}, 0, \pm \frac{1}{2} \Delta z)$$

$$\Delta x_p^\mu = x_{p_+}^\mu - x_{p_-}^\mu = (0, 0, \Delta y, 0)$$

$$\Delta x_q^\mu = x_{q_+}^\mu - x_{q_-}^\mu = (0, 0, 0, \Delta z)$$

$$l_p^2 = g_{\mu\nu} \Delta x_p^\mu \Delta x_p^\nu \text{ is a scalar}$$

$$= (1 + h_+(t, x)) \Delta y^2$$

$$l_q^2 = (1 - h_+(t, x)) \Delta z^2$$

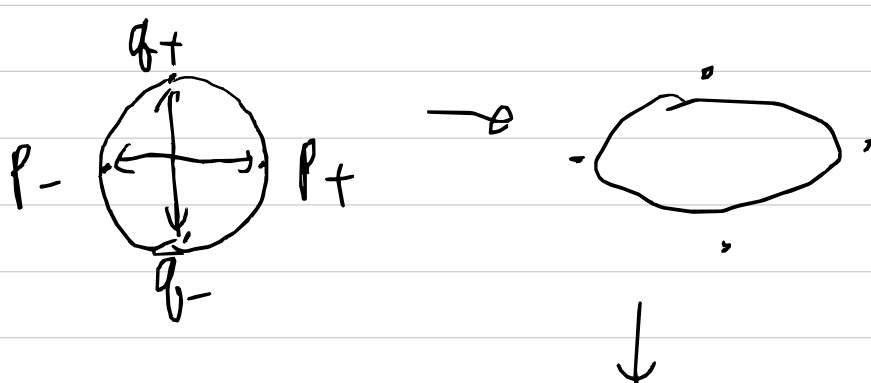
Scalars all observers agree on this
ST distance

→ Scalars
so can be corrected

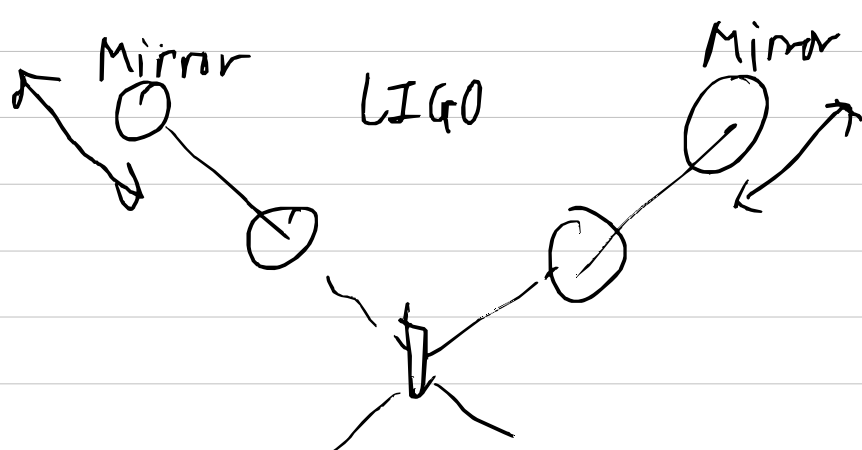
free-falling
→ their geodesic deviation → Riemann tensor R

plus
+ polarization

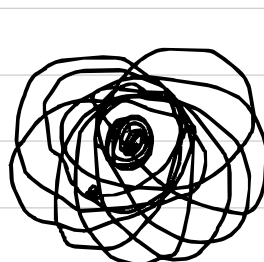
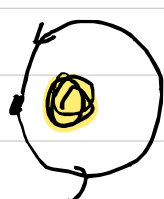
(4)



LIGO
↓
Mirrors
are free falling
particles



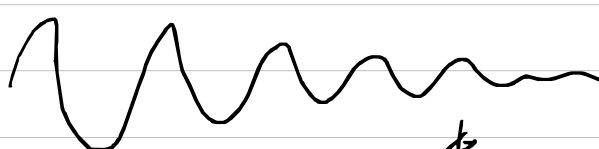
FRW
↑



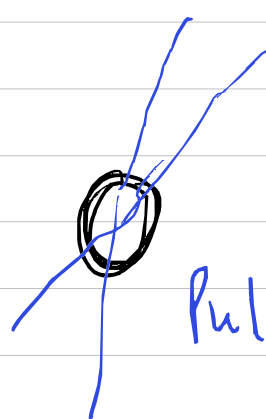
Falling
to one
another



In Newtonian grav.



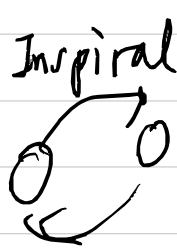
In reality
what we observe



Pulsars

Binary Pulsars

Period T decreases over time



Inspiral

Merger

Ringdown



Gravity in the "Strong" Regime (near BHs)

Pulsar Timing Array (PTA)

Primordial GWS	PTA	LISA	Simons LIGO
-------------------	-----	------	----------------

R O Y G B I V

①

GR PB To-do list

Claudia said: non-zero compo.s of Riemann tnsr.

→ PB Q9.20. & likes

#1. In the Newtonian frame

$$(t, x^i) \equiv (t, x, y, z) :$$

$$ds^2 = -(1+2\varphi) dt^2 + (1-2\varphi)(dx^2 + dy^2 + dz^2) + \dots$$

$$\varphi = \varphi(x^i) \longrightarrow \partial_0 \varphi = \partial_t \varphi = 0$$

(a) Check: $g_{\mu\nu} = \eta_{\mu\nu} - 2\varphi \zeta_{\mu\nu}$

$$\& \ g^{\mu\nu} = \eta^{\mu\nu} + 2\varphi \zeta^{\mu\nu} + \mathcal{O}(\varphi^2)$$

$$\zeta_{\mu\nu} = \text{diag}(1, 1, 1, 1) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

We can write the metric $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$.

s.t. $ds^2 = +g_{tt} dt^2 + g_{xx} dx^2 + g_{yy} dy^2 + g_{zz} dz^2$

So, $g_{tt} = -(1+2\varphi) = g_{00}$

$$g_{xx} = g_{yy} = g_{zz} = 1-2\varphi$$

(or $g_{ij} = (1-2\varphi)\delta_{ij}$ where $i, j = 1, 2, 3$)

So, $g_{\mu\nu} = \begin{bmatrix} -(1+2\varphi) & 0 & 0 & 0 \\ 0 & 1-2\varphi & 0 & 0 \\ 0 & 0 & 1-2\varphi & 0 \\ 0 & 0 & 0 & 1-2\varphi \end{bmatrix}$

$$\eta_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow \text{We now can write:}$$

Using the deviation from flat space $h_{\mu\nu}$ $\eta_{\mu\nu} + h_{\mu\nu} = g_{\mu\nu}$

$$h_{\mu\nu} = \begin{bmatrix} -2\varphi & 0 & 0 & 0 \\ 0 & -2\varphi & 0 & 0 \\ 0 & 0 & -2\varphi & 0 \\ 0 & 0 & 0 & -2\varphi \end{bmatrix} = -2\varphi \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= -2\varphi \zeta_{\mu\nu} \longrightarrow \therefore g_{\mu\nu} = \eta_{\mu\nu} - 2\varphi \zeta_{\mu\nu}$$

Since $g_{\mu\lambda} g^{\lambda\nu} = \delta_\mu^\nu :$

$$g^{\mu\nu} = \begin{bmatrix} -1-2\varphi & 0 & 0 & 0 \\ 0 & 1-2\varphi & 0 & 0 \\ 0 & 0 & 1-2\varphi & 0 \\ 0 & 0 & 0 & 1-2\varphi \end{bmatrix}$$

-1
reciprocal

$$= \begin{bmatrix} -\frac{1}{1+2\varphi} & 0 & 0 & 0 \\ 0 & \frac{1}{1-2\varphi} & 0 & 0 \\ 0 & 0 & \frac{1}{1-2\varphi} & 0 \\ 0 & 0 & 0 & \frac{1}{1-2\varphi} \end{bmatrix}$$

Using $g_{tt} = -\frac{1}{1+2\varphi} \approx -(1-2\varphi)$

& $g_{xx} = g_{yy} = g_{zz} = (1-2\varphi)^{-1} \approx (1+2\varphi)$

$$g^{\mu\nu} \approx \begin{bmatrix} -(1-2\varphi) & 0 & 0 & 0 \\ 0 & 1+2\varphi & 0 & 0 \\ 0 & 0 & 1+2\varphi & 0 \\ 0 & 0 & 0 & 1+2\varphi \end{bmatrix}$$

$$g^{\mu\nu} \approx \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2\varphi & 0 & 0 & 0 \\ 0 & 2\varphi & 0 & 0 \\ 0 & 0 & 2\varphi & 0 \\ 0 & 0 & 0 & 2\varphi \end{bmatrix}$$

②

From the matrix form of $g^{\mu\nu}$, using

$$g_{\mu\lambda} g^{\lambda\nu} = \delta_\mu^\nu$$

we assume: $g^{\lambda\nu} \approx \eta^{\lambda\nu} + h^{\lambda\nu}$

Plugging in $g_{\mu\lambda} = \eta_{\mu\lambda} + h_{\mu\lambda}$:

$$(\eta_{\mu\lambda} + h_{\mu\lambda})(\eta^{\lambda\nu} + h^{\lambda\nu}) = \delta_\mu^\nu$$

$$\text{LHS} = (\eta_{\mu\lambda} - 2\phi \xi_{\mu\lambda})(\eta^{\lambda\nu} + h^{\lambda\nu})$$

$$= \underbrace{\eta_{\mu\lambda} \eta^{\lambda\nu}}_{\delta_\mu^\nu} - 2\phi \xi_{\mu\lambda} \eta^{\lambda\nu} + \eta_{\mu\lambda} h^{\lambda\nu} - 2\phi \underbrace{\xi_{\mu\lambda} h^{\lambda\nu}}_{\text{This term will contain } \phi^2}$$

Ignoring terms of order ϕ^2 or higher:

$$\begin{aligned} \text{LHS} &= \cancel{\delta_\mu^\nu} - 2\phi \xi_{\mu\lambda} \eta^{\lambda\nu} + \eta_{\mu\lambda} h^{\lambda\nu} - \mathcal{O}(\phi^2) \\ &= \cancel{\delta_\mu^\nu} (= \text{RHS}) \end{aligned}$$

$$\eta_{\mu\lambda} h^{\lambda\nu} = 2\phi \xi_{\mu\lambda} \eta^{\lambda\nu} + \mathcal{O}(\phi^2)$$

$$\eta^{\sigma\mu} \eta_{\mu\lambda} h^{\lambda\nu} = 2\phi \xi_{\mu\lambda} \eta^{\lambda\nu} \eta^{\sigma\mu} + \mathcal{O}(\phi^2)$$

$$\delta^\sigma_\lambda h^{\lambda\nu} = 2\phi \xi^{\sigma\nu} + \mathcal{O}(\phi^2)$$

$$h^{\sigma\nu} = 2\phi \xi^{\sigma\nu} + \mathcal{O}(\phi^2)$$

$$\boxed{h^{\mu\nu} = 2\phi \xi^{\mu\nu} + \mathcal{O}(\phi^2)} \quad (\sigma \rightarrow \mu)$$

where $\xi^{\mu\nu} = \text{diag}(1, 1, 1, 1)$

$$h^{\mu\nu} = \begin{bmatrix} 2\phi & 0 & 0 & 0 \\ 0 & 2\phi & 0 & 0 \\ 0 & 0 & 2\phi & 0 \\ 0 & 0 & 0 & 2\phi \end{bmatrix} + \mathcal{O}(\phi^2)$$

so the inverse metric $g^{\mu\nu}$:

$$g^{\mu\nu} \approx \eta^{\mu\nu} + h^{\mu\nu}$$

$$\approx \eta^{\mu\nu} + 2\phi \xi^{\mu\nu} + \mathcal{O}(\phi^2)$$

#1 (b) The Christoffel symbols :

(3)

$$\Gamma_{\nu\rho}^{\mu} = \frac{1}{2} g^{\mu\alpha} \left(\overset{(1)}{\partial_{\nu}} g_{\alpha\rho} + \overset{(2)}{\partial_{\rho}} g_{\alpha\nu} - \overset{(3)}{\partial_{\alpha}} g_{\nu\rho} \right)$$

Using the form $g_{\mu\nu} = \eta_{\mu\nu} - 2\phi \xi_{\mu\nu}$,

$$\begin{aligned} (1) : \partial_{\nu} g_{\alpha\rho} &= \partial_{\nu} (\eta_{\alpha\rho} - 2\phi \xi_{\alpha\rho}) = \cancel{\partial_{\nu} \eta_{\alpha\rho}} - 2(\partial_{\nu} \phi) \xi_{\alpha\rho} - 2\phi \cancel{(\partial_{\nu} \xi_{\alpha\rho})} \\ &= -2(\partial_{\nu} \phi) \xi_{\alpha\rho} \end{aligned}$$

$$(2) : \partial_{\rho} g_{\alpha\nu} = -2(\partial_{\rho} \phi) \xi_{\alpha\nu}$$

$$(3) : -\partial_{\alpha} g_{\nu\rho} = +2(\partial_{\alpha} \phi) \xi_{\nu\rho}$$

$$\text{So, } \Gamma_{\nu\rho}^{\mu} = \frac{1}{2} g^{\mu\alpha} \left[-2(\partial_{\nu} \phi) \xi_{\alpha\rho} - 2(\partial_{\rho} \phi) \xi_{\alpha\nu} + 2(\partial_{\alpha} \phi) \xi_{\nu\rho} \right]$$

$$= g^{\mu\alpha} \left[-(\partial_{\nu} \phi) \xi_{\alpha\rho} - (\partial_{\rho} \phi) \xi_{\alpha\nu} + (\partial_{\alpha} \phi) \xi_{\nu\rho} \right]$$

$$\text{Since } g^{\mu\alpha} = \eta^{\mu\alpha} + 2\phi \xi^{\mu\alpha} + \mathcal{O}(\phi^2)$$

Plugging this into $\Gamma_{\nu\rho}^{\mu}$:

$$\Gamma_{\nu\rho}^{\mu} = (\eta^{\mu\alpha} + 2\phi \xi^{\mu\alpha}) \left[-\phi_{,\nu} \xi_{\alpha\rho} - \phi_{,\rho} \xi_{\alpha\nu} + \phi_{,\alpha} \xi_{\nu\rho} \right] + \mathcal{O}(\phi^2)$$

$$= \eta^{\mu\alpha} [\dots] - 2\phi \phi_{,\nu} \xi^{\mu\alpha} \xi_{\alpha\rho} - 2\phi \phi_{,\rho} \xi^{\mu\alpha} \xi_{\alpha\nu} + 2\phi \phi_{,\alpha} \xi^{\mu\alpha} \xi_{\nu\rho} + \mathcal{O}(\phi^2)$$

$$= \eta^{\mu\alpha} [\dots] - (\phi^2)_{,\nu} \xi^{\mu\alpha} \xi_{\alpha\rho} - (\phi^2)_{,\rho} \xi^{\mu\alpha} \xi_{\alpha\nu} + (\phi^2)_{,\alpha} \xi^{\mu\alpha} \xi_{\nu\rho} + \mathcal{O}(\phi^2)$$

$$= \eta^{\mu\alpha} [\dots] + \mathcal{O}(\phi^2)$$

$$\left(\partial_{\rho} (\phi^2) = \frac{\partial \phi}{\partial \rho} \frac{\partial (\phi^2)}{\partial \phi} \right)$$

$$\therefore \Gamma_{\nu\rho}^{\mu} = \eta^{\mu\alpha} \left[-\phi_{,\nu} \xi_{\alpha\rho} - \phi_{,\rho} \xi_{\alpha\nu} + \phi_{,\alpha} \xi_{\nu\rho} \right] + \mathcal{O}(\phi^2)$$

#1 (c) Riemann Tensor

④

$$R_{\mu\nu\rho\sigma} = \partial_\rho \Gamma_{\mu\sigma}^\lambda - \partial_\sigma \Gamma_{\mu\rho}^\lambda + \Gamma_{\rho\alpha}^\lambda \Gamma_{\mu\sigma}^\alpha - \Gamma_{\sigma\alpha}^\lambda \Gamma_{\mu\rho}^\alpha$$

Since we want to find the leading-order terms in φ , we can ignore the quadratic terms $\Gamma_{\rho\alpha}^\lambda \Gamma_{\mu\sigma}^\alpha$ & $\Gamma_{\sigma\alpha}^\lambda \Gamma_{\mu\rho}^\alpha$:

$$\begin{aligned} \Gamma_{\rho\alpha}^\lambda \Gamma_{\mu\sigma}^\alpha &= \eta^{\lambda\beta} \left(-\varphi_{,\alpha} \xi_{\beta\rho} - \varphi_{,\rho} \xi_{\beta\alpha} + \varphi_{,\beta} \xi_{\rho\alpha} \right) \\ &\quad \times \eta^{\alpha\tau} \left(-\varphi_{,\sigma} \xi_{\tau\mu} - \varphi_{,\mu} \xi_{\tau\sigma} + \varphi_{,\tau} \xi_{\mu\sigma} \right) \\ &= \mathcal{O}(\varphi^2) \end{aligned}$$

$$\begin{aligned} \Gamma_{\sigma\alpha}^\lambda \Gamma_{\mu\rho}^\alpha &= \eta^{\lambda\beta} \left(-\varphi_{,\alpha} \xi_{\beta\sigma} - \varphi_{,\sigma} \xi_{\beta\alpha} + \varphi_{,\beta} \xi_{\sigma\alpha} \right) \\ &\quad \times \eta^{\alpha\tau} \left(-\varphi_{,\rho} \xi_{\tau\mu} - \varphi_{,\mu} \xi_{\tau\rho} + \varphi_{,\tau} \xi_{\mu\rho} \right) \\ &= \mathcal{O}(\varphi^2) \end{aligned}$$

So, we compute only these:

$$(1) \quad \partial_\rho \Gamma_{\mu\sigma}^\lambda = \eta^{\lambda\alpha} \left(-\varphi_{,\sigma\rho} \xi_{\alpha\mu} - \varphi_{,\mu\rho} \xi_{\alpha\sigma} + \varphi_{,\alpha\rho} \xi_{\mu\sigma} \right)$$

$$\text{where } \varphi_{,\sigma\rho} = \partial_\sigma (\partial_\rho \varphi)$$

$$\varphi_{,\mu\rho} = \partial_\rho (\partial_\mu \varphi)$$

We use

$$\left(\eta^{\lambda\alpha} \xi_{\alpha\mu} = \xi_{\mu}^{\lambda} \right)$$

$$(2) \quad \partial_\sigma \Gamma_{\mu\rho}^\lambda = \eta^{\lambda\alpha} \left(-\varphi_{,\rho\sigma} \xi_{\alpha\mu} - \varphi_{,\mu\sigma} \xi_{\alpha\rho} + \varphi_{,\alpha\sigma} \xi_{\mu\rho} \right)$$

$$R_{\mu\nu\rho\sigma} \approx \partial_\rho \Gamma_{\mu\sigma}^\lambda - \partial_\sigma \Gamma_{\mu\rho}^\lambda$$

$$\begin{aligned} &\approx -\cancel{\varphi_{,\sigma\rho} \xi_{\mu}^{\lambda}} - \varphi_{,\mu\rho} \xi_{\sigma}^{\lambda} + \varphi_{,\alpha\rho} \eta^{\lambda\alpha} \xi_{\mu\sigma} \\ &\quad + \cancel{\varphi_{,\rho\sigma} \xi_{\mu}^{\lambda}} + \varphi_{,\mu\sigma} \xi_{\rho}^{\lambda} - \varphi_{,\alpha\sigma} \eta^{\lambda\alpha} \xi_{\mu\rho} \end{aligned}$$

$$R_{\mu\nu} = R^{\lambda}_{\mu\lambda\nu}$$

Plug in $R_{\alpha\mu\lambda\nu} = \varphi_{,\alpha\lambda} \xi_{\mu\nu} + \varphi_{,\mu\nu} \xi_{\alpha\lambda} - \varphi_{,\mu\lambda} \xi_{\alpha\nu} - \varphi_{,\alpha\nu} \xi_{\mu\lambda} :$

$$\begin{aligned} R_{\mu\nu} &= \eta^{\lambda\alpha} R_{\alpha\mu\lambda\nu} \\ &= \eta^{\lambda\alpha} \left(\varphi_{,\alpha\lambda} \xi_{\mu\nu} + \varphi_{,\mu\nu} \xi_{\alpha\lambda} - \varphi_{,\mu\lambda} \xi_{\alpha\nu} - \varphi_{,\alpha\nu} \xi_{\mu\lambda} \right) \end{aligned}$$

#1(d) In a vacuum region, there's no matter, i.e. $T_{\mu\nu} = 0$.

$$\text{So, } G_{\mu\nu} = 0$$

#1(e) $T_{00} = -\rho$

$$h_{00} \approx -2\phi$$

$$G_{00} \approx \nabla^2 \phi$$

$$\nabla^2 \phi = 4\pi G\rho$$

$$\nabla^2 \phi = -K\rho$$

$$K = -4\pi G ?$$